Society in Silico

Simulating futures to build the one we want.

By Max Ghenis. This listen edition drops citations, tables, and editorial marks — the web edition at societyinsilico.com carries them all.

Preface

In 2019 I wanted a number nobody would give me. The question was simple to state: if the United States paid every adult a basic income, how many families would that pull over the poverty line, and what would it cost once you netted out the programs it replaced? Washington had models that could answer it — the Congressional Budget Office's, the Joint Committee on Taxation's, the Treasury's; outside government, the Tax Policy Center had one too. I could read their reports. I could not ask their models my question, because the models lived behind institutional walls, most ran on confidential data, and all answered only to their owners.

So I built. First a scrappy research shop called the UBI Center, which ran analyses on the open-source tools that existed. Then, with Nikhil Woodruff — a college student I found on a subreddit, who turned out to pair real economics with real engineering — PolicyEngine: policy simulation for anyone with a web browser. (Footnote: A disclosure that doubles as a plot point: as of July 2026, Nikhil serves in the UK government at 10 Downing Street. When this book describes British institutions evaluating tools he helped build, read it knowing where he sits now.) Type in your household, move a policy lever, watch your taxes and benefits recompute. No credentials, no paywall, no waiting.

Building it raised questions I had not planned to spend my thirties on. If you can encode tax law, what else can you encode? When an AI assistant answers a benefits question — and it will, because that is where people now ask — who builds the calculator it should be calling instead of guessing? If simulation can say what a policy does to a family's budget, can it say what the public thinks about it? And if machines can hold the rules, the people, and the predictions all at once, what keeps any of it honest?

That last question runs the book. My answer, stated up front so you can hold me to it: a simulation deserves your attention only where you can check it against something real — a statute, a census total, an official number that arrives later and grades the forecast. Everything I describe here either passes that test, fails it in the open, or gets labeled as the speculation it is.

What this book is

A story about tools and the people who built them. It starts in 1957, when an economist named Guy Orcutt proposed simulating a nation household by household, decades before computers could. It runs through the era when such models existed but belonged to institutions — mainframes at the Urban Institute, black boxes at the CBO — and through the open-source turn that put them in browsers. It ends somewhere I did not expect when I started writing: AI agents encoding entire countries' tax codes in a day and proving their work against the reference models governments already trust.

That ending rearranged the book under me. A chapter I drafted in 2025 described infrastructure that did not exist and said so plainly. By mid-2026 the infrastructure existed, and the chapter had to become a field report. I have kept the labels current throughout — this works, this is early, this is speculation — because the boundary between them is the book's real subject, and because the speed at which things crossed it is the story.

What this book is not

Not a textbook; when I include technical detail, it is there to make an idea land, and the references point to proper treatments. Not a memoir; my path supplies connective tissue, and the story is bigger than it. Not a manifesto; I hold views and argue them, but the chapters on simulated opinion and simulated democracy give the strongest counterarguments I could find, because those ideas deserve adversaries.

And not finished. The Axiom Foundation launches this summer. The Thesis Institute follows in the fall, and its first forecasts will take months and years to grade — as I write, not one has resolved. Some of what you read here will look different in two years. Between my first draft and this one, it already does.

Who this book is for

Three readers, any one lens enough. Policy people who want to know how computation is changing their trade, with no code required. Technologists building AI, fintech, or data systems who need to understand why this domain punishes shortcuts. And anyone who has wondered why the chatbot gets tax questions wrong, why budget models are secret, or what it would mean to run society's experiments in silicon before running them on people.

How to read it

Five parts. Part I is history: how government built an analytic machine the public could not inspect, and how you would even know whether such a machine is right. Part II is PolicyEngine: rebuilding that machine in the open, for one household and for a country. Part III is the turn everything else follows from: AI agents writing verified law-as-code at scale, the discipline that makes their work trustworthy, and what happened when the method reached countries that never had public models at all. Part IV crosses from calculation to prediction — the uncertainty that point estimates hide, and a public scoreboard that waits for reality to grade it. Part V walks to the speculative edge, simulated democracy and simulated values, and returns to the choice the introduction opens.

Impatient readers: introduction, then Part III, then whatever pulls you. The history repays the time, but the book tolerates skipping.

I wrote this book because the tools we use to understand society decide what we can see about it, and for fifty years the best tools belonged to the few institutions that could afford them. That era is ending. What replaces it — open machinery anyone can inspect and check, or closed machinery with a friendlier interface — is being decided now, in code, in procurement contracts, and in choices that look technical but are not.

The number nobody would give me in 2019 is now anyone's to compute — in a browser, with the machinery in plain sight.

Max Ghenis Washington, DC, 2026

Introduction: The model and the world

"For the first time, history has an author."

Engerraund Serac says it with pride. He is the villain of Westworld's third season: after watching Paris burn in a thermonuclear strike, he builds Rehoboam, an AI that models every human life and predicts its course — when you will lose your job, when you will get sick, when you will die. Prediction slides into control. The system starts arranging outcomes to match its forecasts, and people who wander off their assigned trajectories get flagged for "reconditioning." The show's premise is that a person is "just a brief algorithm," compressible to code. Its horror is simpler: history has an author, the author has a model, and no one else knows either exists.

I watched this in 2020 and recognized the machinery. I had been building models of society since 2018 — smaller, tamer ones that predict how a tax change lands on millions of simulated households. The writers had taken tools I used every day and followed them to their darkest terminus.

They had also, without meaning to, posed a design question that this book spends seventeen chapters answering. The dangerous thing about Rehoboam was never the simulation. It was the secrecy, the monopoly, and the absence of anything that could tell the system it was wrong. Governments already score citizens for fraud risk and bail decisions with models no defendant can inspect. Insurers price your premiums with algorithms they will not explain. The machinery of prediction exists and is spreading; the only live question is its architecture. Closed, proprietary, graded by no one? Or open, inspectable, and checked — line by line — against the world it claims to describe?

That fork is the book.

The AI can't do your taxes

Start with a fact that surprises people who expect artificial intelligence to be good at arithmetic.

In July 2025, the tax-software company Column Tax published TaxCalcBench, a test of whether frontier AI models could complete real federal tax returns. The models got everything a preparer would get: the taxpayer's full information, the rules, unlimited time to think. The best performer, Google's Gemini 2.5 Pro, produced a correct return less than a third of the time. Claude Opus 4 managed 27 percent. The errors followed a pattern: the models computed tax from bracket percentages instead of looking up the IRS tax tables, drifting $3 to $5 off on return after return; they fumbled credit eligibility; they lost track of supplementary forms. Software that costs fifty dollars does this flawlessly. The most capable AI systems on Earth could not.

Researchers had seen it coming. In 2023, a Johns Hopkins and Maryland team found GPT-4 answered only 67 percent of true-or-false tax questions correctly — better than a coin, useless for a filing. And it has not aged away. By mid-2026, PolicyEngine's own benchmark, PolicyBench, was scoring two dozen frontier and open models on complete household tax-and-benefit calculations; the best model still got roughly one in nine households wrong by more than a dollar, and the typical model missed about one in four. A dollar sounds like a rounding slip until you see what the misses are: the wrong Medicaid eligibility, the wrong food-assistance amount.

The limitation is structural. Language models learn patterns from text; tax law is not a pattern, it is a specification. It changes every year, differs across fifty states, and turns on interactions among dozens of provisions that no amount of pretraining can memorize into reliability. More training data does not fix a specification problem.

Tools fix it. AI systems do not memorize multiplication tables; they call calculators. They do not learn orbital mechanics by reading about rockets; they call physics engines. For the same reason, an AI answering a tax question should call a tax engine — software that encodes the rules exactly, cites the statute behind every number, and returns answers that can be checked. When people ask what this book is about technically, this is the one-sentence version: teaching machines to call verified models of society instead of guessing.

The catch, and the reason there was a book's worth of work to do: the tool did not exist. Avalara computes sales tax. ADP runs payroll. TurboTax files returns. Nothing answered the whole question — for this household, in this state, all taxes and all benefits, every interaction, and what changes if the law does. That question has a sixty-nine-year-old name: microsimulation. Its story starts in 1957 and runs straight into the benchmark failures above.

Why the story turns now

Three developments converged to make this decade different from every previous era of policy analysis.

When someone wonders how much Child Tax Credit they will get, they increasingly ask a chatbot, not IRS.gov: AI became the interface. Whatever the assistant says, someone may act on. The infrastructure those assistants should consult — accurate, current, citable — either exists as public plumbing or every AI company improvises its own, badly.

The rules those assistants must navigate kept compounding. The tax code has roughly doubled in length since 1985, and it interlocks with dozens of benefit programs whose rules shift by state and by year. A single parent earning $35,000 faces genuinely different arithmetic in Texas and New York, and the difference can run to thousands of dollars. No unaided human navigates this; the question is only which computer helps, and who can see inside it.

And the answer to both was growing up in public. What began as academic code became working infrastructure: by 2025 PolicyEngine covered federal taxes plus all fifty states, the United Kingdom, and Canada, built by more than a hundred contributors; EUROMOD spanned twenty-seven European countries; OpenFisca ran on four continents. The question stopped being whether open policy simulation could work and became whether it would be the shared foundation — for AI systems, for governments, for citizens — or a curiosity beside the closed models that still decide budgets.

What a model shows you

Picture a single mother in Ohio earning $45,000 with two kids, offered a promotion to $55,000. Obviously she should take it.

Run the rules and the obvious answer wobbles. Across that income range her Earned Income Tax Credit phases out, her food assistance shrinks, and — depending on her county and her childcare subsidy — a benefit can vanish at a hard threshold. Of each new dollar, the tax-benefit system quietly takes back most of it; cross the wrong line and the raise leaves her family with less than before. Millions of American families sit somewhere on this labyrinth, and none of it is visible in an average. You can only see it by applying every bracket, phase-out, and eligibility test to her household in particular — and then to a few hundred thousand more households, to learn how many share her trap.

That is microsimulation: compute the law on each simulated family, then add up. Guy Orcutt proposed it in 1957 after concluding that models of national aggregates could never answer the questions that mattered, because two economies with identical averages can contain utterly different lives. His idea became the machinery behind every serious answer to "who wins and who loses" — and for fifty years that machinery sat inside the CBO, the Treasury, and a handful of institutions, where you could read its outputs but never check its work.

This book is about what happened when that stopped being true: first slowly, through open source, and then all at once, when AI agents learned to build and verify the models themselves.

The stakes

When Congress debated extending the 2017 tax cuts in 2025, cost estimates ranged from four to five trillion dollars depending on modeling assumptions — a spread wider than most federal agencies' budgets, on the single largest fiscal decision of the decade. The official scorekeepers used models the public cannot inspect. Independent groups produced competing numbers from different data and methods. Members of Congress quoted whichever figure served, and voted; the extension passed that July. Nothing in the process could tell a voter which number deserved belief, or why.

It does not have to work like that. Estimates can come from open models, where a disagreement traces to a named assumption instead of dissolving into "methodological differences." Uncertainty can be printed instead of hidden: "$4.5 trillion, plus or minus $700 billion, and here is the assumption doing the work." A family can run the proposal on their own circumstances before their representative votes on it. An AI assistant can cite a calculation instead of hallucinating one, because a public engine exists for it to call. None of this is hypothetical anymore; the chapters ahead describe each piece running.

But "running" is a claim, and this book holds claims to a rule that chapter 3 states and the rest enforce: a simulation is admissible only where its verification chain ends in ground truth. A rule proves itself against the statute and against independent calculators. A synthetic population proves itself against the census. A forecast proves itself when the official number lands and grades it — and until then it is a bet, labeled as one. The alternative to that discipline is Serac's machine with better branding: fluent, confident, and answerable to nothing.

Four questions

Everything the book covers is machinery for answering four questions, in rising order of difficulty. What does the law say? Who are the people? What will happen? What do we want? The first has exact answers a computer can hold. The second has statistical answers a census can check. The third has probabilistic answers reality eventually grades. The fourth belongs to humans — and the book's final chapters ask, carefully, whether simulation can help us hear ourselves answer it without ever answering it for us.

The five parts follow that ladder:

- **Part I — The closed stack.** Orcutt's idea; the institutional models that ran the country's fiscal arguments from behind closed doors; the question that disciplines everything — how would you know whether any model is right? — and the family arithmetic that turned the question personal.

- **Part II — The open engine.** Building PolicyEngine: what the engine shows one household, what it shows a country, and the three problems every such model must solve.
- **Part III — The agent turn.** AI agents writing verified law-as-code at scale; the gates and oracles that make machine-encoded law trustworthy; five African tax systems encoded and checked in a week; and why the organization rebuilt itself around the epistemology.
- **Part IV — The prediction pole.** The uncertainty point estimates hide, the public scoreboard built to price it, and what language models actually add to opinion research once you benchmark them honestly.
- **Part V — The horizon.** Simulated democracy and simulated values, labeled as the research directions they are, and the return to the fork.

Two warnings before we start. I built many of the tools this book describes — PolicyEngine, the PolicyBench benchmark, the HiveSight and Democrasim experiments of Parts IV and V — and I am not a neutral witness; where the work is mine, I say so, and where it failed, I show the failure. And the boundary between "runs today" and "someday, maybe" moved faster during the writing than in any year of my career — so every chapter marks which side of it each claim sits on, as of the date on the preface.

The technology here will not save democracy or end poverty. What it can do is narrower and worth having: make the invisible visible — the cliff hidden in a benefit formula, the uncertainty behind a confident headline number, the difference between a model someone owns and a model anyone can check. What we do with the visibility is the political question, and it stays ours.

This is the case for the open path.

Part I: The closed stack

Chapter 1: The birth of microsimulation

In 1957, an economist named Guy Orcutt opened a paper with a verdict on his own profession: "Existing models of our socio-economic system have proved to be of rather limited predictive usefulness". The journal was the *Review of Economics and Statistics* — respectable, not glamorous — and the paper, "A New Type of Socio-Economic System," was dense with equations and read by almost nobody. What it proposed was a simulated United States, built one household at a time, on computers that could not yet run it.

The idea arrived in three acts. In 1957 Orcutt proposed it. In 1961 he and three co-authors published a working demonstration — limited, but running. In 1975 the Urban Institute completed DYNASIM, the first full-scale version, eighteen years after the proposal. Whatever Orcutt saw in 1957, he judged it worth a career.

The engineer

Orcutt took a physics degree from the University of Michigan in 1939, finished it in three years, and began graduate work in economics the same fall; his Ph.D. came in 1944. He never really left the lab bench. Inspired by Jan Tinbergen's econometric models of national economies, he designed an analogue electrical-mechanical "regression analyzer" in his dissertation and built the prototype at MIT — a machine of circuits and dials that ground out statistical estimates decades before software did. Near the end of his life he wrote of his early fascination with science and his transition from engineering to economics, and the order of those words is right: the engineer never left the economist.

By his early thirties he had the credentials of a rising star in exactly the tradition he would later attack. In 1949, with Donald Cochrane, he published a method for handling serial correlation in regression — the Cochrane-Orcutt procedure, still taught today. He directed the Littauer Statistical Laboratory at Harvard. He consulted for the Federal Reserve Board and the International Monetary Fund. The historian Chung-Tang Cheng, who wrote the intellectual biography, describes the ambition Orcutt carried through those years as a "Tinbergen dream": one model that could capture an entire national economy.

The dream soured on contact with the instruments. The dominant approach in 1950s economics was macroeconomic modeling — systems of equations describing relationships among aggregates: total consumption as a function of total income, investment as a function of interest rates, employment as a function of output. The Cowles Commission, then the leading center for mathematical economics, was refining Keynesian simultaneous-equation models of exactly this kind, and Milton Friedman was arguing that their forecasts beat naive extrapolation not at all. Orcutt's complaint cut deeper than Friedman's. The models could say something about whether GDP would grow. They could say nothing about what would happen to actual people — which families a recession would break, which a tax change would reach — and that silence covered precisely the questions a government exists to answer. He had spent years refining instruments that answered the wrong question with steadily improving precision.

The aggregation problem

Orcutt located the flaw in the act of aggregation itself — not a limitation of the aggregate models but a mathematical error inside them.

Take a hundred people and a rule that turns each person's input X into an output of X squared. If every person has X = 1, total output is 100. Now split them: fifty people at X = 0, fifty at X = 2. The average input is still 1, so a model built on the average still predicts 100. Compute person by person instead. The fifty zeros contribute nothing; the fifty twos contribute four apiece; the total is 200.

The aggregate is identical; the truth is twice as large.

Whenever the relationship between input and output bends — whenever it is nonlinear — the average of the outcomes is not the outcome of the average. And tax-and-benefit law is nonlinearity in bulk: brackets, thresholds, phase-outs, eligibility tests, cliffs where a dollar of earnings switches off thousands of dollars of support. A rule like that applied to an "average household" produces an answer for a household that does not exist, while the households that do exist land on every side of every threshold. This is the aggregation problem, and Orcutt stated it without mercy: "There is an inherent difficulty, if not practical impossibility, in aggregating anything but absurdly simple relationships about elemental decision-making units."

His alternative was to stop aggregating relationships and start aggregating people. Build the model from "interacting units which receive inputs and generate outputs" — individuals, households, firms — each carrying characteristics drawn from real data, each following rules of behavior. "The most distinctive feature of this new type of model," he wrote, "is the key role played by actual decision-making units of the real world such as the individual, the household, and the firm". The totals economists cared about would still exist, but they would come out of the model instead of going in: "Predictions about aggregates will still be needed but will be obtained by aggregating behavior of elemental units rather than by attempting to aggregate behavioral relationships."

That inversion is the whole breakthrough. Compute the law on each simulated family; sum the results; the aggregate is whatever the people add up to. Orcutt named the method "microanalytic simulation." Practitioners shortened it to microsimulation, and the name stuck.

A science the opposite of Seldon's

Science fiction got to population-scale prediction first, and got it backwards in a way worth pausing on. Isaac Asimov's *Foundation* trilogy, published from 1951 to 1953, imagined "psychohistory," a mathematics that predicted the future of a galactic civilization — with a built-in limit. It worked only on masses. "The reaction of one man could be forecast by no known mathematics; the reaction of a billion is something else again". Psychohistory was also a monopoly: legible to one man, steered by a secret foundation, invisible to the billions it described.

Orcutt's method runs the other way. It starts with the one man — the one household — and reaches the billion only by addition. Nothing about it requires secrecy or priesthood: if any household can be simulated, then anyone can, in principle, ask how a policy lands on families like their own and check the answer against a life they know. The fictional science was aristocratic by mathematical necessity. The real one is granular and, in principle, democratic. Whether it would be democratic in practice is the story of the next two chapters.

What it was in 1957 was impossible. Computers filled rooms, cost millions, and were programmed with punch cards in assembly language. The IBM 704, state of the art when the paper appeared, managed roughly 12,000 floating-point operations per second; a modern laptop does billions. Simulating millions of households, each with its own attributes and transitions, sat far beyond the machines — and Orcutt said so himself, with the driest understatement in the paper: "The problem of keeping track of all possible paths and their respective probabilities appears rather appalling." The ambition had company — John von Neumann had already cast weather forecasting as a computing problem, and Herbert Simon and Allen Newell were writing the first artificial-intelligence programs at the Carnegie Institute of Technology — but company is not feasibility.

Orcutt knew it was hard. He published anyway.

First demonstrations

The first act took four years to reach the second. In 1961, with Martin Greenberger, John Korbel, and Alice Rivlin — Rivlin was his doctoral student — Orcutt published *Microanalysis of Socioeconomic Systems: A Simulation Study*, a book demonstrating a working, if limited, microsimulation of the American economy. Simulated people were born, died, entered and left the labor force, and earned income through time. It was small and it creaked, but it ran: the 1957 paper's central claim — that you could compute a society from its units — was no longer hypothetical.

Hold on to Rivlin's name. Fourteen years after the book, she became the first director of the Congressional Budget Office, the agency whose estimates now govern every serious fiscal argument in Washington; later she served as vice chair of the Federal Reserve. The line from an obscure 1957 paper to the machinery that scores every major American bill runs, in part, through one advisor and one student.

Government got its own first taste almost immediately, from a different direction. Between 1962 and 1965, a young economist named George Sadowsky brought computers to revenue estimation at the Treasury's Office of Tax Analysis. In about three months in 1963 — a Yale graduate student, consulting for the Treasury — he built a microanalytic model to analyze preliminary versions of what became the Revenue Act of 1964. It was far simpler than Orcutt's dynamic vision: it aged nobody forward, simulated no births or marriages. But it did something no one had done before — it ran a proposed federal tax change against a sample of real taxpayers before enactment, so that Congress could see the consequences of a bill while it was still a bill. By the late 1960s, that kind of analysis was becoming standard in budget work. Sadowsky moved on through Brookings and then to a newly founded Washington think tank called the Urban Institute, where this story will find him again.

Eighteen years, start to finish

Orcutt spent the decade after 1958 trying to scale the demonstration into the real thing, and the decade defeated him. He moved to the University of Wisconsin–Madison and founded the Social Systems Research Institute in 1959, intending to build the full model there. The computing was too thin, institutional support wavered, and the ambition outran what a university lab could sustain. Cheng's biography calls the Wisconsin years a "failed trial" — a label worth keeping, because the failure was of capacity, not of the idea, and the distinction decided what happened next.

In 1968 the Urban Institute hired Orcutt to lead the project he had been proposing for eleven years, and by 1969 it had the resources to begin in earnest. DYNASIM — the Dynamic Simulation of Income Model — aimed to simulate every major event in American lives: births, deaths, marriages, divorces, schooling, employment, disability, retirement, taxes, benefits. The team built it on a DEC System-10 mainframe inside a custom framework called MASH, written by George Sadowsky, the same economist who had computerized the Treasury's estimates a decade earlier. The model carried 10,000 simulated people — enough to draw statistical inferences.

The first version was complete in 1975: eighteen years after the proposal, fourteen after the first demonstration. Its architecture holds up as a description of the field today — modules organized by domain, one for demographic events, one for the labor market, one for taxes, transfers, and wealth, the whole run as an integrated system in which a simulated recession changes simulated marriages. And it refused to die. The Urban Institute carried DYNASIM through a narrower retirement-income version in the early 1980s, a major overhaul around 2000 that projected the American population seventy-five years forward on the Social Security and Medicare trustees' assumptions, and a fourth generation still active today — among the longest-lived computational models in the social sciences.

The family it founded

DYNASIM was *dynamic* in the field's vocabulary: it aged its population forward through time, so you could ask what the retirement system looks like in 2050. A *static* model asks a nearer question — take today's population, change the law today, and compute who pays what tomorrow morning. Static models are where legislation gets scored, and the agencies followed the need: the IRS, the Congressional Budget Office, and the Treasury all built static microsimulation models of the tax system. (Budget fights would later attach a different meaning to "dynamic" — whether a tax cut changes the size of the whole economy — and the next chapter keeps the two senses apart.)

Around the original, a family grew. Steven Caldwell built CORSIM at Cornell, a direct descendant, which in turn seeded POLISIM for Social Security analysis and DYNACAN for Canadian pensions. Statistics Canada built SPSD/M, a static tax-and-transfer workhorse with a feature this book will circle back to: it ran on a deliberately synthetic database, records assembled from surveys and administrative data so that no confidential file ever left the agency. Sweden's SVERIGE modeled the country's entire population. And in 1974, Joseph Pechman and Benjamin Okner of Brookings merged data on 72,000 households to ask, for the first time with real coverage, who actually bore the burden of American taxes.

The method also escaped economics entirely. Health researchers project how a population ages, develops disease, and responds to a vaccination program. Climate models pair physical projections with economic ones to trace how warming lands on different households. Transportation planners simulate individual travelers choosing routes and modes; epidemiologists model transmission person to person across contact networks. Wherever aggregates hide the action, someone eventually rebuilds the problem from units.

What the units buy, concretely:

(A table appears here in the print and web editions.)

The last three rows are the reason the method exists. They are also the rows every closed institution in the next chapter will monopolize.

What the units turned out to be for

The constraint that made Orcutt's idea absurd in 1957 dissolved almost completely within his lifetime. DYNASIM simulated 10,000 people on a mainframe that cost a fortune to run; a present-day laptop simulates millions; computing power that cost millions of dollars in 1975 now fits in a pocket. Orcutt died on March 5, 2006, having lived from the regression analyzer to the eve of the smartphone. The work stayed in the family: his daughter, Alice Orcutt Nakamura, became an economist, and his granddaughter, Emi Nakamura, won the 2019 John Bates Clark Medal — awarded to the most promising American economist under forty — for work that answers macroeconomic questions with finely disaggregated data. Three generations spent on the same conviction about averages.

Cheng's summary of what Orcutt built is the right one: microsimulation was "an engine designed for not only scrutinizing the system but reengineering the society". But the deepest consequence of the design took fifty years to surface, and it is the one this book is built on. Orcutt wanted units because averages lied. Units turn out to buy something he never advertised: a model assembled one household at a time can be *checked* one household at a time. This family's income tax can be compared against the statute that sets it, that family's benefit against a determination letter, and only then the sum against the totals a government publishes. The audit can reach all the way down. Aggregate models cannot offer that even in principle — there is no unit to check.

For the first fifty years, hardly anyone could run the check. The models worked, and the institutions that owned them answered to no outside examiner, because outside the institutions there was nothing to examine with. Orcutt's question — can you compute a society from its units? — was answered by 1975. The question that drives the rest of this book is the one his answer created: when the units, the rules, and the checks exist, who gets to hold them?

Chapter 2: The tax model wars

In 1983, a small London think tank did something no outside body had done before: it built a working model of the British tax and benefit system and turned it on the government's own budget. The model was called TAXBEN, and it belonged to the Institute for Fiscal Studies. When the Chancellor announced new rates and thresholds, IFS economists could feed the rules in and simulate their effects on families up and down the income distribution within hours — publishing who gained and who lost before the official gloss had time to set. The institute's annual Green Budget became required reading for journalists, civil servants, and the politicians it embarrassed.

The institution behind the model began, like a surprising number of things in this book, with frustration at a tax law nobody could decode. The 1965 Finance Act had introduced capital gains tax to Britain in a form so opaque that four financial professionals — the banker Will Hopper, the investment-trust manager Bob Buist, the stockbroker Nils Taube, and the tax consultant John Chown — decided the country needed an institution whose job was understanding fiscal policy from outside the government. They settled it over dinner on July 30, 1968, at the Stella Alpina restaurant in London, and incorporated the Institute for Fiscal Studies the following year. Fifteen years from a dinner-table grievance to a model that could check the Treasury's arithmetic: that is roughly the speed at which this kind of independence gets built, and it is worth remembering later, when the building speeds up.

TAXBEN was a tax-benefit model — a microsimulation that computes both the taxes a household owes and the benefits it receives, capturing how the two interact. The distinction from a mere tax calculator matters more than it sounds, because the deepest injustices in fiscal systems live in the interaction: a family that pays little tax can still lose most of a raise to benefit withdrawal, and no model that sees only one side of the ledger will ever show it.

But before the outsiders could fight the government's numbers, the government had to build them. That machine is where the war starts.

Washington builds the machine

In 1974, after Richard Nixon impounded funds Congress had appropriated, Congress created the Congressional Budget Office to give itself analytical capacity independent of the executive branch. Its first director was Alice Rivlin — the same Alice Rivlin who, as Guy Orcutt's doctoral student, had co-authored the 1961 book that first made microsimulation run. She wanted a nonpartisan office willing to publish numbers that discomfited both parties, and she staffed it with economists rather than political operatives. The culture held: CBO built microsimulation models for health insurance, Social Security, and tax analysis, and for long-run questions it built CBOLT, a long-term model of the entire federal budget.

Congress's tax arithmetic ran through an older body. The Joint Committee on Taxation had existed since 1926, and by the 1970s its staff had developed a suite of microsimulation models — an individual income model, a corporate model, an estate and gift model, an international model for cross-border flows. The budget process built on the Congressional Budget Act of 1974 made JCT's numbers the official revenue estimates for tax legislation, and "official" is doing heavy work in that sentence. When a senator drafts a tax amendment, JCT staff feed it into the models, which apply the proposed rules to a large, confidential sample of actual tax returns; within days, sometimes hours, out comes a score — the projected change in federal revenue over ten years. That single number decides whether the amendment fits the budget resolution, survives reconciliation, or is fiscally viable at all. A score too high kills a proposal before any vote. This is budget scoring: a gate legislation must pass through, operated by a model most members of Congress have never seen.

The executive branch had its own. Treasury's Office of Tax Analysis had run computerized tax models since George Sadowsky's work in the early 1960s — Sadowsky later called that first, three-month effort "probably the most useful program I ever wrote", a fair self-assessment for a program that taught the federal government to test tax law before enacting it. By the 1980s his hack had matured into the Individual Income Tax Model, refreshed regularly from IRS returns, producing the revenue estimates behind each Administration's proposals as JCT's served Congress.

And because taxes were only half the ledger, the Urban Institute — Orcutt's old home — ran the TRIM series, updated every year from the Current Population Survey to model the programs the tax models left out: SNAP, Medicaid, TANF, SSI, housing assistance. It captured the interaction between taxes and transfers, the thing TAXBEN had shown mattered. It served government agencies under contract; the public had no access.

By the mid-1980s, then, the United States had a complete analytic apparatus for fiscal policy: models at CBO, JCT, Treasury, and Urban, collectively capable of telling you what nearly any tax or benefit change would do to revenue and to households. It was a genuine intellectual achievement. It was also, from the outside, a set of black boxes — and the walls around them were not primarily made of code.

The moat

Section 6103 of the Internal Revenue Code forbids the disclosure of individual tax-return information, with a narrow set of exceptions. Under section 6103(f), the tax-writing committees and the Joint Committee on Taxation have statutory access to the returns themselves; Treasury's analysts hold theirs as part of administering the tax system, under 6103(h). External researchers have neither.

The consequence is a permanent asymmetry in raw material. The government's models run on large, carefully edited samples of actual returns, drawn by the IRS's Statistics of Income division, carrying detail no outsider ever sees — the joint pattern of incomes, deductions, and credits as filers actually report them. Everyone outside works from public-use files: sampled down, anonymized, and top-coded, meaning the highest incomes are collapsed into a single category or swapped for averages so that no billionaire can be re-identified from the data. Top-coding is a reasonable privacy safeguard with an awkward side effect: it blurs the data exactly where the biggest tax fights are decided. Debates about taxing the top one percent are conducted, outside government, on files designed to conceal the top one percent.

The confidentiality itself is justified — nobody should be able to look up a neighbor's tax return, and the case for that rule does not weaken because analysts find it inconvenient. But its byproduct is an analytical moat. You can publish your code, your methods, and your assumptions, and you still will not have the data.

The moat had costs that compounded quietly for decades. An estimate people disliked could be dismissed as biased, and there was no external check to settle the question. An estimate that was simply wrong could not be reconstructed from outside, so errors survived. Debate narrowed to whatever the scorekeepers were willing and able to model — a proposal the models could not score was, politically, a proposal that could not exist. Expertise pooled inside a few institutions, because nowhere else could a tax economist practice on real machinery. And citizens were asked to take the numbers on faith, which worked about as well as faith usually does in a polarized country: each side kept its own arithmetic.

The rest of this chapter is the fifty-year campaign against that arrangement. The law forbids breaching the moat, so the outsiders rebuilt, in public, nearly everything inside it.

The outsiders arm themselves

The first serious tool arrived from academia. At the National Bureau of Economic Research, Daniel Feenberg began building TAXSIM in the early 1980s: a program that computed federal and state income-tax liability for any household you described to it. By the 1990s TAXSIM was reachable over the internet — one of the first tax calculators anyone could run remotely — and Feenberg maintained it with a craftsman's persistence for the rest of his career. Economists used it in over a thousand published papers. What TAXSIM did not do was politics: it calculated taxes for individual records but produced none of the aggregate revenue estimates and distributional tables that drive legislative fights.

The institution built for the fights came in 2002, when Gene Steuerle, Len Burman, and William Gale founded the Tax Policy Center as a joint venture of the Urban Institute and the Brookings Institution. The founders knew the machine from the inside: Steuerle had helped build the official models at Treasury, Burman at CBO and Treasury. TPC built its microsimulation on the same public-use files available to everyone outside government — less granular than the confidential returns, but enough to estimate the revenue and distributional effects of a proposal within days.

Its arrival changed what a campaign promise could get away with. In 2012, Mitt Romney ran on a tax plan that promised to cut rates sharply while remaining revenue-neutral through unspecified base-broadening. TPC ran the arithmetic and showed the promise was mathematically impossible: no combination of the available base-broadeners could pay for the rate cuts without raising taxes on the middle class or losing revenue. The campaign could dispute the finding; it could not rebut it with numbers, because it had no comparable model and the one that existed said no. Five years later, during the debate over the 2017 Tax Cuts and Jobs Act, TPC's tables showed how heavily the bill's benefits tilted toward high-income households — analysis Congress's official distributional tables were never going to lead with.

Others followed. Kent Smetters launched the Penn Wharton Budget Model at the University of Pennsylvania in 2016, distinguishing it by producing dynamic estimates — scores that let a tax change move the size of the economy rather than holding it fixed. The Budget Lab at Yale arrived later with another lens. In little more than a decade, the number of independent shops able to score a major tax proposal went from essentially zero to half a dozen, each with a discernible temperament: the Tax Policy Center and the Institute on Taxation and Economic Policy leaned left on distributional questions, the Tax Foundation leaned right on growth effects, Penn Wharton positioned itself in the middle. Pluralism had replaced monopoly — at least for anyone senior enough to know which shop's assumptions to discount.

The dynamic-scoring war

With multiple models came a fight about what a score should even measure, and it is the clearest case study in how methodological choices carry political weight.

Traditionally, JCT scored a tax cut against an economy of fixed size: cut revenue by $100 billion and the score says $100 billion. The convention is usually called static scoring, and the label misleads in an important way — conventional estimates already let people respond to the law, shifting the timing of income, reorganizing businesses, rebalancing portfolios (behavioral responses, in the trade's vocabulary: the ways filers rearrange their affairs when the rules change). What the convention holds fixed is the macroeconomy itself: total output, employment, investment. Supply-siders argued this systematically overstated the cost of tax cuts, since lower rates should spur growth that claws some revenue back. Score the same cut dynamically — letting it change the size of the economy — and it might book at $70 billion instead.

The academic verdict, worked through in forums and papers across the 2000s and stated formally by Alan Auerbach in 2005, was that dynamic scoring is conceptually correct and practically treacherous: the macroeconomic models needed to estimate the feedback are themselves deeply uncertain, so the choice of model smuggles in the answer. Congress made the choice anyway. In January 2015, the new Republican House majority required JCT to produce dynamic scores for any legislation with budgetary effects above a quarter of a percent of GDP.

The Tax Cuts and Jobs Act of 2017 became the first major test, and the numbers are worth keeping straight because chapter 3 grades them. JCT's conventional score put the ten-year revenue loss at $1.46 trillion. Its dynamic score came in near $1.07 trillion: faster growth adding roughly $451 billion of revenue, higher interest costs eating $66 billion of that, for a net deficit reduction of about $385 billion — trimming the bill's cost by roughly 27 percent, a real effect and nothing like self-financing. Penn Wharton, running the feedback through its own model, judged the growth effects more modest still: by its estimate the law would add $1.9 to $2.2 trillion to federal debt over the decade even after growth. No credible model found a free lunch. The law passed anyway, its individual provisions scheduled to expire in 2025 — a schedule Congress then overrode, extending them that July in the One Big Beautiful Bill Act.

A decade into the experiment, Douglas Elmendorf, Glenn Hubbard, and Heidi Williams — a former CBO director, a former chair of the Council of Economic Advisers, and one of the country's leading empirical economists — assessed dynamic scoring and concluded it had improved budget analysis at the margins without resolving the fundamental uncertainty about macroeconomic feedback. That is roughly where the fight rests: the models multiplied, the assumptions became somewhat more explicit, and the deepest disagreements moved from arithmetic into premises, where at least they can be argued about honestly. Whether any of the numbers were *right* is a different question, and the next chapter takes it up directly.

The idea crosses borders

While Washington fought over scoring conventions, the method itself was going global, and each country's version tells you something about what independence requires.

Australia got an institution. From 1993, Ann Harding built a microsimulation program at the National Centre for Social and Economic Modelling in Canberra; her edited volume *Microsimulation and Public Policy* served for years as the field's standard reference, and NATSEM's models turned up everywhere from government agencies to parliamentary committees to the evening news, on everything from tax reform to childcare.

Europe got a federation. From 1996, a team led by Holly Sutherland set out to build EUROMOD: one tax-benefit model spanning, eventually, every EU member state, with national systems harmonized so a reform in Spain could be compared with one in Finland on the same terms. It was developed at the University of Essex on European Commission research grants; researchers applied for access, the ethos was sharing even where the code was not yet fully open, and the Commission eventually took over its maintenance through its Joint Research Centre. EUROMOD is the quiet giant of this field — twenty-seven countries' rules, maintained for decades, validated country by country — and later chapters lean on it as a reference point in a way that should be read as respect.

Britain got a spin-off. With funding from the Nuffield Foundation, a project led by the economist Mike Brewer turned EUROMOD's UK component into a standalone model, UKMOD — built and housed at Essex's Centre for Microsimulation and Policy Analysis, which Matteo Richiardi directs, and introduced in a 2021 paper by Richiardi, Diego Collado, and Daria Popova. "We wanted to democratize access to tax-benefit analysis," Brewer said of the project. Its first free public release, in October 2019, made it the UK's first freely available tax-benefit model, and public bodies picked it up — the Scottish Parliament's research service, NHS Health Scotland, the Welsh Government.

And individuals could do it alone. Howard Reed ran TAXBEN inside the IFS from 2000 to 2004, spent a spell as chief economist at IPPR, then founded Landman Economics in 2008 and built his own tax-transfer model, which he pointed at basic income, welfare reform, and the cumulative toll of austerity — work in service of an ambition he described as "a new settlement of the same scale and sustainability as the Beveridge-inspired reforms of 1945". Malcolm Torry, directing the Citizen's Basic Income Trust, used EUROMOD across nearly a decade to cost basic-income schemes in granular detail. Independence of analysis, it turned out, could follow from a freely downloadable model as surely as from a new institution.

The French break

The sharpest conceptual jump came from France, a country that gave its parliament no analytical shop at all — no CBO, no independent scorer; evaluating a proposed amendment meant asking the Finance Ministry to evaluate the proposal it might well oppose.

In January 2011, Camille Landais, Thomas Piketty, and Emmanuel Saez published *Pour une révolution fiscale* — "For a Tax Revolution" — arguing for restructuring the French income tax into a single progressive levy. The book mattered less than its companion. At revolution-fiscale.fr, any French citizen could design a tax reform and watch its budgetary and distributional consequences resolve on screen; hundreds of thousands did. For the first time anywhere, the general public could run the kind of analysis that finance ministries ran, on a question the finance ministry would rather not have publicly analyzed.

Four months after the book appeared, a small team inside the French government's own strategy unit — the Centre d'analyse stratégique, later renamed France Stratégie — began building something quieter and more consequential: OpenFisca, written by Mahdi Ben Jelloul and Clément Schaff, released to the public later that year. It was the source code beneath a simulator, built on the premise that a country's tax and benefit rules should exist as executable code alongside the legal prose. Where the révolution fiscale site opened one model's *results* to the public, OpenFisca opened the *machinery* — anyone could read the encoded rules, correct them, or build a different tool on top.

The premise acquired a name — rules as code, with variants like "legislation as code" and "machine-consumable rules" — and a small international movement. In 2018, New Zealand's Service Innovation Lab spent three weeks translating a piece of legislation into Python and human-readable rules simultaneously, demonstrating that a law could be drafted in both forms at once. The OECD published *Cracking the Code* in 2020, a primer for governments on the approach. OpenFisca itself spread across four continents — powering France's Mes Aides benefits calculator, a New Zealand rates rebate, adaptations in Tunisia and Senegal — and collected an OECD innovation award and recognition from the European Commission as its most innovative open-source software. At the 2016 Open Government Partnership summit in Paris, a team of French and Tunisian volunteers modeled Senegal's income tax in under thirty-six hours and won the hackathon. By 2019, France's National Assembly had built LexImpact on top of OpenFisca, letting legislators simulate the effects of proposed amendments; within two years the tool had informed well over a hundred simulations in parliamentary debate.

Keep the Senegal number in mind — thirty-six hours, volunteers, one tax — because a version of it, scaled to entire countries and verified against reference models, is where this book eventually goes.

The American open-source turn

The United States' contribution to the open movement came from an unexpected quarter: the American Enterprise Institute, a market-oriented think tank not usually accused of undermining institutional authority. In 2013, Matt Jensen founded the Open Source Policy Center there and recruited Martin Holmer, an MIT-trained economist, to write Tax-Calculator: an open-source model of US federal income and payroll taxes, in Python, with its code on GitHub where anyone could read the formulas, run the model, and check the results. The provenance did the project a strange favor. Open modeling pitched from the left would have been read as advocacy in disguise; coming from AEI, it was harder to dismiss as partisan, and contributors arrived from across the spectrum.

Tax-Calculator grew into the Policy Simulation Library, a loose confederation of independent open-source models sharing transparency and interoperability standards while each specialized — overlapping-generations macro models for dynamic scoring, a capital-cost-recovery model from the Tax Foundation, computational-economics tools from QuantEcon — with contributors ranging from CBO staff to the City of New York. Even the Federal Reserve joined the drift toward glass boxes, releasing a Python implementation of FRB/US, its several-hundred-equation macroeconomic model, in 2022.

By the early 2020s, the open movement had produced something remarkable and something incomplete. Remarkable: working, credible, inspectable models of major tax systems, free to anyone. Incomplete: every one of them demanded a programmer. The engine was open. The last mile — to something an ordinary person could use, and rely on — was not yet built. A tool called PolicyEngine would eventually put the open approach in front of anyone with a web browser; it enters this story in Part II.

What the moat still held

Step back from the fifty-year campaign and audit what it won. TAXSIM, TPC, Penn Wharton, OpenFisca, Tax-Calculator: every one of these was a gain in *method* — open code, published assumptions, reproducible results. None was a gain in *data*. Section 6103 stood exactly where it stood in 1976, and it should: the confidentiality of tax returns is not a bug to be routed around, because you cannot open-source a dataset you are legally forbidden to hold.

And rules as code, for all its promise, has a second limit hiding inside the first. Encode a country's rules and you get an engine that can compute any individual scenario — this household, that income, those benefits. But the questions legislation turns on are aggregate: what does the reform cost, who gains, how many fall below the poverty line. To answer those, the engine needs something it does not contain — a representative population to run the rules over — and building one runs straight back into the moat. The returns that would make the population faithful are the returns nobody outside may touch.

There is exactly one answer the constraint permits, and it is the strategy the rest of this book builds on. Take the public files you are allowed to hold. Synthesize from them an artificial population — households that exist only in silicon. Then calibrate it: adjust each record's weight, the number of real households it stands for, until the weighted totals reproduce the administrative aggregates the government does publish — total returns filed, total wages by bracket, total SNAP enrollment by state. The government will not show you the microdata, but it publishes thousands of true totals about it, and each one is a constraint an artificial population can be forced to satisfy. Statistics Canada had proven a version of the idea decades earlier, shipping its SPSD/M model on exactly such a synthetic database; what remained was to build it at the scale and openness the moment now demands.

You do not breach the moat. You reconstruct, statistically, what stands behind it.

Whether the reconstruction can be made faithful enough is a question that occupies much of Part II. But it sits inside a harder and older question, one the closed apparatus spent fifty years half-avoiding and the open movement now had to face about its own work: how would you know whether any of these models is right?

Chapter 3: The accuracy question

The 2017 tax fight ended with a pile of contradictory numbers and a vote. The Joint Committee on Taxation said the Tax Cuts and Jobs Act would reduce revenue by $1.46 trillion over ten years; Penn Wharton's dynamic analysis pointed to roughly $2 trillion of added debt; congressional supply-siders waved both off and predicted the cuts would pay for themselves through growth. Somebody had to be wrong. The unusual thing — the reason this chapter exists — is that this time, reality eventually filed a report.

By 2024 the receipts were in. The growth windfall never arrived: revenue did not come in above what the scorekeepers had projected under the law, and the self-financing prediction failed on its own terms. The models fared differently. The Committee for a Responsible Federal Budget compared real, inflation-adjusted revenue for 2018 through 2024 against CBO's 2018 projections and found it within 0.5 percent — setting aside 2022, a year CRFB attributes to a surge of capital-gains realizations and inflation rather than to the tax law. That comparison deserves its caveats stated plainly: receipts landing near a projection grades the *forecast*, not the law's true cost, because the world without the TCJA was never observed, and inflation, a pandemic, tariffs, and later legislation all moved revenue too. With those confounds acknowledged, the microsimulations were approximately right about direction and magnitude, and the free-lunch prediction was not. The One Big Beautiful Bill Act extended the law's individual provisions in July 2025, so the 2018–2024 window stands as a completed natural experiment rather than the end of the story.

But "approximately right" is the best anyone can honestly say.

Three ways to be right

Microsimulation validation happens at three levels, in rising order of difficulty.

The first is component validation: does each individual calculation match the rules? A married couple earning $100,000 with two children owes a specific federal income tax, derivable from the statute, checkable against IRS worksheets or commercial tax software. Either the model computes it or it does not. This is the tractable case — tedious at scale, but tractable.

The second is aggregate validation: summed across the population, does the model reproduce reality's totals? Simulated SNAP enrollment against USDA administrative records; simulated income-tax revenue against IRS collections. A model can pass every component check and still fail here, because the aggregate depends on the population you ran the rules over, not just the rules.

The third is predictive validation: when the model forecast the effect of a change, did the forecast hold? This is the hard one. A policy change never arrives alone; it lands bundled with economic shifts, other reforms, and behavioral responses nobody anticipated, and the model's error is tangled with everything else that happened.

Good models pass the first two levels reliably. The third is where humility enters, and the right posture toward it comes from weather forecasting: tomorrow's forecast is reliable, next week's roughly right, next month's a best guess, and the responsible move is to calibrate confidence to the horizon rather than abandon the forecast.

The ACA test

The Affordable Care Act supplies the best natural experiment we have on the third level, because the CBO made a specific, dated, checkable prediction and then a decade happened to it.

In March 2010, the CBO projected that the ACA would cut the share of nonelderly Americans without insurance from more than 18 percent to about 7.6 percent by 2016, on the assumption — universal in every model at the time — that every state would expand Medicaid. Two years later the Supreme Court made expansion optional, and nineteen states declined: a legal shock no microsimulation had a module for. FactCheck.org later reworked the arithmetic to ask what the 2010 projection implies once you account for the states that opted out; the adjusted figure is 9.4 percent. The measured 2016 rate, from CDC data, was 10.4 percent — within a point of the adjusted projection, across a six-year horizon interrupted by a constitutional ruling.

That is the headline, and it flatters the model. The components tell a different story:

(A table appears here in the print and web editions.)

Source:

The CBO overestimated exchange enrollment by more than half and underestimated Medicaid enrollment by nearly half. The total came out close in part because the two errors ran in opposite directions and partly cancelled. As the FactCheck.org retrospective put it, the CBO's error "was in estimating *where* the uninsured would get covered, not *how many* of them would gain coverage".

Both readings of this episode are true, and the book needs both. Read generously: predicting a national coverage rate within a point, six years out, through a Supreme Court shock, is a real achievement no rule of thumb could match. Read strictly: an aggregate that lands while its components miss has passed the second validation level by partial cancellation while failing the first — and a decomposition performed years later, with the answers in hand, is not the same thing as the original forecast being right. A number that comes out right for the wrong reasons has not been validated; it has been lucky, and luck does not generalize. The only way to know which you have is to check the components — which is exactly why the levels are ordered the way they are.

The forecaster grades itself

The CBO does something most forecasting institutions never do: it grades its own work in public, publishing systematic retrospectives that compare past projections against what happened. The retrospectives show real improvement. For deficit projections six years out, the average absolute error fell from 3.2 percent of GDP over 1989–2001 to 1.0 percent over 2002–2019 — a threefold gain in two decades, built on richer IRS administrative data, more capable computing, and the retrospectives themselves, which exposed systematic biases that could then be corrected. Self-grading is the feedback loop that makes improvement possible; the agencies that skip it stay exactly as wrong as they started.

The same retrospectives also set the ceiling. CBO's projections are about as accurate as the Blue Chip consensus of some fifty private forecasters, and about as accurate as the Administration's — everyone converges on a similar wall. And the wall moves. The 2021 projection was CBO's largest overestimate on record; the 2023 was its largest underestimate, off by 3.9 percent of GDP, more than triple the historical average. Models tuned on ordinary decades met a pandemic economy and missed in both directions, two years apart.

Then there is the finding I like least and trust most. Kevin Kliesen and Daniel Thornton, economists at the Federal Reserve Bank of St. Louis, analyzed CBO forecasts published from 1976 to 2007 and found that a random walk — assume next year's deficit simply equals this year's — would have beaten the CBO on average, over both short and medium horizons. I cite this against my own interest, since this book spends several chapters on forecasting infrastructure. The right reading says less about CBO than about the problem: deficit forecasting is saturated with irreducible uncertainty — recessions, crises, and pandemics are precisely the events no model foresees — and beating a naive baseline is genuinely hard. Computing a household's taxes under known law and predicting next year's deficit are different problems wearing the same word "model," and an honest accuracy question keeps them apart. The first has an exact answer. The second has a distribution, and institutions that publish it as a point are hiding the interesting part.

Markets, and the people who beat polls

If forecasting is the weak flank, who does it well? One answer that actually gets graded: markets where people bet. An NBER study of prediction markets found their forecasts "weakly more accurate" than survey forecasts — the study's own careful phrase — across GDP, inflation, and employment. The Monday before the 2024 presidential vote, with polls showing a coin flip, Polymarket had Donald Trump at 58 percent. Academic comparisons of market prices against FiveThirtyEight's model found the markets competitive or better.

The other answer is a particular kind of person. For Federal Reserve decisions, Good Judgment's superforecasters reportedly beat financial futures markets by about 30 percent in 2024–2025. Behind that result sits Philip Tetlock's two decades of evidence: most experts forecast little better than chance, but a small minority — roughly 2 percent of Good Judgment Project participants — consistently outperform, by updating often, thinking in probabilities, and refusing to marry a prediction to an ideology.

Markets and superforecasters share one discipline the scorekeepers lack: they are graded, publicly and unavoidably, every time. But they have a structural blind spot that keeps them from replacing anything in this book. A market needs a question that resolves — an event that either happens or does not. It can price *will the bill pass*; it cannot price *what would the bill have done had it passed*, because a never-enacted counterfactual resolves nothing. Counterfactuals are exactly microsimulation's terrain. The interesting move, which Part IV builds an institution around, is the synthesis: structural models to compute the counterfactuals, market-style grading to keep the forecasts honest — every estimate published with an interval and scored when the official number lands. As this book went to press, that scoreboard's first forecasts were still waiting on reality; nothing there is validated yet, only staked.

The data are broken too

Everything so far assumed the model's inputs describe the country. They do not, and the person who proved it is Bruce Meyer, an economist at the University of Chicago who has spent two decades documenting a quiet crisis in household survey data.

Meyer and colleagues linked survey responses to administrative records — what people told the Census Bureau against what the program files show — and the gaps are not rounding errors. Roughly 40 to 50 percent of SNAP recipients did not report receiving benefits in the Current Population Survey. More than 60 percent of Temporary Assistance for Needy Families and General Assistance went unreported. About a third of housing-assistance recipients did not mention it; even Social Security was missing from about a tenth of recipients' answers. Feed those surveys to a flawless model and it will confidently understate the safety net's reach, because the people it simulates deny receiving what they receive.

The rot compounds. Survey response rates have fallen since the 1990s, and the people who stop answering skew low-income, young, and mobile, so the Census Bureau increasingly fills the gaps by imputation — statistically inferring the answers people never gave — which means growing stretches of the "data" reflect the imputation model rather than any respondent. Retirees misreport in their own direction: a Census Bureau study linking survey responses to IRS and Social Security records found median income for Americans 65 and older was 30 percent higher in the administrative data than in their survey answers, enough to put measured senior poverty at 9.1 percent when the validated records showed 6.9. And at the top, the public files are top-coded by design, compressing exactly the incomes that high-end tax reforms target.

Administrative data fixes much of this and introduces its own holes: tax returns miss non-filers, omit non-taxable transfers, and describe legal tax units rather than the households people actually live in; IRS research files run about two years behind; and no common identifier links records cleanly across agencies. Nor do the rules themselves tell you who shows up: EITC take-up runs around 78 to 80 percent, SNAP around 82 percent with wide state variation, and SSI take-up among eligible seniors may run as low as 50 to 60 percent. A model that assumes everyone claims what the law owes them overstates every program it touches — and take-up moves when policy does, as the 2021 Child Tax Credit expansion showed by reaching non-filers in patterns no prior model would have forecast.

The lesson generalizes: accuracy is as much a property of inputs as of equations. Getting the rules exactly right — chapter 2's project — buys nothing if the population you run them over is a funhouse mirror of the country. That is why the synthesize-and-calibrate strategy the last chapter ended with is a load-bearing wall, and why so much of Part II is about data rather than rules.

When models disagree

In 2017, four institutions put numbers on the TCJA and the spread was enormous: JCT's conventional $1.46 trillion revenue loss; its dynamic $1.07 trillion; the Tax Foundation's far sunnier $448 billion; Penn Wharton's $1.9-to-$2.2 trillion — and even reading that list takes care, because the figures measure different objects. The first three are revenue losses under different feedback conventions; Penn Wharton's is added *debt*. Part of the apparent chaos is institutions answering subtly different questions.

The rest decomposes into three sources, and naming them is more useful than despairing over the spread. First, data: JCT computes on confidential IRS returns, TPC on less granular public files, and different shops assume different baselines — the assumed path of current law a reform is measured against, a choice that can move an estimate by hundreds of billions before the reform itself is even modeled. Second, behavioral assumptions: how much people rearrange work, saving, and business form when rates change. The Tax Foundation's large assumed growth response versus Penn Wharton's modest one drove more of the gap between them than any other factor — and reality supplied a live example of the stakes, when the TCJA's new 20 percent deduction set off a surge of restructuring into pass-through form, exactly the kind of response every scorekeeper had to guess at in advance. Third, modeling choices: how income shifts between corporate and individual returns, how growth is projected, how expiring provisions are treated.

Model disagreement, decomposed this way, stops looking like failure. It is evidence that policy analysis rests on judgment about data, behavior, and method — and running several models turns a shouting match between conclusions into a comparison of stated premises, which is the only kind of policy argument that can actually be adjudicated.

There is also a quieter mechanism that keeps shared tools honest. Daniel Feenberg maintained TAXSIM for more than four decades, and the thousand-plus published papers that used it formed an informal validation network: every author had an incentive to catch and report the bug that would embarrass their own results, and the model improved under collective scrutiny. For computing federal tax from given inputs, TAXSIM's accuracy is high — the rules are deterministic and documented — with discrepancies clustering in state edge cases and credit interactions; it computes taxes only, record by record, and produces no aggregates. In software testing's vocabulary, TAXSIM became an *oracle*: an independent implementation of the same rules against which another model's answers can be checked, one record at a time — an idea that chapter 10 promotes from convenience to method.

What can never be checked

In 2017 the CBO estimated that repealing the ACA under the American Health Care Act would cost 23 million people their coverage by 2026. The bill failed, partly on the strength of that number. So the prediction was never tested — and never can be. The genre is built that way. Enact a reform and the world without it goes unobserved; reject it and the world with it does. A counterfactual cannot be checked against reality, because being unrealized is what makes it a counterfactual.

Which forces the question of what a counterfactual's claim to trust can rest on. The answer is the parts. The whole — "this reform would have cost X and covered Y" — is untestable. But each rule in the model can be proved against the statute; the population can be checked against the census and the administrative totals; the behavioral assumptions can be labeled, sourced, and varied to show what swings on them. The pieces can be verified even when the whole cannot, and a counterfactual assembled from verified pieces is a different epistemic object from one that is turtles all the way down.

Occasionally reality even grades a piece of the whole. The 2021 Child Tax Credit expansion arrived with model-based predictions that it would cut child poverty by roughly 40 percent. Census data then showed child poverty falling from 9.7 to 5.2 percent in 2021; Columbia's poverty center tracked the decline month by month as payments went out; and when the expansion lapsed, child poverty climbed back. No controlled experiment — the economy was reopening, other pandemic programs were moving — but direction and rough magnitude landed, on one of the few occasions when a modeled counterfactual became enacted law fast enough to watch.

George Box gave the field its permanent epigraph: "all models are wrong, but some are useful". The record of this chapter sharpens it. The TCJA models were approximately right; the ACA model was right in total and wrong in composition; the deficit forecasts lost to a random walk; the survey data under everything was quietly rotten in measured, correctable ways. Wrong, useful, improvable — and the difference between responsible modeling and false precision turned out, in every case, to be the same thing. Not accuracy. Checkability.

So here is the discipline this book runs on, stated once and enforced everywhere: a simulation is admissible only where its verification chain terminates in ground truth — a component checked against the statute, an aggregate checked against administrative totals, a forecast graded when the official number finally lands. Where the chain holds, a number can be trusted no matter who produced it, agency or amateur. Where it breaks, the number is assertion wearing the costume of arithmetic, and no amount of computational sophistication redeems it.

The rest of this book is an attempt to build things that pass that rule, and an honest accounting of where they do not. It begins, though, somewhere much smaller than a national model: with one household, one benefit cliff, and the frustration that turned a family's spreadsheet into a reason to build any of this at all.

Chapter 4: A wall of frustration

My interest in economic policy became personal through my brother Alex.

In June 2004 — a month after my high school graduation, two months before I left our Menlo Park home for UC Berkeley — Alex suffered a spinal cord injury in a mountain biking accident. He was sixteen; I was seventeen. He became a quadriplegic, dependent on attendant care for the daily mechanics of living, from cooking and cleaning to getting in and out of bed.

Like me, Alex went to Berkeley, earning an undergraduate degree and then a master's in public policy. And when he entered the workforce, our family ran headlong into the question of how his benefits would interact with his earnings.

Medicaid covered his In-Home Supportive Services — attendant care that would otherwise have cost tens of thousands of dollars a year. But Medicaid had an income limit. If Alex earned more than roughly $70,000, he would lose eligibility; his medical expenses would then become tax-deductible, but the trade was brutal, and he would have had to earn something like $160,000 for the added income to make up for the coverage he had lost. Across that whole range, the effective marginal tax rate exceeded 100 percent — the arithmetic our spreadsheets kept producing, no matter how many times we rebuilt them looking for a way through. Earning more would leave him worse off, and the system offered no way around it.

We modeled scenario after scenario, and the complexity was overwhelming: tax brackets, benefit phase-outs, deductions, the interaction of state and federal programs, each rule sensible on its own and the combination punishing. The tools to understand how policy actually landed on one person's particular circumstances simply did not exist; we were building them from scratch, badly, in a spreadsheet, for an audience of one.

And none of it was visible from the outside. A poverty statistic or an average tax rate would have shown nothing of what Alex faced; his cliff lived in the interaction of specific programs at a specific income for a specific person, and you could only see it if you modeled that person directly. The aggregates were silent about exactly the thing that was reshaping my brother's decisions about whether to work.

This was my introduction to what economists call means-tested benefit cliffs and implicit marginal tax rates. For the person living inside the system, it was just a wall of frustration.

Individuals moving through states

The strange part is that I already knew how to model systems like this. The method had followed me from a Berkeley classroom to a factory floor to Google's hiring pipeline — everywhere, that is, except home.

In 2008, as a Berkeley undergraduate, I took IEOR 131, Discrete-Event Simulation. The premise of the course was the working engineer's version of the idea chapter 1 traced through economics: model a complex system not as a set of equations but as a collection of individuals moving through states. To simulate an emergency room, you do not estimate an average wait time; you track each patient's arrival, each nurse assignment, each treatment decision, then change something — one more nurse on the night shift — and watch the system respond. The course ran on Excel and Visual Basic for Applications, and its examples were operational: healthcare facilities, manufacturing lines, service queues.

My first internship put the method on a factory floor. Finelite, a lighting-fixture manufacturer in Union City, California, had decisions that sound trivial until you cost them: whether to pre-cut wire to standard lengths or cut it to order; how to sequence assembly lines to minimize changeover time. I simulated them in Matlab and watched the answers emerge from the individual units rather than from any formula about the factory as a whole. What the method kept delivering was surprise — the behavior of the system, computed from its parts, was routinely one the averages had hidden. The syntax changed over the years that followed, VBA to Matlab to R to Python, and the frame never did: individuals, states, transitions, and a sum at the end.

By that description, the spreadsheet my family kept rebuilding had been a simulation all along: one household, moving through the states a statute defines, with a sum at the end that kept coming out negative. Knowing how to model a system, it turned out, was different from having a model of it.

Project Lorenz

After two years in consulting I joined Google's People Analytics team in 2010, a group premised on bringing the same standard of evidence to decisions about people that the company brought to decisions about products. In 2012 a colleague and I founded a small data-science team inside it, working on natural-language processing of employee feedback, social-network analysis of the org chart, and — my piece — simulation models for workforce planning.

The planning problem was concrete and enormous. Google's staffing group processed between one and two million job applications a year; hundreds of recruiters managed thousands of open roles across divisions with different hiring needs. Leadership wanted to project headcount growth against targets, accounting for recruiting capacity, candidate pipeline dynamics, internal mobility, and attrition. The existing tool was a set of spreadsheets.

I built a simulation from the bottom up and named it Project Lorenz, after Edward Lorenz, whose weather research showed how micro-level dynamics drive macro-level phenomena. It modeled candidates entering through different channels and estimated their probabilities of moving between hiring stages with survival models, the same machinery health researchers use to predict disease progression. It accounted for variation in recruiter productivity, for attrition, for transfers between divisions. It ran in R, used Monte Carlo methods to quantify uncertainty, and produced a distribution of outcomes rather than a single number. It was a microsimulation of one company's labor market, though I did not yet know the word and had never read Orcutt.

Project Lorenz never fully materialized. It had too many moving parts, and wiring them together made the whole thing unstable: a small change to one transition probability would ripple through the pipeline and produce a headcount forecast nobody believed. The idea was right. My execution was not. We ran the spreadsheets for another cycle.

Cash as a benchmark

Around the same time, conversation inside Google kept turning to technological unemployment — what AI might eventually do to the labor market, and whether society would need new institutions to guarantee basic needs. Some people talked about universal basic income: unconditional cash payments, no means test, and so none of the cliffs that means-testing produces.

The idea had a live experiment attached. In 2012, Google.org gave GiveDirectly a $2.4 million Global Impact Award. GiveDirectly made unconditional cash transfers to extremely poor households in Kenya — families living on roughly a dollar a day received about a thousand dollars, no strings attached — and its founders ran randomized controlled trials to measure the effects. The results were encouraging: gains in earnings, assets, nutrition, and children's education. I volunteered with GiveDirectly on the side, helping them use their data more efficiently and hosting their researchers for talks at Google.

I did not know whether basic income was good policy; I knew it was a clean benchmark. Safety-net design runs on a tradeoff between targeting and universality: means-tested programs concentrate money on the poorest and, in exchange, build the phase-outs and cliffs my family had been mapping, while a universal payment costs far more and builds none. Set the two designs side by side and the cliff stops hiding — the same move my Berkeley course had drilled, hold everything constant, vary one thing, watch the system respond. Against a benefit that never phases out, every phase-out shows. Seeing the comparison, though, takes a model that can run both designs over the same families, household by household. The benchmark was, quietly, a demand for a microsimulation.

The policy turn

In 2015 I moved to YouTube's data-science team, working on growth models, experiment analysis, and the launch of YouTube Go, a product built for markets with poor connectivity and lower-end phones, primarily in India and sub-Saharan Africa. The work was good. My attention kept drifting to policy.

Senators Michael Bennet and Sherrod Brown had introduced the American Family Act, an expansion of the Child Tax Credit, and it raised questions with computable answers: what the expansion would cost, which families it would reach, how many children it would pull over the poverty line. The models that could compute them lived behind the walls chapter 2 described, and I was a data scientist at a video platform — not a constituency the Joint Committee on Taxation serves.

Then I found Tax-Calculator, the open-source model of US federal income and payroll taxes that chapter 2 traced to the American Enterprise Institute. It was written in Python and maintained by experienced economists, and its code sat on GitHub, where anyone could read the formulas, run the model, and check the results. Using it required no one's permission.

So I did. I started using Tax-Calculator, then contributing to it, then spending my evenings and weekends on policy analysis with open-source tools while working full-time at YouTube. A question about a Senate bill could become a distributional table by the weekend, produced on a laptop, checkable by anyone who cared to look. Serious analysis took a good model, public data, and the willingness to do the work — no think tank, no agency. This was how policy analysis *should* work.

In 2018 I took three months off from Google to work on policy full time. I enrolled in MIT's MicroMasters program in Data, Economics, and Development Policy — graduate coursework that could ladder into a full master's — and spent the leave working with the Open Source Policy Center at AEI, contributing to Tax-Calculator's technical infrastructure and running distributional analyses of tax reforms. Then I went back to YouTube, and the return did not take. In July 2018, after eight years, I left Google to do policy research on my own, living on savings.

The UBI Center

In 2019 I founded the UBI Center, a think tank for basic income research with one structural commitment: all of the code and all of the data public. The commitment came from the previous three years, in which everything I had done as an outsider had been possible because someone else's model was public. You do not climb that ladder and then pull it up.

The first discovery was that you cannot model basic income by itself. A $1,000-a-month basic income for the United States costs something like $3 trillion a year, and paying for it means changing taxes or benefits — so an honest UBI analysis is an analysis of the entire tax-and-benefit system, because it has to show what would fund the thing: which taxes rise, which programs shrink, and what both do to the same families the dividend helps. And holding the benchmark against the real system did what benchmarks do. Stacked phase-outs pushed implicit marginal tax rates above 50 percent for low-income families; cliffs meant that earning a little more could forfeit thousands of dollars in benefits. Alex's cliff had been no anomaly.

The tools were not ready. Tax-Calculator handled federal income and payroll taxes but no benefits — no SNAP, no Medicaid, none of the programs whose phase-outs build the cliffs. OpenFisca, the French framework from chapter 2, offered a way to encode the rules, but OpenFisca-US was young and incomplete. Neither had an interface a non-programmer could use. State-level tools barely existed, and the programs that had produced Alex's cliff ran through the states.

What the UBI Center had before it had tools was people. The first researcher was Nate Golden, a middle-school math teacher in Washington, DC, who wanted to fight poverty with evidence and would later found the Maryland Child Alliance to push child-poverty policy at the state level. The second came from the internet: I posted on the Basic Income subreddit asking for help, and a college student in the UK named Nikhil Woodruff replied. A message board is where you recruit when no institution hires for the job you think needs doing. Nikhil had a rare pairing — serious interest in economic policy and serious software-engineering skill. (Footnote: As of July 2026, Nikhil serves in the UK government at 10 Downing Street. The public trace: PolicyEngine's own account describes an "Innovation Fellow with 10DS" — No 10's data-science team — for roughly six months from the summer of 2025, building microsimulation tooling on PolicyEngine. I note it here as a disclosure — much of what follows in this book touches UK policy and the institutions that model it.)

The whole operation, in those first months, amounted to a few people — a math teacher, a college student from a message board, me — working nights and weekends on a model of the American tax-and-benefit system. It did not look like infrastructure. It looked like a hobby with unusually high stakes.

The analyses came anyway. When Andrew Yang's Freedom Dividend — $1,000 a month for every American adult — put basic income into a presidential primary, we ran the arithmetic all the way through: the plan would cut poverty by 74 percent (from 7.3 percent of the population to 1.9), and it would cost about $2.8 trillion a year while the five taxes Yang proposed would raise $1.2 trillion, leaving a $1.6 trillion gap. Both findings went into the same report at the same volume; we were modelers rather than advocates, and the job was to show the tradeoffs and let the numbers say what they said. Nate worked out whether a basic income should go to adults, children, or both, and found that for a fixed budget, child allowances cut poverty most. Every analysis meant cobbling partial tools together and working around their gaps.

And there was no open-source model of the UK tax-and-benefit system at all. If Nikhil and I wanted to do comparative UBI analysis across the US and the UK, we would have to build the UK model from scratch.

So we did.

Building openfisca-uk meant encoding an entire national system nobody had put in the open before — Universal Credit, the personal allowance, the benefit taper — line by line from the legislation and guidance. The exactness was the point, and the family spreadsheet had taught me why: a cliff lives or dies on one threshold, so a model wrong about one parameter lies to every household near it. We tested against every case we could check by hand. It was the spreadsheet ritual again, at national scale, with one difference: a rule we got right stayed right for everyone who came after us.

Nikhil built on the UK's Family Resources Survey; I worked on enhancing the US Current Population Survey. By 2020 the UBI Center had grown to ten researchers, its biggest year, and its center of gravity had moved from writing reports to building what reports need: microdf, for analyzing survey data, and openfisca-uk, for simulating UK policy, both accepted into the Policy Simulation Library catalog beside Tax-Calculator. The models started answering questions: a £100-per-tonne carbon tax with the revenue returned as a flat dividend would cut UK poverty by 14 percent and deep child poverty by 33 percent, and the US companion put a $100-a-ton version at a 10 percent poverty cut. It was the two-country comparison we had built the UK model to make.

Open in principle

All of it took Python. A journalist on deadline was never going to install a programming language to check a claim about the Child Tax Credit, and neither were most advocates or policymakers — the people who needed the answers most. The work was open-source in principle and inaccessible in practice: the code was public; the ability to use it was not.

The frustration, I finally understood, was infrastructure-shaped. The tools were fragmented — tax models that didn't talk to benefit models, federal systems that didn't connect to state ones — and running any of them meant installing software, preparing data, and writing code. The gap was not a missing model. It was that no one had assembled rules, data, and an interface into a single thing that a person without a programming background could sit down and use to ask a real question and get an answer they could trust. Rules, data, and an interface, welded into one instrument and made reliable enough to believe — that was the whole of it, and it had gone unbuilt because each piece belonged to a different discipline and a different institution, and putting them together was nobody's job. That was the wall, and it was the same wall my family had hit around Alex's kitchen table, scaled up to a country. Someone would have to build the thing that tore it down.

Part II: The open engine

Chapter 5: Proof of concept

For a few years around the turn of the decade, open-source policy modeling went through a quiet boom — and I spent most of it unable to answer a simple question.

The boom was real. In October 2019, UKMOD became the first freely available tax-benefit microsimulation model of the United Kingdom, built on EUROMOD's UK component, funded by the Nuffield Foundation, open to anyone who asked. Britain had good models long before that; you just couldn't touch them. The Institute for Fiscal Studies had run TAXBEN since 1983, HM Treasury had IGOTM, and the Department for Work and Pensions had its Policy Simulation Model, all behind institutional doors. Then COVID hit, and demand for policy analysis spiked because the rules changed weekly — the CARES Act, expanded unemployment insurance, recovery rebates. NBER updated TAXSIM for the pandemic provisions. Tax-Calculator shipped a pandemic release, version 3.2.1, in August 2021. In December 2020, EUROMOD — the backbone of European tax-benefit modeling for two decades — went fully open source, and the European Commission's Joint Research Centre took it over the following month. That same December, the International Microsimulation Association convened a conference devoted entirely to pandemic policy responses, and the Policy Simulation Library kept adding models to its open-source catalog.

Almost none of it was usable by anyone who could not write code. I knew, because I had spent the boom running the UBI Center — the open-source research shop I started in 2019 when no public tool could say with any specificity what a basic income would cost or whom it would help — on exactly these tools: Tax-Calculator, custom Python, Census survey data. Two lessons kept arriving. The first was fragmentation. Tax-Calculator handled federal income tax but ignored benefits; TAXSIM handled state taxes but ran in batches; nothing captured the full tax-benefit interaction that decides whether a basic income is progressive, regressive, or poverty-reducing. Fund a basic income by tapering an existing benefit and you can lift some families while stranding others at a cliff — and no available tool showed both effects at once. The second lesson was the access gap. The audience for policy analysis was far larger than the audience that could run the tools. Our analyses lived in Jupyter notebooks, which meant a reader could not change a parameter, swap in their own household, or model their own group's variant. The code was public; the ability to use it was not — the wall chapter 4 ended on, met again from inside the notebook.

Putting the model in a browser

By late 2020, Nikhil Woodruff and I had a working microsimulation model of the UK tax and benefit system, built at the UBI Center. Nikhil — the college student chapter 4 introduced — had led the build; he was a University College London undergraduate reading mathematics and computer science, and the stronger engineer of the two of us. The model, OpenFisca UK, could calculate taxes, simulate benefits, and estimate what a reform would do. It could do all of that only inside Python scripts that a programmer could run, which made it one more instance of the pattern we saw everywhere we looked: TAXSIM wanted batch files, Tax-Calculator wanted Python, UKMOD was built for academics, TAXBEN was proprietary. Serious tools, locked behind technical barriers. So we set the goal that defined the next five years: put the full power of microsimulation in a web browser, so that anyone could design a reform, see what it costs and who it helps, and check what it does to their own household, without writing a line of code.

We called it PolicyEngine.

The architecture had three layers. At the bottom, the model: OpenFisca UK running in Python on cloud servers, returning exact taxes and benefits before and after any change. In the middle, an API wrapping the model in HTTP endpoints, so the interface and the rules could evolve independently. On top, a React web application that turned a user's answers — income sources, number of children, housing costs — and their reform choices — sliders, toggles — into API calls, and drew the results. The load-bearing decision was that the model had to be usable without the interface. The API was the product; the web app was merely its first consumer. At the time this looked like overengineering. It later meant that any researcher, journalist, or piece of software could call the same engine that ran the website.

Nikhil handled most of the engineering — API design, the front end, state management — while I worked the policy logic: which parameters to expose, how to present results, what comparisons mattered. Exposure was its own design problem. A tax-benefit system has thousands of parameters; show them all and a user drowns, hide too many and the tool is gutted. We exposed a curated set of the frequently debated ones — income tax rates, benefit levels, credit amounts — plus a path to reach any parameter in the model, and we staged the output: headline numbers first, net cost and poverty change, then breakdowns by income decile, household type, and region.

PolicyEngine UK went live on September 1, 2021. A user could raise income tax, increase child benefits, or introduce a carbon dividend, and read off the cost to the Treasury, the change in poverty, the shift in inequality, and the effect on their own household. Within weeks, advocacy groups were using it to design proposals. When the Chancellor's Autumn Budget changed Universal Credit that October, we published distributional analysis within a day — and learned what it feels like when a website goes viral faster than two people can fix the bugs the traffic exposes. We weren't a media operation. We were two people with a model, fielding interview requests with one hand and patching the code the traffic had broken with the other.

That October we spun PolicyEngine out of the UBI Center as its own nonprofit, and the mission statement shifted from "make everyone a policymaker" to "help people understand and change public policy." Incorporating as a 501(c)(3) was a trust decision as much as a legal one. The code stayed open for anyone to inspect, and the organization had no shareholders whose interests could bend a result — which is what lets groups who agree on nothing else trust the same engine. On paper we were now an institution. In practice we remained two people plus a rotating cast of volunteers and interns.

Crossing the Atlantic

PolicyEngine US launched in March 2022, and the United States turned out to be a different shape of problem. The UK has one national tax system — Scotland sets its own income tax rates — and Universal Credit had folded most means-tested support into a single program. The US has fifty state income tax codes layered over a complicated federal one, and benefits scattered across agencies: SNAP at the Department of Agriculture, SSI at the Social Security Administration, Medicaid at Health and Human Services, housing vouchers at HUD, childcare subsidies at the states. Each program defines income its own way, counts the household its own way, and sets eligibility its own way. Modeling the US wasn't harder than modeling the UK by some constant factor; the complexity grew multiplicatively rather than additively.

We started with what we could model — federal income and payroll taxes, SNAP, the EITC — and shipped the household calculator first. In July 2022 we added population-level impacts, computed on the Current Population Survey, the same microdata foundation behind official government estimates. State coverage came one state at a time. Some states piggyback on federal adjusted gross income and fall into place almost automatically once the federal model exists. Others are miniature ecosystems — New York alone has multiple income tax schedules, city taxes, and supplemental credits — and encoding one rarely transferred to the next. By the end of 2022 we had modeled six states. Forty-four remained. The constraint was structural: we wanted comprehensive, always-current coverage, and we had a small volunteer team encoding rules on weekends — a contributor might encode Oregon's working-family credit only to find the legislature had amended the eligibility rules mid-session, with no updated documentation to encode from. Anyone could submit a pull request adding a state, and we reviewed for accuracy and code quality; the model worked wherever a local expert volunteered and failed wherever none did. That is what open cost: the map filled in where the enthusiasm was, not where the need was.

What open bought

On September 23, 2022, Chancellor Kwasi Kwarteng stood up in the Commons and announced the Truss government's mini-budget: abolish the 45p additional rate of income tax, cut the basic rate, reverse a National Insurance rise, all in the name of growth. We published household-level distributional analysis the same day. The numbers said the package was overwhelmingly a high-earner event: the top income decile would gain about £2,738 a year, or 2.5 percent of net income; the bottom decile, £45, or 0.4 percent. Ours was among the only independent estimates available in those first days, and the post noted that its figures broadly resembled the IFS's and the Resolution Foundation's. Media outlets cited it. Within weeks the government reversed the 45p abolition, and within a month the prime minister had resigned. I don't claim a chart did any of that — markets and mortgage rates did the heavy lifting — but for the first time, anyone arguing about the mini-budget in public could point at an open model instead of waiting for an institution.

Quieter uses accumulated underneath the news cycle. A congressional aide could take a child tax credit proposal, discover that phasing the credit in over the first few thousand dollars of earnings missed the poorest households entirely, reshape the legislative language, and only then request a formal CBO score — arriving with a refined proposal instead of leading with "what does this cost?" That did not replace the official process; it changed the quality of the questions arriving at it. Advocacy groups and academic researchers who had relied on government estimates or expensive consultants ran their own simulations; UBI Lab Northern Ireland modeled a recovery basic income without commissioning a think tank. And developers began building on the API rather than the website — the Fund for Guaranteed Income wired PolicyEngine into a tool that showed cash-pilot participants how their other benefits would change as their income rose. We had set out to build a product and were turning into infrastructure.

Whose bug is it?

Open source means anyone can inspect the code. Inspection is not correctness. From the beginning we invested in checking outputs against official calculators and published statistics, and each model carried a test suite comparing its answers to official calculations across hundreds of household scenarios. Every discrepancy raised the same three-way question: is the bug in our code, in the official calculator, or in our reading of an ambiguous rule?

Sometimes the answer was plainly the first. An outside contributor comparing results against IRS tables caught a transposed digit in a phase-out rate — 21.60 percent entered where the EITC's two-child schedule specifies 21.06. Sometimes there was no clean answer at all: when a regulation says income is figured "net of applicable deductions," different official sources sometimes disagree about which deductions apply. We kept a running log of those cases, and the log was itself a finding — the tax-benefit system is complex enough that the agencies administering it sometimes disagree about its own rules.

One limitation would not yield to more tests, because it wasn't a bug. Our aggregate revenue estimates ran below official statistics, for a reason built into the inputs: we computed on household survey data, governments compute on administrative tax records, and surveys under-sample the very rich — the CPS might hold a few hundred households above $500,000 where the IRS has vastly more. High earners generate a disproportionate share of revenue, so the sampling gap cascades into the totals. We learned to lead with relative impacts — the change a reform makes against current law — because the survey's error moves baseline and reform alike and largely cancels. The next chapter puts a number on that gap and on what it does and does not contaminate; here it enters as the first crack of light between the rules, which were exact, and the data, which was not.

From tool to infrastructure

In 2024 we finally closed the map: income tax models for all fifty states and the District of Columbia, the work of more than a hundred open-source contributors. Flat-tax states with a handful of credits took days; New York, California, and Hawaii took weeks of iterative debugging each. Then the institutions arrived. On August 15, 2025, the National Science Foundation awarded PolicyEngine a Pathways to Enable Open-Source Ecosystems Phase I grant — award #2518372 to the PSL Foundation, $299,974 — a bet that this should become permanent public plumbing rather than a project.

In September 2025 we signed a memorandum of understanding with the National Bureau of Economic Research to build an open-source emulator of TAXSIM, the calculator that had powered academic tax research since the 1970s; more than 1,200 published papers had relied on it. Daniel Feenberg, TAXSIM's creator, joined our advisory board and served as external mentor under the NSF grant. The emulator's purpose was partly preservation — researchers keep TAXSIM's capabilities whatever happens to any one institution — and partly something more interesting: a framework in which several independently developed models cross-check one another.

A month later we signed a second MOU, with the Federal Reserve Bank of Atlanta, whose Policy Rules Database, built with the National Center for Children in Poverty, covers the benefit side TAXSIM never touched: SNAP, Medicaid, housing vouchers, childcare subsidies. The database powers the Atlanta Fed's CLIFF tools, which Colorado's Workforce Development Council and New Mexico's Caregivers Coalition use to help families see how earning more would affect their benefits. With three independently developed models — PolicyEngine, TAXSIM, the Policy Rules Database — agreement builds trust and divergence isolates where a discrepancy originates: it stops being "the models disagree" and becomes a specific question about a specific rule, sometimes a bug and sometimes a genuine dispute about what the policy means. PolicyEngine's role in that arrangement was the integrator's: cross-checking across Python and R, across taxes and benefits, across academic and government implementations of the same law.

And then came the UK.

On December 17, 2024, HM Treasury published an Algorithmic Transparency Record disclosing that it had piloted PolicyEngine UK as a possible supplement to IGOTM — the closed model that had produced the government's policy estimates for decades. A pilot, to be precise about it: the record states the model is not currently deployed, describes an evaluation rather than an adoption, and leaves further evaluation pending. The evaluation's headline was that National Insurance calculations proved very accurate, with "almost 60% of datapoints falling within 0.5% of one another." Read that carefully, because it is easy to over-read: it measures agreement between two models, not a grade against reality. Two implementations of the same statute should agree, and where they don't, one of them has a bug or the law has an ambiguity. That is exactly what made the income tax comparison, in the record's words, less promising: the two systems had trouble even identifying which variables corresponded, with edge cases everywhere housing benefit met legacy benefits or a household crossed into Scottish provisions. The hard part of checking a model against a government model is definitional: two systems can define "income," aggregate units, and handle household composition in quietly incompatible ways. The record also singled out the machine-learning approach we had built for survey error — combining multiple years of the Family Resources Survey, filling in missing incomes with random forests, reweighting households by gradient descent — as the thing under evaluation inside government.

There is a disclosure that belongs beside this passage. From the summer of 2025, Nikhil served roughly six months as an Innovation Fellow with 10DS, the Downing Street data-science team, building a package called 10ds-microsim on top of PolicyEngine — work described, in PolicyEngine's account of the fellowship, as "intensively validated against external projections and official forecasts". As of this writing he serves in the UK government at No 10. When this chapter says British institutions evaluated tools he helped build, weigh it knowing where he now sits — and knowing the shape of the arc, which I have given up trying to improve on: the co-founder I had recruited off a subreddit, now working inside the machinery an outside tool had helped pry open.

Government microsimulation had been closed by default — proprietary code, restricted data, results no outsider could reproduce. Nothing about a pilot overturns that. But an outside model good enough for the Treasury to test against its own machinery weakens the argument that the machinery must stay closed, one evaluation at a time. In 2019 I could not find a tool to answer a basic question about a basic income. Five years later, HM Treasury was piloting one we had built.

Three clocks

By late 2025 the tool was real and unfinished. Household calculations were sharp; unusual cases still exposed gaps; revenue totals still ran below official statistics. And building it had produced the distinction that the rest of this book is organized around, one we did not design so much as keep tripping over. A policy question that looks like one question is really three, and they come apart the moment you try to compute them. What does the law say — the exact tax owed and benefit due for this household, in this place, in this year? Who are the people — how many households look like that one, and how do they sum to a country? What will happen — who changes their behavior, how the economy moves, what next year looks like?

Each is answerable, and each is checkable — but against a different kind of truth, on a different clock. An encoded rule is right or wrong against the statute, and against another calculator built from the same statute: provable to the dollar, today. A population estimate is right or wrong against surveys and administrative totals: calibrated, with the residual measured, today. A prediction is right or wrong only when the future arrives and the official number lands: committed now, graded later. We ran all three inside one engine and one organization, because that is how the tool had grown. But the seams were visible if you looked. The revenue gap was a data problem, not a rules problem. The rules stayed exact precisely where the data was uncertain. The behavioral responses belonged to neither. Three concerns, three verification loops, three different clocks.

What we had built as one thing was, underneath, three things — each buildable, checkable, and trustworthy on its own terms. Pulling them apart is a later chapter's story. First the engine has to show what it can do: for a single household, and then for a country.

Chapter 6: The household and the society

A single parent in New York earns $30,000 a year. Her child has a disability. How much does she owe in taxes, how much does she receive in benefits, and what happens if she takes an extra shift?

The answer runs through federal income tax, state income tax, payroll taxes, the Earned Income Tax Credit, the Child Tax Credit, Supplemental Security Income, SNAP, and a dozen other programs, each with its own definition of income, its own idea of who counts as the household, and its own phase-out schedule, none of them designed to fit together. No one can calculate this in their head. Most accountants would struggle. Yet the answer shapes every financial decision the family makes.

This is the problem PolicyEngine's household calculator was built to solve, and this chapter is about the two questions it turned out to answer. The first is the family's: what would this mean for me? The second belongs to anyone writing the rules: what would this mean for everyone? The same engine answers both, at two scales, and the connection between the scales is the closest thing this book has to a keystone.

What you owe, what you get

A typical low-income family might simultaneously draw on the EITC, a partially refundable Child Tax Credit, SNAP, SSI for a disabled member, Medicaid, a housing voucher, a state childcare subsidy, and free school meals — eight programs, written independently, administered by different agencies, each with its own arithmetic. The fragmentation runs through the professions that serve them: caseworkers specialize in one program, tax preparers work one side of the ledger, benefits counselors know eligibility but not tax consequences. It is why eligible families leave benefits unclaimed, and why a raise can quietly cost more than it pays.

The calculator's promise is to put the whole ledger on one screen. It walks through household composition — adults, children, ages — then income by source: wages, self-employment, investment, retirement. Then the specifics that drive eligibility, the questions a tax form never asks but a benefit office always does: housing costs, childcare expenses, disability. From those answers it computes dozens of outputs under current law — each tax owed, each benefit due — ending in a single net figure for what the family keeps. The US model spans federal income and payroll taxes, the EITC and CTC, SNAP, SSI, WIC, TANF, and dozens of state programs; the UK model spans income tax, National Insurance, Universal Credit, Child Benefit, and the full run of means-tested support. Each program is a separate module, drawn from statute, regulation, and agency guidance.

The interactions force that breadth. A family's SNAP benefit depends on net income after taxes and deductions; its taxes depend on credits tied to the same children who might also generate SSI or a childcare subsidy. Change one input and the effects ripple through programs that have never heard of each other. The only way to capture the ripple is to evaluate every program at once, for the same household — which no single-program calculator, however good, can do.

Rates above a billionaire's

Plot the New York family's net income against its earnings and the line has a hidden structure. There are flat stretches, where more work barely raises take-home pay. There are vertical drops — cliffs, where crossing a threshold loses a benefit worth more than the raise. Chapter 4 met this arithmetic in a single family — my brother's cliff, where earning more would have cost him the attendant care he lives by — and the calculator's first job is to make that same structure visible to any family before it stumbles over it.

The mechanics stack. As the parent's earnings climb from \$20,000 toward \$30,000, SNAP withdraws about thirty cents of each additional dollar of net income. The EITC phases out at 15.98 cents on the dollar for a family with one child, 21.06 for two or more. SSI takes back roughly fifty cents per dollar of countable income. Income and payroll taxes begin to bite. Stack them and the combined marginal rate can approach eighty percent — more than double what a hedge-fund manager pays on his last dollar. A low-income worker often faces a higher marginal rate than any billionaire, because benefit phase-outs pile on top of taxes; and this is the ordinary structure of a low-income budget, not an edge case. It also disappears into any average: blend this parent's marginal rate with her neighbors' and the eighty-percent stretch dissolves into something unremarkable, visible only at the resolution of one household — which is why chapter 1's argument for computing household by household was never really about computational taste. The same stacking happens wherever means tests overlap: in Britain, Universal Credit's taper interacts with income tax, National Insurance, and council-tax support to push some working families' marginal rates above 70 percent.

At a true cliff the marginal-rate chart spikes past 100 percent — earn more, keep less — and the calculator shades the earnings range where extra work barely moves net income: a dead zone, drawn on the family's own chart. Whether the New York parent's extra shift pays at all depends on where in that structure she stands, which is exactly what no pay stub, no program manual, and no national average will tell her. That visibility serves two audiences at once. A family can plan around a cliff it can finally see. And a policymaker can see work disincentives that no single program creates, because each cliff is an interaction — the fault of the combination, which means the fault of nobody, which is how it survived.

Three families

Three worked cases show what the calculator catches that intuition does not. Each is an interaction between rules that are defensible alone, which is why no program's own manual warns about any of them.

The first is a cliff created by a courtesy. Receiving SSI confers categorical eligibility for SNAP — qualify for one and you qualify for the other, regardless of SNAP's usual income test. As a disabled worker's earnings rise, SSI tapers gently toward zero; the day it reaches zero, categorical eligibility vanishes, and the household must suddenly pass SNAP's own income test, which it may fail outright. The gentle slope ends in a drop, positioned exactly where someone is working their way off the program.

The second is the marriage penalty. Take two single parents, each earning $25,000, each with one child. Separately, each qualifies for a substantial EITC and Child Tax Credit. Marry them, and a combined $50,000 income pushes into phase-out ranges that two $25,000 incomes never touched; the merged household can end up with less than the two households had apart. No line of the tax code says "penalize marriage." The penalty emerges from programs' differing treatment of household structure, and the calculator surfaces it the only way it can be surfaced: run the two adults separately, run them as one household, and compare.

The third arrives on a birthday. A family receiving SSI for a disabled child hits a discontinuity when the child turns eighteen: the program stops counting the parents' income and starts counting the child's own. The SSI payment itself often rises. But the same birthday can move the child out of pediatric Medicaid provisions, school meals, and childcare subsidies — family-based benefits keyed to age, not need. Nothing about the child's disability changed overnight. The rules did. For the New York parent from the opening of this chapter, that third case is a date on her calendar, and the calculator can show her the whole discontinuity years before it lands.

The what-if machine

Everything so far describes current law. The calculator's second mode edits it. Open the policy editor and the parameters of the system become movable: raise the EITC, smooth the SNAP cliff, add a child allowance, make the Child Tax Credit fully refundable, replace the lot with a basic income, flatten the income tax. The current-law world draws in gray, the reform in blue, and the charts update live for the family on the screen: two net-income lines, two sets of dead zones, stacked marginal-rate curves showing where the reform flattens the benefit system — and where it carves a new cliff somebody has not noticed yet.

Run it on recent history. In 2021, Congress temporarily raised the Child Tax Credit to $3,600 for children under six and $3,000 for children six to seventeen, and made it fully refundable. For a single parent earning $15,000 with two young children, the engine computes roughly $1,800 under the old rules — held down by a phase-in that only credited earnings above $2,500 — against $7,200 under the expansion. When the expansion lapsed at the end of 2021, that family's income fell by more than $5,000. Today the same editor measures any proposal against the $2,200-per-child credit that 2025's One Big Beautiful Bill Act set. A legislative aide who once proposed a reform and waited weeks for a score can model dozens of variants in an afternoon — the benefit at $3,000 or $4,000, the phase-in from the first dollar or from $2,500, the cap at $75,000 or $150,000 — and read a different household chart for each.

It's your life, under different rules.

Estimates, not determinations

Every household calculation ships with a caveat, displayed rather than buried: PolicyEngine provides estimates, not benefit determinations. The model encodes the rules as written; an actual award turns on caseworker discretion, documentation, asset tests, and local variation that no rules engine sees. The team checks continuously against official calculators and published tables and treats every discrepancy as a bug to investigate — while admitting that perfect accuracy in a system this complex is unattainable, and that claiming it would be worse than naming the gap.

Within those limits, the calculation became a building block for other people's tools. The household view had been born in the UK — the 2021 launch chapter 5 described — and it matters most exactly where the system is most tangled, which is why the applications found it. Build the computation once and let everything call it: the API argument chapter 5 made, vindicated by who showed up. In 2025, MyFriendBen — a benefits-screening service — launched in Colorado, North Carolina, New York, and Illinois on PolicyEngine's API: plain-language questions in a dozen languages, and in about six minutes a household learns which programs it likely qualifies for. In Colorado, users surfaced an average of $1,500 a month in benefits they appeared eligible for but had not claimed. PolicyEngine's own note reported that the estimates matched expected amounts more than 90 percent of the time — a self-reported figure, not an independent audit, and I flag it as such. A newer prototype reorganizes the whole calculation around life events — a birth, a move, a marriage, a job loss, turning 65 — because that is how people actually meet policy.

The household view, it turned out, was a primitive — and in two senses at once. The obvious sense: benefits screening, financial coaching, guaranteed-income design, and legislative analysis all separately need the same basic computation, so build it once and let everything call it. The deeper sense took longer to see. The household calculation is the atomic unit of ground-truth verification — the level at which an encoded rule can be proven against a reference calculator, an independent implementation of the same statute, or against what a caseworker actually determined. There is no such test for a national total; there is for one family's SNAP benefit.

A national estimate is nothing but a weighted sum of household calculations. If the household calculation is wrong, everything built on top of it is wrong in ways no aggregate will reveal.

That is what separates microsimulation from macro summary. A macroeconomic model might say a tax cut costs $100 billion and lifts GDP by 0.3 percent. Microsimulation can say that and also tell you the same cut hands $12,000 to a high-income household in Connecticut and $200 to a low-income household in Mississippi — and that the Mississippi family now faces a higher marginal rate, because the cut nudged it into a phase-out. The rest of this chapter is about the summing: how one family's arithmetic becomes a country's.

From one household to a country

Distributional analysis has a founding text. In 1974, Joseph Pechman and Benjamin Okner of Brookings published *Who Bears the Tax Burden?*, the first comprehensive attempt to allocate US tax burdens across the income distribution using household-level data. They merged records for 72,000 households and found the tax system roughly proportional across most of the distribution — a result that irritated everyone, challenging conservatives who called the system punishingly progressive and liberals who called it a regressive sham. Annoying both sides at once is roughly what you would expect from the first method that forced the argument through household records instead of anecdotes. Their method — run micro-records through explicit incidence assumptions, then sum — became the template for the distributional tables that Treasury's Office of Tax Analysis, the Joint Committee on Taxation, and the CBO produce to this day.

The method's first constraint is that you cannot survey everybody. The monthly Current Population Survey samples about 60,000 households; its annual income supplement reaches seventy-five to ninety thousand — either way, a sliver of some 130 million American households. Each surveyed household therefore carries a weight saying how many others it stands in for: a rural Wyoming household might represent 5,000 similar ones, a Manhattan household 500, weights the Census Bureau computes to correct for sampling design and non-response. To score a reform, the engine computes the change for every sampled household, multiplies each by its weight, and sums. Simple arithmetic — and an inheritance mechanism, because whatever is wrong with the records passes straight through the weights into every national number built on them.

The data underneath

Plenty is wrong with the records. The survey does not merely blur the picture; it biases it. High incomes are top-coded and under-reported. Benefits are undercounted — Bruce Meyer's linkage studies, which chapter 3 introduced, found 40 to 50 percent of SNAP recipients tell the survey they receive nothing, with the same pattern across programs. The sample runs too thin for most state-level work, and the asset questions are sparse. Feed that survey to a flawless rules engine and the biases choose a direction: you understate the cost of reforming means-tested programs, because the data is missing benefits, and you understate the revenue from taxing top earners, because the data is missing their income. The errors land exactly where policy debates live.

The fix is to calibrate the survey to administrative reality — adjust it until its totals match what the tax and benefit agencies actually recorded. PolicyEngine's milestone on that road was the Enhanced CPS, released in August 2025: five datasets integrated, reported taxes and benefits replaced with computed amounts for internal consistency, income distributions corrected against IRS records, and the survey weights re-solved to match 9,168 administrative totals — cutting deviations from those targets by about 97 percent. Matching the targets you optimized against is fit, not proof — the honest reading of that 97 percent, and the reason the next chapter pauses on what calibration cannot guarantee.

That data work has since been rebuilt as populace: an open commons of calibrated microdata, run the way software infrastructure is run rather than as the private appendix to somebody's tax model. Where the survey is thin, it synthesizes records — a synthetic population, artificial households assembled from public data and calibrated so the whole behaves like the country without exposing any real person's record. Where variables are missing, it fills them in — imputation, predicting what the survey never asked, a method the next chapter takes apart. It calibrates weights to administrative aggregates, and it publishes certified releases the way software ships versioned builds, so an analysis names the exact population it ran on. One consequence reshapes local analysis: a city or state estimate becomes a filter on one national calibrated dataset, not the fifty-first bespoke dataset built by the fifty-first team.

Scoring a reform

With rules encoded and a population in hand, scoring a reform is two runs of the same engine: simulate every household under current law, simulate again under the reform, and sum the weighted differences in tax revenue and benefit spending. Each of the roughly 100,000 records in the enhanced dataset — more records than the survey it started from — triggers thousands of eligibility checks and bracket calculations, and the whole run finishes in seconds. That speed is bought up front: building the engine took years; scoring the marginal reform takes hours because the fixed cost is already paid. Compare the traditional pipeline — read the legislation, code it up, run the model, write the memo, clear review — which takes weeks or months, by which time the legislative moment has often passed. Speed changes what analysis is for, and so does form. A think-tank report is an argument about whether to trust the authors.

Infrastructure anyone can run moves the argument to whether the model is sound, and that fight can actually be settled: model quality is testable in ways a report's credibility is not.

A score is also a bundle of choices that a single headline number hides. Scores are static by default — behavior held fixed, the reform's day-one effect. Layer in labor-supply elasticities and the estimate moves: assume people work more in response to a reform and the cost comes down, assume they work less and it goes up, which is one reason two honest analysts publish different numbers for the same bill. The scoring year matters. And most subtly, every score is measured against a baseline: the world the reform is compared to, which is itself an assumption about what current law would become. Before July 2025, scoring an extension of the 2017 tax cuts required first deciding whether the baseline assumed those provisions would expire on schedule or continue — a choice worth hundreds of billions of dollars before the reform itself was modeled. OBBBA settled that particular question by making the provisions permanent, but the general lesson stands: a cost estimate without its baseline is a number without a denominator. And a cost figure without its distribution is barely a number at all — fifty billion dollars concentrated on families in poverty is a different policy from fifty billion spread evenly across the population, and only the distributional table tells you which one a bill is.

Here is what a score looks like when I run one myself. Take the Child Tax Credit under current law and remove its refundability cap and its earnings phase-in, so that families who owe no federal income tax receive the whole credit. Scored against 2026 law, my own static calculation puts the net cost at roughly $23 billion for 2026, with child poverty falling by about 2.6 percentage points — on the order of 1.9 million children. That is an author-run model output, not an independently verified fact, and it carries every caveat this chapter has raised. Its distributional shape, though, follows from the mechanics: the gains go to families whose credit currently exceeds the income tax they owe — low- and moderate-income working families with children, who forfeit part of the credit today — while higher-income households, already claiming the full credit, see nothing change.

Poverty, and what to show

Scores need outcome measures, and the most consequential is poverty. PolicyEngine uses the Supplemental Poverty Measure, which — unlike the older official measure — counts taxes and in-kind benefits like SNAP, adjusts for geographic differences in housing costs, and nets out work and medical expenses; for two adults and two children renting, the 2024 SPM threshold is about $39,400. The choice of measure decides what the model can see: a poverty measure that ignores SNAP cannot register a SNAP reform; the SPM can, which is what makes it the right yardstick for a model whose whole subject is taxes and transfers. In a 2023 analysis, PolicyEngine put roughly 9.6 percent of Americans below their threshold, with children poorer than working-age adults and seniors, and women poorer than men]. Splitting the microdata exposes what a single rate conceals: WIC, a program that looks gender-neutral on its face, lowers overall poverty by about 0.8 percent but child poverty by 2.6 percent, deep poverty by 2.2 percent, and women's poverty by 0.9 percent against men's 0.7 — the program lands where the mothers and children are.

The same runs slice by income decile, by wealth decile — using wealth imputed onto the survey — by age, by sex, by geography, and by race where the data supports it. Even inequality comes in flavors: the Gini coefficient responds most to the middle of the distribution, so a reform transforming the very bottom can barely move it, while the Atkinson index, tuned by an aversion parameter, weights the bottom heavily and moves sharply. The two can disagree about the same reform, instructively.

Which raises the question of what the tool should say about any of it. PolicyEngine reports the outcomes — budget, poverty, inequality, the distribution across every cut — and adds no editorial verdict. It documents its behavioral assumptions and data limits and keeps the methodology open to challenge. I will not pretend that makes it neutral: choosing to display a poverty rate implies poverty matters, and framing results by income decile frames the debate a particular way. The honest claim is narrower — a deliberate separation of analytical infrastructure from advocacy, so that people who disagree about what to value can at least argue from the same arithmetic.

Solid and soft

Not every number the engine produces deserves the same confidence, and the tiers are worth stating plainly.

Household calculations are the most reliable output. They depend on the encoded rules, not on the data distribution: for a family with stated characteristics, the answer is exactly as good as the statute encoding, no better and no worse for whatever the survey missed. Directional and comparative results come next — whether a reform helps low-income families more than high-income ones, whether Reform A costs more than Reform B — because distribution shapes and reform ratios hold steadier than absolute levels, and the data's limits press on both sides of a comparison. Absolute budget scores deserve the most caution: PolicyEngine's aggregate revenue estimates have historically run well below official totals — by roughly a third — because survey microdata undercounts top incomes and under-reports benefits, and a static model omits behavioral and macroeconomic feedback. For an order-of-magnitude figure or a cross-reform comparison, that is workable; for a precise revenue estimate, the Joint Committee on Taxation and the CBO remain the standard, because they compute on confidential return data with the top of the distribution intact.

Calibration narrows the gap and is itself checked — hold out some administrative targets, calibrate to the rest, and test the reweighted sample against what it never saw. But a 97 percent improvement on a large gap still leaves a gap, and the tiers above are where it lives. What separates an open model from a closed one was never the absence of limits. An open model prints its limits where anyone can read them; a closed model's limits are whatever institutional authority says they are.

Two reforms, seen clearly

Two episodes show what the society view catches while a debate is still live — and where its confidence should stop.

The first is the 2021 Child Tax Credit expansion. The American Rescue Plan raised the credit and, more consequentially, made it fully refundable — removing the earnings phase-in that had excluded the poorest families, the same phase-in the what-if machine priced for one family above — and paid half of it monthly from July through December. Before enactment, Columbia University's Center on Poverty and Social Policy projected the broader relief package could cut child poverty by more than half. Then the Census data arrived: SPM child poverty fell from 9.7 percent in 2020 to 5.2 percent in 2021, a record low. The attribution demands precision, because the debate routinely conflates two numbers the microdata keeps distinct: the credit as a whole kept 2.9 million children out of poverty that year, and the expansion alone accounted for 2.1 million of them. The one-year cost ran to roughly $105 billion. When the expansion expired in January 2022, Columbia's researchers tracked child poverty nearly doubling within months. The forecasts landed for a reason worth naming: the expansion worked through mechanical channels — direct cash, on simple eligibility rules — where the model's arithmetic is nearly the whole story and behavioral uncertainty barely enters. That is microsimulation's best case; high-end tax changes, which turn on behavioral assumptions the data cannot pin down, are its worst. And one subtlety belongs in the record: because the expansion also raised the maximum credit for every eligible family, the distributional tables showed the largest percentage gains at the bottom but real dollar gains well up the distribution — a broad coalition or loose targeting, depending on values no model can adjudicate.

The second episode is British, and it is a diptych. One panel is the September 2022 mini-budget — the same-day analysis chapter 5 recounted. The society view is what gave that analysis its content: beyond the decile table, the engine put the package's inequality effect in numbers, a 1.2 percent rise in the Gini coefficient and a 2.2 percent rise in the top-1-percent income share.

The other panel is the Universal Credit taper cut from the Autumn 2021 Budget: Chancellor Rishi Sunak cut the rate at which UC is withdrawn as earnings rise from 63 to 55 percent and raised the work allowance by £500 — a reform costing about £2.8 billion a year, cutting poverty by around 3.1 percent, with the gains flowing to roughly 12 percent of the population. For a single parent working 25 hours a week at minimum wage, the change meant keeping about £1,000 more a year, while a two-earner couple gained less.

Set the taper cut against the pandemic's £20-a-week UC uplift and the shapes invert: the uplift reached everyone on Universal Credit while the taper cut reached only working recipients, so per pound the uplift cut poverty roughly 40 percent more effectively — while the taper cut did what the uplift could not, dropping the share of workers facing marginal rates above 70 percent from 26 to 9 percent. Similar money, the same program, opposite signatures: one buys poverty reduction, the other buys work incentives, and the society view is what lets a country see the trade before choosing.

By 2026 the machinery had started watching legislation as it moved — flagging which bills the model could already score and routing the rest toward encoding, the on-ramp to a much larger encoding project that Part III describes. But every capability in this chapter, household and society alike, is one engine running one computation at two scales, and it holds up only because three things beneath it hold: the rules encoded correctly, the data actually representing the people, and the behavioral responses — the seam where the model stops describing law and starts predicting humans — handled honestly. Those are the three ingredients of any microsimulation. The next chapter takes them one at a time.

Chapter 7: The three ingredients

Every microsimulation model rests on the same three foundations: rules encoded as executable logic, microdata constructed into a population, and dynamics — the assumptions about how people respond. You will not find that trio stated in the textbooks, which organize around techniques rather than architecture, but it is what every team building such a model must solve, and each ingredient fails in its own way. Get the rules wrong and the calculations are incorrect. Get the data wrong and the estimates are biased. Ignore dynamics and the projections miss how people respond. PolicyEngine, Tax-Calculator, EUROMOD, and TAXSIM each answer the same three questions differently, and the differences are the most revealing thing about them.

The trio also explains why this chapter sits where it does. Each ingredient has become a separate construction project — a layer that AI agents can increasingly build at a fraction of its former cost. That changes nothing about the standard each layer must meet, and everything about who can meet it. A cheaply built layer is worth nothing if you cannot tell whether it is right, so the question that matters for each ingredient is the same: what can its output be checked against?

The rules

The first ingredient is encoding policy rules as something a computer can evaluate — every tax rate, benefit threshold, eligibility condition, and phase-out schedule.

This sounds simple. It isn't.

Take the Earned Income Tax Credit, one credit among dozens. Its phase-in rate depends on the number of qualifying children — and "qualifying child" carries its own age, relationship, residency, and joint-return tests. The credit depends on earned income, which excludes some income types and includes others. The phase-out threshold differs for single and married filers. Many states layer their own EITCs on top, each with its own conformity rules. Encoding this one credit takes hundreds of parameters and dozens of functions, and a single misreading produces wrong answers for millions of households. A comprehensive US model must do the same for federal income tax and its brackets and credits, payroll taxes, fifty state income tax codes, SNAP, Medicaid, SSI, TANF, housing subsidies, WIC, school meals, and more — on the order of a few thousand parameters, each traceable to a legislative or regulatory source, each changing on its own schedule. And because programs interact — eligibility for one benefit often turns on income as defined by another — a change to a single rule ripples through the rest. It is the edge cases and the interactions, far more than the headline rates, where encodings go wrong: a model can get every bracket right and one definition of earned income wrong, and it will be wrong for exactly the families the credit exists to serve.

How to encode the rules is the interesting question, and the field has produced three answers — three architectures, of rising vintage. They share more than their partisans admit: each separates the logic from the parameter values, and each has to answer for time, because policy is a time series and last April's threshold is not this April's. Where they part ways is in who writes the rules, who can read them, and what the encoding is checked against.

Three architectures

The first architecture is a structured document behind a graphical interface, and EUROMOD is its most developed example. Policy rules live in XML: parameters, a "spine" defining the order of calculation, switches for which policies are active. Researchers edit them through EUROMOD's own application, which presents each policy as an expandable tree with dialogue boxes for the values. The advantages are real. A researcher with no programming background can change a rate or a threshold; the interface enforces structure; the files are openly licensed. The costs arrive at scale: the country models run large, navigating them depends on knowing where in the tree to look, tracking changes across versions means comparing XML, and the skill a contributor builds is EUROMOD's tooling rather than anything transferable.

The second architecture is code-native, the approach Tax-Calculator and PolicyEngine take. Policy rules are ordinary source code — readable in any editor, browsable on GitHub, changed through the same version-control workflow developers use for everything else. A calculation is literally a function you can step through in a debugger and cover with unit tests. In PolicyEngine's design the logic lives in Python and the values live separately in YAML, keyed by their effective dates. Here is a UK National Insurance contribution, whole:

```python
def formula(person, period, parameters):
    earnings = person("employee_earnings", period)
    ni = parameters(period).gov.hmrc.national_insurance
    pt = ni.class_1.thresholds.primary_threshold
    uel = ni.class_1.thresholds.upper_earnings_limit
    main_rate = ni.class_1.rates.main

    liable_at_main = max_(0, min_(earnings, uel) - pt)
    return liable_at_main * main_rate
```

```yaml
gov:
  hmrc:
    national_insurance:
      class_1:
        thresholds:
          primary_threshold:
            values:
              2024-04-06: 242  # weekly
        rates:
          main:
            values:
              2024-04-06: 0.08
```

The separation is the design: when a new tax year changes the threshold, you edit the YAML and the code never moves; the code changes only when the structure of the policy does. Both this and the EUROMOD approach are genuinely open — what differs is the experience and the mental model. One says “use this application to view these configurations”; the other says “read this code like any other software project.” That is a community difference, not a verdict. EUROMOD serves researchers who work inside its ecosystem for years, where investment in its tooling pays off in depth; code-native tools serve people who might contribute once, work across many projects, or wire microsimulation into other software, where investment in general programming skill pays off in breadth.

The last of the chapter’s opening four tools sits outside this axis entirely, and it matters for what comes next. TAXSIM — maintained by Daniel Feenberg at the National Bureau of Economic Research since the early 1980s — is neither a spreadsheet nor an open repository. It is a compiled program behind a web interface, and its distinguishing feature is decades of validation against actual IRS return data, which made it the benchmark other tax calculators are measured against. You cannot step through its code. You can trust its answers. Opaque internals, validated outputs: hold that combination in mind.

The third architecture is the newest, and at first glance it looks like a small variation on the second. Rules are still versioned, testable files — a YAML format called RuleSpec, in which the logic and the parameters carry their own effective dates — but the author is usually an AI agent rather than a person, and nothing the agent writes merges on its say-so. The agent drafts against a corpus of the actual source law — statutes, regulations, and agency guidance ingested as anchored provisions — rather than from its training memory, and its output is treated as a proposal, not a result. Every proposal passes through merge-blocking gates. The code must compile and the tests must pass, as in any software project. Then come two checks specific to law. First, money-atom grounding: every monetary amount in the encoding must trace to a quoted passage of the governing statute or regulation, or the build fails — no dollar figure floats free of a source. Second, oracle comparison: the encoding's outputs are compared, case by case, against a reference calculator — an independent implementation of the same law, TAXSIM for US federal tax, EUROMOD and UKMOD abroad, official calculators where they exist — and mismatches block the merge until they are explained. Human review sits on top of the automated gates, not in place of them. The Axiom Foundation is building this layer, and at the time of writing it spans US federal law plus dozens of state codes and several other countries, each encoding carrying a signed manifest recording what was built and how it checked out.

What makes machine-written law admissible — what lets you trust an encoding no human wrote — is not the agent and not the YAML. It is the criterion the gates enforce: every unit's output is checkable against ground truth. A statute has a fact of the matter; an encoding either reproduces it, household by household, or it doesn't, and on every case you put to them, the reference calculators say which. TAXSIM earned trust through decades of validation; the third architecture generalizes that bargain and runs it before every merge.

Agent scale without it is just confident fiction at volume.

How the verification actually works — the oracles and their tolerances, the conformance gates that only ratchet toward correctness, what happens when the reference model itself turns out to be wrong, the countries encoded in days rather than years — is the subject of Part III. Here it is enough to name the criterion, because the same criterion is about to reappear in a place it is much harder to satisfy.

The data

Rules encoding determines what the model computes. Microdata construction determines who it computes for. A microsimulation calculates for tens of thousands of records standing in for the whole population, each needing a realistic income, family structure, location, and program participation, and the model sums the weighted results into national estimates — so every revenue score, poverty rate, and distributional table inherits whatever is right or wrong about the records underneath.

Most records start life in a household survey: in the US, the Current Population Survey — the annual supplement of seventy-five to ninety thousand households that chapter 6 weighted into a country — in the UK the Family Resources

Survey, across Europe EU-SILC. No survey has everything. The CPS captures wages well and capital income poorly; people under-report dividends and gains, the very wealthy rarely land in the sample, and very high earnings are top-coded — compressed into a ceiling exactly where high-earner policies bite. It omits tax-relevant detail like itemized deductions and retirement contributions entirely, and benefits go unreported at the rates chapter 3 measured. The blind spots are survey-specific: the Family Resources Survey lacks the council-tax detail UK local-tax modeling needs; EU-SILC misses the within-year timing that drives benefit calculations. Whatever you want to model, the survey you have is missing part of it.

So the data ingredient is a pipeline, and its stages deserve to be understood, because national statistics ride on them. For decades this was the least glamorous work in the field — done separately inside every modeling shop, rarely published, seldom reused — which is precisely why it repaid being rebuilt as shared machinery.

Statistical matching combines surveys. Take a CPS household — $80,000, two children, California — and find a similar record in the Survey of Consumer Finances, which carries the wealth and asset detail the CPS lacks; merge their characteristics into one richer record. The craft hides in the word "similar." Match on too few variables and you weld a tech worker's income to a retiree's asset portfolio; match on too many and no two households ever pair. Tax-Data, part of the Policy Simulation Library, pioneered this for US tax modeling by matching CPS records to IRS Public Use File data, recovering the capital-gains realizations and itemized-deduction patterns the CPS alone would miss. Matching recombines existing information rather than creating any; its validity rests entirely on whether the matching assumptions hold.

Imputation predicts a missing value from observed ones. The traditional version fits a regression and applies its average relationship, so every $100,000 earner in California receives roughly the same predicted deduction — a tidy fiction. Quantile regression forests instead estimate the whole distribution of plausible values and sample from it, so a high-income California household might draw a deduction anywhere from $15,000 to $50,000. That spread does real work: a reform that caps itemized deductions has winners and losers only if the model's households actually vary, and mean imputation quietly deletes the variation the reform turns on.

Calibration adjusts the weights — each record's stand-in count, from chapter 6 — so that weighted totals reproduce known aggregates. If the survey implies $9 trillion in wages and IRS records show $10 trillion, calibration re-solves the weights until the total comes out right. The mathematics is optimization: define a loss measuring how far the weighted aggregates sit from the targets, add a penalty for moving any weight too far from its survey value, and minimize the sum. The targets are specific and numerous — federal income tax by bracket, SNAP benefits by state, Social Security payments by age group, wages by industry — and the penalty is what keeps the reweighting honest, holding the new weights close to the survey's original design instead of letting a few records balloon to carry the country.

Aging moves the snapshot forward, because the 2023 survey describes 2023 and the policy question is usually about next year or the next decade. The simple version scales by indices — wages by projected wage growth, investment income by projected returns, everyone a year older. The sophisticated version models who retires, who enters the workforce, how individual incomes evolve, buying accuracy at a much higher build-and-check cost. PolicyEngine takes its projection factors from the Congressional Budget Office's forecasts, so its estimates line up with the baselines Washington already argues over.

These stages used to live as one-off scripts inside each modeling team, which meant every result depended on code nobody else ran. They are now reusable libraries — imputation methods behind a common interface, weight calibration with holdout checks and diagnostics, synthesis and projection alongside — consolidated into populace, the calibrated-microdata commons chapter 6 introduced. The methods are the point; the name just means they ship as tested, versioned infrastructure rather than as the private appendix to somebody's model.

One risk survives every stage, and it is the data ingredient's version of the edge-case problem in rules. Calibration matches margins: after it runs, the published totals come out right by construction. Nothing in that guarantees the joint structure — that the right households hold the right combinations of income, assets, and benefits. A population can reproduce every published total and still assemble those totals out of people who don't quite exist. Joint structure is exactly what many reforms turn on: a benefit that phases out with income but only for households above an asset threshold depends on how income and wealth are paired, not on either total alone. Testing the pairing — holding a synthesized population against ground truth it was never fitted to, rather than against the totals it was — is per-unit verification work again, in harder terrain, and Part III returns to it.

The dynamics

Rules handle the arithmetic and data handles the who; the third ingredient is how people respond. Raise a marginal tax rate and some people work less. Cut a benefit phase-out and some work more. Tax carbon and some drive less. These responses move revenue, distribution, and effectiveness, and they are harder to model than everything above, because they are not written down anywhere.

Most microsimulation is static: compute what each household owes and receives under current policy, compute it again under the reform with behavior held fixed, and report the difference as the day-one effect. Static analysis answers who gains and who loses and by how much — answers that barely depend on behavioral assumptions — but it misses responses that change the totals. A carbon tax scored at $100 billion on the assumption that nobody drives less will raise less than $100 billion, because people drive less. The static-versus-dynamic gap is largest exactly where behavior is most sensitive, which is also where honest estimators most disagree.

The most-studied response is labor supply: how much hours and reported income shift when the after-tax wage changes, summarized as an elasticity. The broadest summary number is the elasticity of taxable income — the ETI — which folds in not just hours worked but tax planning, income shifting, and avoidance. Carina Neisser's meta-regression of the literature — 1,720 estimates from 61 studies — finds a mean around 0.30, meaning a 10 percent rise in the net-of-tax rate raises reported income by roughly 3 percent. The mean is the least interesting thing about it. The estimates spread from near zero to above 1.0, varying with income level, tax system, time horizon, and method — and the spread is money: raising the top federal rate from 37 to 39.6 percent brings in something like $50 billion more per year under a low elasticity than under a high one. Within the spread, some patterns hold. Primary earners in married couples barely move. Secondary earners and single parents respond more. The reported income of the very wealthy moves most of all — though much of that is tax planning rather than any change in actual work, as the post-2017 restructuring surge chapter 3 described made vivid, when a new 20 percent deduction redrew business forms within a couple of filing seasons.

PolicyEngine's answer to the spread is to refuse to bury it. A user scores a reform with no behavioral response, with CBO-style elasticities, or with parameters of their own choosing, and watches the revenue estimate move as the assumption does. The same machinery carries the other margins: capital-gains realizations respond in their timing, swinging revenue between years as holders wait out a high rate or cash out under a low one; taxpayers restructure around new rules, shifting income between years or reclassifying its type; and program take-up — already well below 100 percent for SNAP and the EITC — rises and falls with generosity and complexity, so a reform can change costs as much by moving enrollment as by changing the benefit.

There is a more ambitious road. Structural models do not take elasticities as given; they specify what people maximize — utility from consumption and leisure, under a budget constraint — and let behavior fall out of the optimization, so that changing the tax schedule and re-solving produces the new behavior directly. The appeal is reach: a structural model can price a 90 percent marginal rate no one has observed, where an elasticity estimated at lower rates may not carry. The price is assumption — that people truly maximize, that the chosen functional forms are right — plus a steep climb in computational complexity. PolicyEngine does not yet implement structural responses; the architecture leaves room for them as modules over the same household data. But no amount of structure changes what kind of claim a behavioral response is, which is the last thing this chapter has to say.

The seam

Everything in this chapter up to the behavioral response is deterministic. Given the rules and a household's inputs — under a stated reading, where the statute admits more than one — the tax owed and the benefits due are computed, exactly, and they can be checked household by household: against the statute, against a reference calculator, against what a caseworker actually determined. There is a fact of the matter, verifiable now, per unit, before the policy ever takes effect. Chapter 6's standing caveat — estimates, not determinations — marks the domain's outer wall: past the encoded rules lie documentation, discretion, and local practice, where administration takes over from arithmetic.

Ask instead how people will change their hours, their portfolios, or their take-up, and you have left that domain. No statute contains the answer; no reference calculator can adjudicate it; the only test is to wait and watch. Structural models do not escape this — their outputs still land on the prediction side of the line, answerable only after the behavior happens. The same model wears two epistemic hats: one output verified against the rules as written, the other against outcomes as they arrive. Chapter 6's reliability tiers were this seam wearing budget clothes — household calculations on the deterministic side, behavioral adjustments to a revenue estimate on the other. Keeping the two straight — being exact where exactness is possible and honestly uncertain where it isn't — is most of what separates a trustworthy model from a merely confident one.

The three ingredients also supply the questions to ask of any microsimulation model, including every one this book profiles: whether the calculations are right, checked against authoritative sources and legible enough that you can see why they produce a given answer; whether the microdata represents the population, with missing variables filled defensibly and weights calibrated to real totals; and what behavioral assumptions are in play, stated or hidden, and how much the results move when they change. Held together, the trio explains why building this took years. The rules you want dictate the data you need: model capital gains and you need assets; model childcare subsidies and you need childcare expenses. The data you have bounds the rules you can honestly simulate: without deduction detail, you cannot credibly score a deduction reform. Dynamics reach into both, since a behavioral response changes effective rates and effective rates change the distributional story. Different models are strong in different columns — TAXSIM in rules, validated against decades of returns; Tax-Data in microdata construction; OG-USA in dynamics, with a full model of overlapping generations behind it — and PolicyEngine's aim was to be strong across all three at once, which is precisely what made it hard. No single tax formula is difficult. What takes sustained engineering is making comprehensive rules work with realistic data while supporting explicit dynamics, all at once, across the whole system.

Now run the chapter backward with the agent question in mind. Rules: an agent can draft them, and gates can check every unit against statute and reference calculators. Data: agents can build the pipeline, and calibration can be checked against administrative totals — with the joint structure still waiting on harder tests. Dynamics: agents can implement the responses, and reality grades them, later, on its own clock. Each ingredient is a layer an agent can build and a machine can check; no single ground truth serves all three, and the verdicts do not arrive together. That decomposition, and what happened when it met a frontier model with a corpus of law, is the story Part III tells.

Part III: The agent turn

Chapter 8: The AI can't do your taxes

In March 2023, four months after ChatGPT launched, we shipped a button I half expected to regret. It sat beside a household's results on PolicyEngine and said "Explain with AI". Consider the kind of case it existed for — hypothetical here, but of a shape the engine computes constantly. A family of four in Connecticut earning $47,000 a year qualifies for about $475 a year in WIC benefits. The number is right, and the *why* is buried: an income threshold expressed as a percentage of a poverty guideline that itself depends on household size, a categorical-eligibility rule that can qualify the family through a different program entirely, participation windows, documentation requirements — dozens of intermediate steps the model tracks and no ordinary person will ever read. Open source made that logic visible. It did not make it legible. Anyone could inspect the eligibility code on GitHub; a determined user could in principle trace the calculation tree from inputs to answer. In practice, almost no one ever would, and a right answer nobody can follow fails the family almost as thoroughly as a wrong one — they cannot tell which fact about their lives is doing the work, so they cannot tell what would change it.

The button closed the gap by handing the whole tree — every intermediate value, every parameter — to a language model, which restated the result in sentences a person could use: you qualify because your youngest child is under five and your income falls below 185 percent of the poverty line for a household your size.

The model did not compute the answer. It read the answer back.

We chose that division of labor and meant it to last. A deterministic engine produces the numbers — encoding the rules exactly, returning identical outputs for identical inputs, auditable to the line of statute — and a language model explains them, never touching the arithmetic, calling the engine instead of consulting its own weights. The arrangement had a second dividend we did not fully appreciate at the time: because every explanation started from the engine's calculation tree, every explanation was checkable against it. Later in this chapter we would learn, expensively, what happens when an explanation floats free of the tree.

In the spring of 2023, though, the architecture looked to some smart people like a failure of nerve. The models were improving weekly; surely they would soon just do taxes. What settled the argument was a series of measurements.

Three benchmarks

In 2023, researchers at Johns Hopkins and the University of Maryland gave GPT-4 nine curated, self-contained sections of the Internal Revenue Code — the SARA benchmark, built so that every answer is derivable from the supplied text — and asked it 276 true-or-false questions about them. It got 186 right: 67 percent. Better than a coin, useless for a filing. The detail worth keeping is the shape of the ninety errors. Not one was arithmetic. Every miss was a misreading of the law the model had been handed.

Pause on what a misreading looks like from the receiving end, because it is the failure mode that makes this domain dangerous rather than merely hard. A model that misreads a statute does not stammer. It produces a fluent, structured, confident derivation — the same register in which it produces correct ones — and simulated comprehension looks exactly like comprehension. A bare wrong number invites a second look. A wrong number wrapped in a plausible explanation actively discourages one, which is why the failures in this section matter more, not less, as the models' prose improves.

Two years later, the tax-software company Column Tax raised the stakes from reading to filing. TaxCalcBench, published in July 2025, put frontier models to work on hand-built federal returns for tax year 2024 — synthetic cases with the taxpayer's complete information supplied, each filing scored line against line. Under strict scoring, the best performer, Google's Gemini 2.5 Pro, produced a fully correct return fewer than a third of the time; Anthropic's Claude Opus 4 managed 27 percent. Loosen the standard to within five dollars per line and the best model still cleared only about half. The failures had a signature: instead of looking up the IRS tax tables the law requires, models computed tax from the bracket percentages, landing three to five dollars off on return after return — a mechanism behind 15 to 20 percent of the failures by itself. And note what the test was not. These were bounded, synthetic, federal-only cases, simpler than what a storefront preparer sees in March. The models failed the easy version.

By 2026 we had built the harder version ourselves. A federal return is one slice of a household's situation; most families meet taxes and benefits at once, an EITC phasing out while SNAP phases down while Medicaid eligibility flips near a threshold. PolicyBench, our public benchmark, draws realistic household scenarios from calibrated microdata — so the cases cluster where actual households are complicated, with multiple earners, mixed-age children, stacked benefits — and asks two dozen frontier and open-weight models for the complete tax-and-benefit result, scored against the engine's answer to within a dollar. As of mid-2026, the best model still got roughly one in nine household calculations wrong by more than that dollar, and the typical model missed about one in four.

The dollar threshold earns its keep: the misses are not rounding. They are a household ruled ineligible for Medicaid that qualifies, a food-assistance amount off by real money, a credit at the wrong value. A benefit determination is a number that decides whether a family gets help this month.

Three benchmarks, three years, one verdict. The obvious rejoinder at every step was that the next model generation would grow out of it, and the dates are the answer: several generations of frontier models came and went between the first test and the last, and the failure survived them all, migrating from reading comprehension to table lookups to program interactions as the tests grew more realistic. Each benchmark measured the same collision from a different angle — a machine trained on the patterns in text meeting a body of rules that is not a pattern. The rules change every January. They differ across fifty states. They turn on interactions among dozens of provisions, and next year's thresholds appear in no training set, because no one has written them yet. The next chapter takes up why this is permanent; what mattered in 2023 was the practical inference. If frontier models cannot reliably compute a return, they need a tool. AI systems do not memorize multiplication tables; they call calculators. An AI answering a tax question should call a tax engine.

The bottleneck moves

Our first attempt at the tool was crude enough to be funny now. In 2023 PolicyEngine generated a structured block of text — a reform's parameters against baselines, its budgetary and distributional results, a note on style — for the user to copy into ChatGPT, which spun it into readable analysis. The numbers were right because they came from the engine. The integration layer was a human being with a clipboard, ferrying data between two systems by hand.

Tool calling ended the ferrying. GPT-4, Claude, and their successors gained the ability to invoke a function mid-response: hit an API, take the result, keep going. Ask what happens if the Child Tax Credit becomes fully refundable, and the assistant translates the question into policy parameters, calls the engine, receives the quantitative results, and explains what came back. Language on the outside, computation on the inside. Then the wiring standardized. In November 2024 Anthropic open-sourced the Model Context Protocol, a common way for AI systems to discover and invoke external tools; OpenAI adopted it in March 2025, and Google followed within months. A PolicyEngine endpoint became findable and callable by any compatible assistant, no bespoke per-provider integration required. The plumbing stopped being the project.

What that buys in practice is a change in who can ask a question. A researcher writes, in plain English, that she wants to know how extending premium tax credits would affect insurance coverage among middle-income families in several states. The assistant translates that into computational steps — define the reform, select the population, run the simulations, aggregate the results — the engine does the analysis, and the assistant renders the findings back in her language. She never sees the parameters file. The engine never sees her prose. Each side does the work the other cannot.

Which moved the problem. In 2023 the binding constraint was the model: could it pick the right tool, format the call, map a person's question onto the tool's inputs without garbling them? By 2025 the models had cleared that bar so thoroughly that nobody discussed it anymore, and the constraint migrated to the tools — whether they were accurate, comprehensive, current, and legible to whoever stood downstream. An AI that flawlessly calls an inaccurate calculator is worse than useless, because it launders a wrong number through a confident, well-formatted sentence. And a calculator can be arithmetically correct and still fail its callers: slow to update, undocumented, full of integration traps.

Tool quality turns out to be several properties, each its own work. Accuracy. Comprehensiveness — a tool covering federal tax and three benefit programs returns the wrong household total the moment a fourth program is in play. Currency — when Congress rewrote the Child Tax Credit in July 2025, a tool answering family tax questions was wrong until someone updated it, and currency is that race, run each time a legislature meets. Legibility to the developer, the auditor, the caseworker downstream. None of these yields to more training. Each has to be built, and kept built, inside the tool.

Where authority lives

The architecture draws four lines, and the AI crosses none of them. It does not set policy parameters: the Child Tax Credit maximum in our engine is $2,200 because Congress set it at $2,200 in the One Big Beautiful Bill Act, and every parameter carries its legislative or regulatory provenance. It does not estimate behavioral responses: elasticities come from the economic literature and from explicit, recorded methodological choices, never from a model's in-the-moment hunch. It does not judge whether a policy is good: the engine can report that a reform cuts poverty 3 percent and costs $50 billion, and whether that trade is worth making belongs to humans. And it does not overrule the engine: if the computation says a household owes $10,000, the explanation explains $10,000. Four boundaries, one rule seen from four sides — authority for a number belongs with whoever can be held to account for it. A legislature for a parameter. A documented literature for an elasticity. A voter for a value judgment. The engine for a computation. Never the system that is merely good at describing the number.

One place upstream deserves a careful exception, because it looks like a violation and is the opposite. A production model should never let an AI silently pick an elasticity and bury it in the code. But an AI can be a fast, auditable way to survey what the plausible values even are before a human commits. One recent experiment put twenty-six economic quantities — canonical labor-supply and tax elasticities among them — to eleven frontier models, repeating each elicitation fifteen times to recover a distribution of answers rather than a single number. A related project with the economist Jason DeBacker probes what language models imply about the elasticity of taxable income, down to replicating lab experiments on simulated taxpayers. These elicitations are diagnostics. They show what distribution a model carries, where models disagree, and where a change of wording moves the answer — and the useful finding is usually the scatter, not the consensus, because a spread wide enough to flip a reform from progressive to regressive names exactly which assumption the result is hostage to. The parameter a model actually uses still gets written down, sourced, and left open to review.

Day to day, what the interface adds is reach, with a seam. When we tested letting several specialized agents coordinate on a research task, they handled the standard distributional runs and stumbled where programs interacted — implementing each correctly in isolation, then missing how a benefit taper in one collides with a phase-out in another. The mechanical translation transfers; the coordination logic, so far, does not, and the seam where programs interact is where a human analyst still stands. The line between augmentation and automation runs through that seam, measured rather than asserted.

The reach cuts both ways. A congressional staffer who knows policy but not programming, an advocacy group with no data team, can now run analyses that used to require an economist — and so can anyone hunting for a cherry-picked scenario or a confident misreading. Those risks existed under the old system too, concentrated among the few who could run the models. Broader access invites broader checking, and the explanation is the part that scales best: show a worker that she faces a 67 percent marginal tax rate, and the engine hands over the pieces — federal tax, payroll tax, an EITC phasing out, a partial credit clawback — for the model to assemble into the one plain sentence explaining why an extra dollar of her earnings mostly vanishes. The old system could compute that rate too. It could not put the explanation in front of the person living it.

Three builders, one wall

We were not the only ones who ended up here. In November 2025 the IRS deployed Salesforce's Agentforce agents across three of its offices — the Office of Chief Counsel, the Taxpayer Advocate Service, and Appeals — for case summarization, document search, and policy navigation. The deployment followed a workforce reduction, and its boundaries were explicit: every final decision stayed with a human, and the AI could neither disburse funds nor make an eligibility determination.

California had started earlier. Poppy, the state's digital assistant for its own workforce, began its pilot on September 29, 2025, across 67 departments — built in-house, run on state servers rather than an external cloud, drawing on roughly ten models (the state's technology department lists Claude, Gemini, GPT, and Nova among them), and grounded on public state data to hold down hallucination. More than 2,800 employees used it during the pilot; the statewide rollout came in July 2026. Poppy helps a state worker find and understand a rule. It computes no one's entitlement.

Look at what both deployments protect. In each, the rules, the statutes, and the calculations stay human-maintained and human-auditable underneath; the AI sits on top, summarizing, retrieving, navigating — the access layer to an authority it does not hold. That is the same wall we built into a benefits calculator, drawn independently by a federal agency spending on commercial software and a state government writing its own.

An open-source project, a commercial vendor inside a federal agency, and a state building for itself, none copying the others, landed on the same division of labor: deterministic systems for the answers, AI for the access. Three builders under three sets of incentives hit the same wall in the same place. The problem put it there.

When the explanation lied

The last lesson of the benchmark years came from our own board, and it stung. PolicyBench originally published an AI-written narrative beside each scored scenario — an explanation layer, meant to make the results legible to auditors. The narratives read well. Some of them lied. One described a household of retirees as enrollees in a "non-elderly expansion" category that could not apply to them: the mechanism was invented, the prose was fluent, and the combination sailed past human review precisely because it sounded like an explanation. An auditor who read it inherited the fabrication whole.

The fix went deeper than a prompt. We stopped letting the narrative come from the model's imagination and made it come from the engine's internals — every explanation regenerated from the actual variables and intermediate values the computation produced, with validators that reject any mechanism the underlying trace does not support. An ungrounded claim now fails the build before it can reach a reader. The general lesson runs through the next two chapters: even the audit layer needs ground truth. If you want an AI to check work — its own or anyone else's — you give it traces, the numbers the system actually computed. You do not give it prose.

So the models became the interface, and by 2026 the interface was good. That only sharpens the question of what it is an interface *to*. Part II ended with a criterion — an encoding of the law earns trust at the unit of one household, the level where its answer can be proved against something real — and this chapter has been the criterion's negative image: three benchmarks showing what happens when the answer comes from a system with no unit anyone can check. A tool worth routing the public's tax and benefit questions through has to clear a bar most software never faces: not merely accurate but accountable — every number traceable to a line of statute, checkable against an independent calculator, reconcilable to totals a government actually counted. At the scale of the whole law, no such tool existed.

Building one turned out to be possible. That is the next chapter.

Chapter 9: Encoding the law

The question sounds too basic to be unanswered: what is a household's real income — after taxes, after benefits, after the two collide? A lending platform needs it to underwrite a gig worker fairly. A benefits screener needs it to tell a family what help exists. An AI assistant needs it the moment someone asks about their Child Tax Credit, unless the plan is to invent the number. Through the mid-2020s, none of them could simply buy the answer.

Call the engineer who discovers this Priya. She is a composite, drawn from fintech teams I have sat across from; her dead ends are not. Her company advances cash to gig workers, and to lend responsibly it needs each worker's true take-home pay. She tries a tax library off GitHub: federal brackets, nothing else — for her low-income users, whose finances run through credits and benefits, it misses by 30 to 40 percent. She tries a commercial tax API: priced per call, and it cannot touch benefits at all. She tries prompting a frontier model, and gets a hallucination rate no financial product can ship — chapter 8 measured exactly this. She prices an in-house build: eighteen months of engineering, obsolete the first January a legislature meets. Every path ends the same way, in a partial, fragile solution that every company facing the question rebuilds privately, again.

A market of slices

Plenty of companies sell tax software. Specialists thrive on every slice. April embeds tax filing inside other companies' apps; Column Tax has filed more than a million returns through its API and built an AI agent to help maintain its engine as the law shifts year to year. Avalara computes sales tax, and was acquired for $8.4 billion in 2022. ADP and Gusto run payroll. MyFriendBen and Benefit Kitchen screen for benefits. PolicyEngine, EUROMOD, and Tax-Calculator serve policy research. The global tax-technology market was climbing toward $60 billion — balkanized into verticals, each re-encoding the same overlapping rules behind its own walls.

What had no home was the whole: income tax plus benefit eligibility plus the population-level simulation that turns one household into a national estimate. The pieces do not separate cleanly, which is the point. SNAP falls as earnings rise while the EITC climbs; the net depends on household composition, state of residence, and thresholds that interact. Because no one sold the whole, every company that needed it rebuilt the overlap privately — the same EITC phase-in, the same SNAP deductions, encoded again and again — and each private copy was a fresh chance to be wrong.

Nor is this a gap that model scale will close, for reasons that are structural rather than temporary. Tax law changes every January, so last year's training data encodes rules that no longer apply, and a model cannot extrapolate to a bracket it has never seen. Fifty states each run their own income taxes, credits, and benefit programs — combinatorial sprawl past what pretraining memorizes reliably. Eligibility turns on dozens of interacting variables, where one wrong input silently yields a wrong result. And financial calculation tolerates neither 95 percent nor 99: right nineteen times in twenty means one return in twenty filed wrong. Column Tax's own engineers, who profit when models improve, put it flatly: language models "cannot 'do taxes' on their own because tax calculations require 100% correctness". On our benchmark, the best 2026 models still got roughly one in nine household calculations wrong by more than a dollar. The way through is the tool chapter 8 specified: a deterministic, auditable engine the model calls, and this chapter builds it.

Administration is the product

Before the building comes a correction, learned the hard way, to what "good" means here. The hardest problems in benefits software are not mathematical. Whether a formula is legally correct is one question; how fast a rule change becomes publishable, how many misunderstanding loops open between the policy expert and the engineer, how much hand-holding a partner developer needs, and how often a tool hands back a misleading answer because its interface cannot express a real-world exception — these decide whether the software is any good, and none of them is a math problem. They surface as operational metrics: time to publish, discrepancy rates against authoritative examples, onboarding time, the rate at which a support ticket exposes an unwritten assumption.

Households, meanwhile, experience policy through its administration. A credit that exists in statute but takes six months to reach a screener is not fully real to the family it was written for. A calculator that silently misses an immigration rule or a state exception erodes trust, burns caseworker time, and suppresses take-up. And when the administration fails at scale, the failures have names and dates. In 2023, during the great post-pandemic Medicaid redetermination, CMS ordered states to restore coverage to roughly half a million children and families after finding a defect in the automatic-renewal logic of their eligibility systems. KFF Health News documented how long fixes take inside the eligibility systems Deloitte runs for states: in Kentucky, one system limitation that cost a resident, Beverly Likens, her Medicaid coverage took about ten months, more than 3,500 hours of work, and over $500,000 to resolve; in Georgia, officials were still untangling a defect affecting more than 25,000 SNAP and TANF cases nearly two years after it was first reported. Deloitte-built eligibility systems span roughly twenty-five states, under contracts that had reached some $6 billion by late 2025. Virginia Eubanks and colleagues showed the quieter version of the same failure: a screening tool can look authoritative while its logic stays opaque, so people act on wrong outputs without thinking to challenge them.

Run through those cases and one trait repeats. The defect was buried in a vendor system that no outside party — often no inside party — could inspect, trace to a rule, or fix quickly. The rule was right in statute and wrong in software, and invisible until it produced a headline.

Pricing the error rate

Then policy started billing for it. The One Big Beautiful Bill Act — Public Law 119-21, enacted July 4, 2025 — sharpened a decades-old lever: quality-control penalties for high error rates go back decades, but for the first time states pay a share of the benefit costs themselves. Beginning in fiscal 2028, that share is tied to the state's own payment error rate: nothing while the rate stays under 6 percent, then 5, 10, or 15 percent of benefits as it crosses 6, 8, and 10 percent — and for that first year, a state may elect whether its fiscal 2025 or 2026 error rate is the one that counts. The national payment error rate for fiscal 2025 was 10.62 percent, an average that would put many states straight into the top tier; one outside estimate put first-year state exposure at roughly $9 billion. The same law raises the state share of SNAP administrative costs from 50 to 75 percent in fiscal 2027. Medicaid's new work requirements, meanwhile, oblige expansion states to stand up systems by the end of 2026 to verify that able-bodied adults are meeting an eighty-hour-a-month engagement requirement. A payment error rate used to be an audit statistic. It is now a line in the state budget — and lowering it is precisely the job of rules infrastructure that is verified, current, and provenance-carrying.

The stakes of getting this wrong run past budgets, and the record is not hypothetical. Australia's Robodebt scheme raised automated debts against welfare recipients on faulty income-averaging; the courts unwound it, and a royal commission put ministers under oath — its report called the scheme "a crude and cruel mechanism, neither fair nor legal". Michigan's MiDAS system accused tens of thousands of unemployment claimants of fraud; of the determinations it issued without any human review, roughly ninety-three in a hundred were wrong. Arkansas replaced a nurse's judgment with an algorithm for allocating Medicaid home-care hours, cut care for disabled residents, and drew years of litigation. In each case the arithmetic was automatable and the judgment was not, and the system automated both.

American benefits law now draws that line explicitly. USDA's Food and Nutrition Service has held that AI may not replace the state merit personnel who make SNAP eligibility determinations — its framework states that "All AI must be used in compliance with program requirements for the use of merit systems personnel, such as those applicable to SNAP", and its companion automation memo draws the same line at the job description: "bots are considered non-merit staff and prohibited from performing these merit staff functions". That boundary matches the architecture this book has been arguing for from the other direction. A human makes the determination. The tool's job is to make the determination checkable — and everything in the failures above, from Robodebt to the Kentucky renewal logic, is a case where the determination was neither human nor checkable.

The foundation

The Axiom Foundation exists to build that tool as a public good. It is a Delaware nonprofit, fiscally sponsored by the PSL Foundation and anchored by funding from Ballmer Group; Ariel Kennan became its president on July 1, 2026, and it launches publicly on July 28. Its charter is narrow and immodest at once: encode the law itself, exactly, as open infrastructure — in a form anyone can run and anyone can check. PolicyEngine had proved that tax and benefit rules could be encoded accurately at all. Axiom is the attempt to encode all of them, and the wager underneath it is the subject of the rest of this chapter: that agent-drafted encoding, behind checks that block a merge, is what makes that scope tractable for a nonprofit. A sibling institute, Thesis, takes the other half of the problem — forecasting what governments will actually do — and belongs to later chapters.

Rules that carry their sources

An Axiom encoding is not an ordinary program. Its format, RuleSpec, is a set of YAML files that hold the law's logic and the law's numbers apart — each file versioned, each testable, each stamped with the dates over which it takes effect. The logic that computes a credit lives in one place; the parameters it reads — a rate, a threshold, a phase-out amount — live in another. An act of Congress changes the logic; an agency's annual inflation adjustment changes only a parameter value. Both land as versioned files with effective dates, so an encoding answers more than "what does the law say?" It answers "what did it say last March, and when did we learn it had changed?"

Take the Earned Income Tax Credit. The statute defines a credit that climbs with earned income at a set rate up to a ceiling, plateaus, then phases out above a higher threshold that shifts with filing status and number of children. In RuleSpec that becomes a formula referencing named parameters — phase-in rate, earned-income ceiling, phase-out rate and threshold — and each parameter carries its own citation, its own effective date, its own link to the source. Watch the two kinds of change move through it. Each fall, the annual inflation adjustments arrive, and the parameter files gain new dated values while the formula — which Congress has not touched — does not move; the diff is a page of numbers with effective dates. When Congress restructures the credit, the formula changes, and the diff says so in logic rather than in numbers. A reader of the repository's history can tell, at a glance, which years the law changed and which years only its amounts did — a distinction that takes a tax lawyer to extract from the statute books.

Beneath the rules sits the corpus: the source text itself. Axiom ingests statutes, regulations, agency manuals, and sub-regulatory guidance as anchored provisions, each clause individually addressable, organized against a registry of roughly 41,000 legal concepts. The corpus is what makes provenance possible — an encoding earns trust by pointing back to the exact governing words, which live in a service anyone can open and read. The anchoring matters more than it sounds, because the rule a caseworker actually applies is often buried in an agency manual or a guidance letter, not the statute; the corpus holds primary law and its sub-regulatory layers at clause resolution, so the pointer lands on the words that govern.

The gauntlet

Encoding law by hand is slow and does not scale; a team can keep one country current or attempt fifty, not both. The pipeline, axiom-encode, turns encoding from analyst-months into agent-runs. It pulls the relevant provisions from the corpus, scaffolds a prompt around them — the statute text, the concepts it touches, the shape of the encoding to produce — and hands the draft to a language model. Then, before anything merges, the draft runs a gauntlet in which every gate can block the merge: the code must compile; its tests must pass; a proof step must validate the logic's internal consistency; the output must be compared against independent reference calculators; and every monetary obligation must be grounded in its source. Only a draft that clears all of it yields a signed manifest — a cryptographic record of what was encoded, from which sources, having passed which checks — so a later reviewer, a skeptical agency, or a court can reconstruct the provenance rather than take it on faith. Human review sits deliberately at the top of this stack: the machine drafts and the gates screen, while a person adjudicates the judgment calls the statute leaves genuinely open — the cross-references and exceptions no gate can settle.

Each gate defends against a different failure. Compilation catches an encoding that does not run. The tests catch a change that breaks a case that used to work. The proof step catches logic that contradicts itself. Comparison against an independently built calculator catches the failure no self-check can see — an encoding that runs, passes its own tests, and is still wrong — which is the subject of the next chapter. And the last gate catches a number with nothing behind it.

That last gate deserves its own paragraph, because it is the pipeline's sharpest idea. The money-atom gate demands that every monetary obligation an encoding produces — every dollar figure the law commands — trace to a quoted excerpt of the governing source: the actual authorizing words, pulled from the corpus, not a section number in a comment. It targets money rather than every clause because a wrong eligibility flag is a bug, but a wrong dollar figure is a household underpaid or a program overexposed — the failure mode with a face on it. If a figure cannot be grounded that way, the build does not warn. It fails. Ordinary software cites its sources in documentation, and documentation drifts out of date the moment someone edits the code; a merge-blocking check cannot drift, because nothing merges until it passes.

Provenance stopped being a promise and became a merge condition.

Notice, finally, what the gauntlet does not trust: the model that wrote the draft. The pipeline runs several language-model backends and stays indifferent to which produced a given encoding: wherever the draft comes from, the gates hold the guarantee. A better model drafts faster and trips fewer checks. It never earns the right to skip them.

The whole of the law

By July 2026 the US repository held on the order of 3,000 rule files with matching tests, covering federal law plus twenty-eight state codes — absorbed with their full git history, so the encodings inherit years of prior fixes and the reasoning behind them, a change of representation rather than a fresh start with fresh bugs. Separate monorepos carried the United Kingdom, Canada, New Zealand, and Belgium, alongside a set of African lanes validated against established reference models — the next two chapters take up the reference-model races and the African week. A single pipeline produced all of it, which puts the marginal cost of the next jurisdiction in units of agent-time rather than institution-years.

The scope is deliberately, almost provocatively, total: not "the parts of law that compute a number" but all public policy — statutes, regulations, agency manuals, statutory guidance, grant conditions. Among the early UK encodings is the council-tax-reduction policy of the Royal Borough of Kingston upon Thames: a local scheme binding on a few tens of thousands of residents, encoded with the same machinery as federal income tax, because a resident is governed by it as concretely as by any act of Parliament.

Bindingness is metadata, never a scope filter.

Whether a provision is hard law or soft guidance is recorded as a property of the encoding; it plays no part in deciding what is worth encoding. And the encodings model authority explicitly: the chain by which a statute delegates power to an agency, and an agency binds itself through its own published policy. The doctrines are old — in US law an agency must follow its own regulations; in English law it must follow its own stated policy absent good reason — and encoding them lets a system answer not only what a rule computes but whether the body applying it had the authority to. That matters because much of law computes no amount at all; it asks whether a conclusion holds, fails, or remains undetermined, given a history of events: is the notice valid, the deadline met, the procedure followed? Modeling authority, delegation, and that second shape of question is the difference between encoding a calculator and encoding the law. Everything ships under permissive licenses — Apache 2.0 for the code, Creative Commons Attribution for the content — so no one need ask permission to run, fork, or build on the encoded law.

A commons, governed

Building this as a public good removes the sharpest objections — no shareholders, no paywall on the law — without removing the need for governance, and honesty requires naming the failure modes. First, whose rules come first: funders and large institutional users have priorities, and the risk is a roadmap that tracks whoever pays rather than where accurate rules matter most; deprioritize one jurisdiction's benefit code and you have decided which vulnerable households get accurate answers. The guardrails are transparent governance, a published roadmap, and the standing ability to fork. Second, capture: even a nonprofit can be captured by a dominant funder or a maintainer monoculture, until "open" is a brand rather than a fact; independent governance and a genuine plurality of contributors are the only durable defense. Third, staleness — the characteristic failure of public goods is not bankruptcy but quiet drift, laws changing while no one notices. The conformance ratchets of the next chapter exist to turn "keep it current" into a gate.

Openness is the mechanism behind each defense. A fintech can read exactly what the encodings compute. An agency can confirm they match statute. A citable, inspectable rule is more defensible than a model's unexplained output when a regulator or a court asks for the audit trail. And the ability to fork is the ultimate check on capture. Underneath these tactics sits a structural commitment: the encoding of law is reference infrastructure — closer to a dictionary or a map projection than to a product — and reference infrastructure is not well served by a private gatekeeper sitting between the public and the statute. So the rules layer is a commons, and the commercial activity growing around it — managed runtimes, service-level guarantees, applied products — is built on the commons, never owner of it. One such operator already exists in formation, organized around graded simulation; it is not yet public.

The foundation commits to the dividing line in public: the commercial service tier will always be a separate entity, so the foundation never competes with builders on its own commons. The law stays free to run. Only the convenience of not running it yourself has a price.

The cache that wasn't

The honest way to end a chapter about a verification regime is with the time it turned on us. In July 2026 we ran an experiment on a premise I found elegant: if every rule is deterministically derived from source text, then no encoding needs to be a durable artifact — encodings are just cache. Discard any module and regenerate it from the corpus on demand. Internally the test was called the B1 probe, and it put the premise to twenty-five modules.

It failed. Regeneration was genuinely cheap — about eight cents of compute per module — but in twenty-three of the twenty-five cases the original encoding was the correct one, and the freshly regenerated version introduced naming instability that would have silently broken every downstream consumer. Same source, same pipeline, different output. We adopted the conclusion the evidence forced, against the elegant version of our own story: an encoding is a durable artifact with provenance — stored, versioned, migrated with care — and regeneration went back on the shelf as a research problem. I count the episode as the discipline working. A project convinced of its own story would have shipped regeneration and met the breakage in production.

Which points at the gap this chapter has left open on purpose. Encoding all public policy at agent speed is worth nothing unless the encodings are right, and "a language model wrote it and the tests passed" is no reason to trust a number that decides whether a family makes rent. Compilation proves the code runs. The money-atom gate proves each dollar traces to the law. Neither proves the encoding matches the world. That proof is a separate discipline — comparison, case by case, against independent calculators built by other people with other tools, and an explanation owed for every place they disagree. How to trust law encoded by machines is the subject of the next chapter.

Chapter 10: The verification problem

The first time we raced our Belgian encoding against EUROMOD, the two disagreed by €643. The encoding had computed roughly €20.9 billion in employee and employer social-security contributions across a simulated population of workers; EUROMOD — the European Union's reference tax-benefit model, decades of institutional work deep — came out €643 higher. Six hundred forty-three euros, on twenty-one billion. Is that success?

The honest answer is that the number alone cannot say, because €643 is consistent with several different worlds. It could be floating-point dust — hundreds of thousands of sub-cent rounding differences, summed. It could be a single catastrophic error: one worker off by €643 while every other matches to the euro. It could be a small systematic bias, a rule applied a hair too generously to everyone. Or it could be the worst case, the one that looks best from a distance: large errors in opposite directions, canceling almost perfectly on their way to a reassuring total. This book has already met that case. When CBO scored the Affordable Care Act in 2010, its projection of the total uninsured rate landed near the outcome while — as later decomposition showed — overstating exchange enrollment by more than half and understating Medicaid by nearly as much, two errors partly canceling. The total was approximately right for the wrong reasons. If the only test is whether two totals sit close together, you cannot distinguish a right model from a flatteringly wrong one.

So we did not test the totals. We compared workers. For each of the 78,479 simulated workers, the pipeline lined up the encoding's contribution against EUROMOD's and took the difference, and the largest single-worker discrepancy in the entire population was three cents a year. Not on average — three cents at the very worst. Every worker matched to within a rounding error, and the

€643 was exactly what it looked like once examined unit by unit: dust.

Aggregate agreement can hide anything. Per-unit comparison hides nothing.

That sentence is the book's discipline compressed, and this chapter is its mechanics. Chapter 9 ended by conceding that an encoding can compile, pass its tests, and ground every dollar in quoted statute while still being wrong about the world. The question of whether to trust law encoded by machines comes down to whether that last gap can be closed — and the answer is that "trust" is the wrong frame. You do not trust an encoding. You verify it, one unit at a time, against something that is true independently of the machine, and you wire the verification in so tightly that a failing encoding cannot ship. An encoding no one has checked, determination by determination, against ground truth is not a cheaper version of a validated model. It is confident fiction with good production values.

The ladder of checks

The ladder starts where chapter 9 left off, with the money-atom gate — merge-blocking, remember, not advisory. Every rate, threshold, taper, and cap that actually moves money must be grounded in a quoted excerpt from the governing source; an amount with no provision behind it fails the build the way a syntax error does. The finished Belgian encoding carried 539 distinct monetary obligations, and the gate demanded a source excerpt for every one; the count of ungrounded obligations stood at zero. The interesting part is not that the count reached zero but that the gate is configured so it can only ever fall.

Grounding is necessary and nowhere near sufficient. A number can be faithfully quoted from the statute and wired into the wrong formula, or into the right formula with the wrong sign. So the encoding must also compile, pass the companion tests that ship beside every rule, and clear a proof pass that checks the logic holds together — that its pieces refer to quantities that exist, in units that match, along paths that resolve. All of that establishes internal coherence and external sourcing. None of it establishes correctness, because a system can be coherent, sourced, and wrong.

For correctness, the encoding is raced against an oracle: an independent implementation of the same law, usually built by other people, often over many years, whose answers function as a second opinion we did not write and cannot quietly influence. The oracles in use are the ones the field already trusts. TAXSIM, the NBER calculator that has anchored academic tax research for four decades. The classic PolicyEngine engine. UKMOD for the United Kingdom. EUROMOD for EU member states. The SOUTHMOD family that UNU-WIDER maintains for a growing list of developing economies. And wherever a tax authority publishes an official calculator, the authority's own.

Chapter 3 laid out three levels on which any model can be validated — component, aggregate, predictive. The oracle program industrializes the first. Component validation used to mean spot-checking one case against an IRS worksheet; here it means replaying entire populations through two independent engines and demanding agreement on every record. Component validation grown teeth. And the whole-population part is load-bearing: a hand-written test suite checks only the situations its author imagined — the married couple, the pensioner, the three-child family — and says nothing about the case nobody imagined, the worker crossing two thresholds in the same year, the household in the seam between two rules. A full calibrated population contains those cases whether or not anyone anticipated them. Racing it through both engines tests them too.

What parity looks like

A word about whose numbers the rest of this section reports: ours. The conformance boards are public and the comparisons are built to be rerun, but what follows is our program measuring its own encodings against reference models other people built — a builder's report, not an outside audit. That is precisely why the machinery has to be this strict.

The United Kingdom is where the encodings and the oracle agree most exactly, and "exactly" is the operative word. For a deterministic calculation there is a single right answer, and a tolerance band is mostly a comfortable place for a wrong answer to hide. UK income tax matches UKMOD to the pound — not within a tolerance — across five test suites, including the withdrawal of the personal allowance, the taper that claws back the tax-free band from high earners and produces the marginal-rate spike that trips up informal tax advice. The Scottish income tax, with its own six-band structure, matches on nine cases out of nine. Savings income and dividend income, each taxed on its own schedule with its own allowances, match exactly. National Insurance matches or is explained: where the two systems diverge, the divergence traces to a documented convention — UKMOD computes contributions weekly and then annualizes, while the encoding works in annual terms — and once you account for it, the two reconcile.

Universal Credit is the instructive case, because the raw score looks bad: of nineteen test cases, eight matched. The other eleven are not errors. UKMOD applies a stochastic take-up draw — deciding at random, household by household, whether an eligible family actually claims, because in reality many eligible families never do — while the encoding computes statutory entitlement, what the household is owed if it claims. Those are different quantities by construction, and every one of the eleven mismatches resolves precisely to that difference. The two systems are answering two different questions, and both are answering theirs correctly.

Which is why the load-bearing word in this program is not "match" but *disposition.* A raw mismatch is neither a failure nor a pass; it is an open question, and it stays open until someone closes it with evidence. An explanation, here, is a checked artifact rather than a shrug: an arithmetic reconciliation that reproduces the gap to the penny, a citation to the statutory provision the oracle implements differently, or a documented account of the oracle's own modeling convention, like the take-up draw. Every mismatch is either dispositioned or unexplained, and only unexplained mismatches count against the encoding. The standing UK result — the number that matters, not the raw match rate — is that across every surface we have compared, 100 percent of mismatches are explained and zero are unexplained.

Belgium tells the same story with different furniture. The regional and municipal surcharges stacked on Belgian income tax match nine cases out of nine. A greenfield encoding of student study allowances — a body of rules with no prior implementation anywhere to copy from — matches six out of six. The contributions match per worker to three cents. Every mismatch on the Belgian conformance board carries its disposition, and none is attributed to an error on the encoding's side. "Clean," on these boards, means everything that disagreed was run to ground and written down.

When the oracle is wrong

The dispositions discipline has a consequence we did not design for. If every mismatch must be chased to evidence, then sometimes the evidence exonerates the encoding and implicates the reference model. Race a fresh, exhaustively grounded implementation against a model maintained by hand for decades, and now and then the fresh one is right where the mature one is wrong.

Exhaustiveness makes that possible. Nobody maintaining a mature model by hand replays tens of thousands of records against a second independent engine on every change; until recently the compute and the tooling made it prohibitive. When you do, an inconsistency that surfaces in a handful of unusual cases stops hiding in the tail. It becomes a specific record with a specific wrong number that a person can investigate.

Chasing one mismatch through the UK savings rules, we found the discrepancy was not in our encoding. UKMOD was mishandling an interaction involving the personal savings allowance, a corner where two provisions meet and the combined behavior is easy to get subtly wrong. We wrote it up the way the discipline demands — the inputs, the statutory result expected and why, the reference model's actual output, the provision each system implemented — and filed it publicly as an issue on the UKMOD-PUBLIC repository for its maintainers to weigh. A second case was stranger. Running the EUROMOD comparison, identical input cases sometimes scored differently depending on where they sat in a batch — the same household, the same year, two answers, determined by the case's neighbors in the queue rather than by anything about the case. That is the fingerprint of state leaking between records that are supposed to be independent. We root-caused it to a batch-processing contamination bug in the adapter layer that runs cases through the model, and reported it to the maintainers with a reproduction in hand.

These are not gotcha stories, and the framing matters more than the finding. The reference models are why our encodings can be trusted at all; strip away the independent second opinion and "the machine says so" is exactly the epistemic position this project exists to escape. TAXSIM has run for forty years. EUROMOD and its national variants represent decades of institutional work. That a model of that size contains a bug is not a scandal — every large model does, our encodings emphatically included — and finding one is no victory over its authors. The collaboration runs in both directions: the reference model gets free, specific, evidence-backed quality assurance from a system precise enough to notice a three-cent gap and disciplined enough to chase it to cause. Openness means the errors flow both ways, bugs filed upstream and downstream, both models better for it.

A one-way valve

The hardest honest objection to a public rules layer is rot. Laws change every year, and the characteristic failure of shared infrastructure is quiet staleness — encodings drifting out of date, coverage eroding at the edges, a model accurate on the day it shipped and subtly wrong three budgets later with no one the wiser. A verification suite that runs once and gets admired is no defense. A suite wired into a ratchet is.

The conformance gates are configured as one-way valves. Coverage — the number of statutory surfaces under test against an oracle — may only rise; a change that would drop a jurisdiction or program below its current tested coverage fails CI and cannot merge. The count of unexplained mismatches may only fall; a change that introduces a new unexplained divergence, or quietly un-explains a previously dispositioned one, fails too. The count of ungrounded monetary obligations may only fall toward the zero it already reached. The valves hold because the full conformance suite runs on every change, not on a schedule someone has to remember. When a new budget moves a threshold, the new value lands as a dated addition beside the old one and must pass against the oracle for its own year before merging — last year's cases checked against last year's law, this year's against this year's. Nothing is overwritten in place, so nothing quietly falls out of coverage. The moment the encodings drift from the law is the moment the build stops going green.

Two honest limits. A rule can still go un-encoded, and a brand-new program can still escape coverage entirely; the ratchet preserves what has been won and checked, and it does not discover law it has never seen. What it converts is the failure mode: staleness stops being an anxious, permanently-behind human maintenance problem and becomes a mechanical invariant on the tested surface — correctness, once won and checked, cannot be lost again without the build going red, and silence was always the dangerous part. That promise is weaker than "always current." It is much stronger than "trust us to keep it current."

The gates are for us first

It would be comfortable to describe all of this as a wall against untrustworthy AI. That is true, and it is not the honest emphasis. The gates catch our own mistakes first, and the most useful report I can give is the times they turned red on us.

The B1 probe from chapter 9 is the cleanest example, and it is worth restating as a verification story rather than an engineering one. We took twenty-five modules in production, regenerated each fresh from its source, and compared the output against the version we were running — regeneration cost about eight cents of compute per module, and twenty-three of the twenty-five came back with the old version correct and the regenerated one worse, mostly through renamed variables that would have silently broken every downstream consumer. Nobody reasoned their way to that in advance; we know it because a gate compared regenerated output to a known-good baseline, and the baseline won, 23 to 2. The deeper lesson outlived the experiment: an encoding is an answer with a history — which source, which version of the law, which call was made where the text was genuinely ambiguous — and regenerating from scratch discards the history while producing an artifact that looks every bit as authoritative. Durability is what lets a consumer trust that yesterday's number still means the same thing today, and that when it changes, it changed because the law did.

The same lesson had already arrived at the benchmark. Chapter 8 told it: an early layer that narrated each PolicyBench result in prose was caught inventing derivations — describing elderly enrollees as beneficiaries of a "non-elderly expansion" — and any audit that read those confident narratives inherited the fabrication whole. The fix grounded every explanation in the engine's actual internals and added a validator that rejects any mechanism the underlying trace does not support. Even the audit layer needs ground truth; you give the judges traces, not prose.

And once, the gate caught a lie aimed at readers of this work. When we adapted a US-calibrated population to the United Kingdom and wrote up the results — the next chapter tells that story properly — a summary statistic that no run had actually produced found its way into a draft: fluent, plausible, fabricated. A verification gate caught it before publication, and it was struck.

None of these episodes is an embarrassing exception to a spotless record. They are the record. A verification regime that only ever vindicates its builders is a marketing department with test coverage; the gates earn their standing by how often they turn red on their own authors.

The admission rule

One rule sits under everything in this chapter, and the rest of the book enforces it. Simulation is admissible only where its verification chain terminates in ground truth — a forecast score, an oracle parity, a statutory exactness, a calibrated marginal.

Each terminus is a different kind of true thing, checked a different way. An oracle parity: an independent implementation agreeing on every record, this chapter's subject. A statutory exactness: the money-atom carried to its end, an obligation traced to the quoted provision that fixes it. A calibrated marginal: an input distribution reconciled against a number the government actually counted, so the population under the rules is pinned to something real. A forecast score: a prediction published with an interval and graded only later, when the official figure lands. They are not interchangeable, and they share the one property that makes any of them admissible — each bottoms out somewhere the model's own confidence gets no vote. Where the chain exists and holds, the estimate is admissible. Where it does not, the estimate — however fluent, however cheap, however fast — is confident fiction, and the honest move is to mark the boundary rather than blur it.

Which exposes the assumption this whole chapter stands on. Everything here rests on the existence of an oracle: a UKMOD to disagree with, a EUROMOD to file bugs against, an official calculator to match to the pound. The chain terminates cleanly for the UK, for Belgium, for the US, because in those places somebody already spent years or decades building a second opinion. Most of the world's tax and benefit law is not like that. For much of it there is no reference model at all — no calculator to race, no mature implementation to disagree with, no independent answer to check. What becomes of the discipline where there is nothing to check against?

The second week of July 2026 put that question to the test.

Chapter 11: Microsimulation anywhere

On July 8, 2026, two findings ledgers went up on GitHub a few hours apart. Ghana's listed thirteen findings. Uganda's listed none. By the end of that week there were five ledgers — Zambia's on the ninth, Ethiopia's after a single overnight run, Rwanda's closing the set — and behind each one stood a country whose tax and benefit system our agents had encoded, provision by provision, and raced against the reference model that UNU-WIDER maintains for it.

Those reference models deserve their billing before anything else in this chapter, because nothing in it works without them. SOUTHMOD is the family of tax-benefit models that UNU-WIDER has built and maintained with national research teams for a decade, covering fourteen developing economies — GHAMOD for Ghana, UGAMOD for Uganda, MicroZAMOD, ETMOD, RWAMOD. They are the product of exactly the slow institutional work this book has described elsewhere: a national tax-benefit model has historically taken years — a team, a budget, a data-sharing agreement, a maintenance commitment measured in decades — which is why such models have lived inside finance ministries and a few well-funded institutes, and why most of the world has none at all. The fourteen SOUTHMOD countries are the exception, and the exception is what made the week possible: they were the places where a second opinion already existed. Chapter 10 ended by asking what the verification discipline becomes where there is nothing to check against. The first answer is that you begin where there is something.

Each country already had a model. What the week added was a second, independent implementation of its statutes — permissively licensed, open to anyone for any purpose, built from the gazetted law itself — plus the thing I have come to think matters more: a public ledger recording every place the two implementations disagree and why. What the reference models' maintainers gained runs the other way, and I will get to it. Everything here, as in the last chapter, is our own program reporting on itself; the ledgers are public and the comparisons are built to be rerun.

Two things had to be true at once for any of this to matter: that a model could be built this fast, and that each one arrived carrying the evidence for whether it was right. Most of this chapter is about the second condition, because without it the first is a liability.

What a finding is

Thirteen and zero, produced the same day by the same method. Neither is an obvious headline, and both are good news — but only once you know what a finding is.

A finding, in these ledgers, is a dispositioned divergence: a specific case where the encoded statute and the reference model produce different numbers, chased until someone can say why, and recorded together with the evidence for the explanation. "Explained" carries the same weight it carried in chapter 10 — an arithmetic reconciliation, a citation to the governing text, or a documented convention of the reference model, never a shrug. The one kind of result the process treats as unfinished is a divergence nobody can account for.

The explanation can land on either side. Sometimes the encoding is wrong — a misread threshold, a missing phase-out, a rate applied to the wrong base — and the fix goes into the encoding. Sometimes the reference model is out of date, or has made a defensible modeling choice the statute does not require, and the finding travels upstream: written up with its arithmetic and its citation and reported to UNU-WIDER, whose team maintains the SOUTHMOD models. Ghana's thirteen findings are thirteen such adjudications, posted where anyone can read them. The reference models are instruments, and like any instrument they are improved by being read carefully against their source. There is a quiet reversal of trust inside that sentence. The natural assumption is that the established, decade-old model is the authority and the freshly generated encoding is on trial — and most of the time that is right, and most findings are the encoding's to fix. But not all. When a divergence is chased to ground and the statute plainly says one thing while a model does another, the encoding becomes the instrument that caught it. Neither implementation is presumed correct. The statute is, and both are measured against it. Nobody grades anybody, and everybody grades the law.

Which brings me to Uganda, the result I keep returning to because of what one day used to buy. PolicyEngine UK went live in the fall of 2021 after Nikhil and I had encoded, by hand, every major tax and benefit in Britain — income tax and National Insurance, Universal Credit and its taper, child benefit and its claw-back. We typed thresholds off gov.uk and argued about rounding. Uganda took one day. On July 8, the agents reused the recipe Ghana had just proven: read the gazetted acts, write the rules, race them against UGAMOD across the full test surface. The ledger recorded no findings at all — in its own phrase, the encoding was "statutorily exact everywhere tested," and the widest disagreement anywhere in the suite was four-tenths of one Ugandan shilling per year, itself examined and dispositioned as rounding: dust rather than a tolerance band. I checked the arithmetic myself that evening, looking for the catch. There wasn't one. The catch was in my assumptions about how long careful work takes.

Zero findings is not the absence of a result. It is one. Uganda's empty ledger says UGAMOD turned out statutorily exact everywhere our suite reached, and that the encoding matched it to the shilling — established by evidence, not assumed from silence. A model checked against an independent reference and found to agree is in a different epistemic state from a model no one has checked, even when the two would print identical numbers. Uganda's zero is a compliment paid to a reference model, in evidence, and of the five results that week it is the one that reassures me most: the method can tell the difference between right and merely unexamined.

The rest of the week

Zambia landed on July 9, encoded in about a day and checked against MicroZAMOD: eight findings, and the first ledger with divergences on both the tax and benefit sides rather than the tax side alone — among them a definition of the pay-as-you-earn base and a set of excise rates carried a year stale. Eight findings on a first pass across an entire system is low, and the both-sides detail is informative: the seam where taxes and benefits meet is where implementations drift apart most easily, and Zambia was the first lane with enough surface tested to see the drift from both directions.

Ethiopia ran in a single overnight pass against ETMOD and produced two findings, both in the same place: the minimum alternative tax introduced only months earlier by Proclamation 1395/2025. A statute that new has barely been implemented by anyone, and two first-pass disagreements about how it works is close to what you would predict — both of the understandable kind, reasonable readings of a fresh law that differ, rather than arithmetic gone wrong.

Rwanda closed the week with five findings, and one procedural first for us: the comparison ran live rather than replayed from a fixture — the encoded model and RWAMOD were handed the same eighty-six cases and asked to agree in real time. The run surfaced the five findings, and once they were dispositioned the two agreed across all eighty-six.

Tanzania is under way as I write. And Nigeria has begun — and Nigeria is different in the way this book cares about most. Nigeria is not a SOUTHMOD country. No reference model exists for it, and no administrative ground truth is available to check an encoding against. The rule of chapter 10 does not bend for enthusiasm: a simulation may be believed only where its chain of verification ends in something real — a reference model it matches, a statute it reproduces exactly, or a number the world will later publish. Nigeria's chain does not yet terminate in any of those. The encoding work can proceed, and it has; the claim that the result is correct stays weaker than the same claim for Ghana or Uganda, and the honest thing is to say so in the same breath that reports the work exists.

The boundary of the method is part of the method.

The recipe

Across five countries and five different statutes, the recipe barely changed, and it fits in a paragraph. Capture the governing acts from the official gazette, each provision anchored to a page in a signed source manifest, so every rule traces back to the exact text it came from. Encode the system one instrument at a time. Put every change through the same merge-blocking gates the previous chapters described — if it does not compile, does not pass its proofs, does not clear the oracle comparison, it does not merge. Run the oracle suites first case by case, then across a whole simulated population, so that agreement holds not just on the examples someone thought to write but on the shape of the distribution. Publish the findings ledger. Report what it turned up to the people who maintain the reference model. And once a country reaches parity, the gates ratchet: coverage can only rise and unexplained gaps can only fall, so the model cannot quietly rot back to wrong.

The cost of a country, run this way, is measured in days of agent time. For the entire prior history of the field it was measured in years of institutional effort. But the whole week was designed to make a point that cuts against celebrating the speed: a model that costs days and that no one can check is only a faster way to be confidently wrong. What traveled from Ghana to Uganda to Zambia was the discipline bolted to the speed: the gazette manifests, the merge gates, the oracle suites, the public ledger of everything that did not match. The verification shipped in the same box as the model.

The moat and the vacuum

Chapter 2 called it the analytical moat: government modelers work from confidential tax microdata with detail no public-use file carries, and outside analysts make do with files that are sampled, top-coded, and blurred. That was told as an American story, with an American statute as its cause. But the moat was never only American, and in most of the world the better word is vacuum: no gap between the government's model and a challenger's, because no public, inspectable model exists on either side. A citizen of most countries cannot check what a tax change would do to a household like theirs, because no one outside a ministry has built the tool, and often no one inside one has published it. The asymmetry Washington lives with — half a dozen institutions arguing over a tax bill within days of its introduction — is, across most of the map, closer to silence.

Consider what a public model changes around a single decision. A finance ministry proposes to broaden a VAT, or to replace a fuel subsidy with a cash grant. The question everyone cares about — who gains, who loses, by how much, across the income distribution — lives inside a model. Where the model is the ministry's alone, the answer arrives as an assertion, and the opposition, the press, civil society, and the affected household can only accept it or reject it whole. Where the model is public and checkable, the same people can run the reform themselves, disagree about assumptions in the open, and be caught when they are wrong. The ministry keeps its tool. What ends is its monopoly on being able to answer at all.

What the July week changed is the marginal cost of ending that silence. An independent, verified encoding of a country's tax-benefit law went from a multi-year undertaking to a job of days — with fourteen SOUTHMOD-family reference models standing ready to check the work, a decade of UNU-WIDER's institution-building converted into the very ground truth that makes fast encoding trustworthy. The encodings ship under Apache and Creative Commons licenses, with the findings ledgers public beside them. Openness runs in both directions here or it is not openness: the countries get an independent open implementation, and the reference models get evidence-backed quality assurance from the most exhaustive reader they have ever had.

When the data can't travel

Rules, it turns out, are the easy half. A statute is public by definition; encoding it requires the text and a way to check the result, and both have now become cheap. The harder half is the people. A tax-benefit model is only as good as its picture of who lives under the rules — at what incomes, in what households, drawing which benefits — and that picture comes from microdata, which is where openness runs out. The asymmetry is structural, and it runs opposite to intuition. A statute is written to be public: promulgated, gazetted, binding precisely because anyone can read it, so encoding it in the open takes nothing that was meant to be hidden. Microdata is the reverse — a record of actual people, protected by law and by promise, its value coming from exactly the granular detail that makes it impossible to release. Law is portable because it is already public. Microdata is stuck because it is already private.

So the summer's second experiment asked a harder question: take the calibrated synthetic population our data layer builds for the United States — populace, the commons Part II introduced — adapt it to a country it was never built for, and grade it honestly against that country's real data. The United Kingdom was the natural test case, because it has both an open reference model and a high-quality household survey to grade against. The adapted population ran through a pre-registered evaluation harness — scoring rules fixed and published before any answers were known — and was scored against held-out survey ground truth from Britain's Family Resources Survey.

The verdict was mixed, and we reported it mixed: "a credible first pass for aggregate, earnings-centric analysis — not a substitute for native microdata on benefits, distribution tails, or subnational detail." Two specifics carried the grade. One defect was found and fixed: a stale data feed had zeroed out state pensions, and the error cascaded downstream into pension-credit calculations — caught, traced to its source, corrected. One defect was found and deliberately left open: capital gains had been carried across one-to-one into a survey surface that captures almost none of them, and the honest fix is not to transplant realizations from one country to another but to model asset positions and the rules under which gains are realized — a rebuild, designed and not yet done. The transferred population is good at what surveys are good at and blind where surveys are blind, and pretending otherwise would have been easy and wrong. Part of why I trust the verdict is the episode chapter 10 already confessed: a summary statistic in an early draft of the write-up turned out to be fabricated — fluent, plausible, unsupported by any run — and a verification gate caught it before publication. The gates aim at our own output first.

Where both halves can be built, they close tightly. In Belgium, the same machinery was calibrated to twenty-one administrative targets and hit all twenty-one within 1.8 percent; on that same population, per-record social-security contributions agreed with EUROMOD to within three eurocents per person per year. Each is a small, checkable number stating exactly how close the model came, worth more than any promise that it is right.

An open referee

Belgium points at the harder case, which is most cases. For many countries the microdata cannot leave the building: national statistical offices hold survey and administrative records under legal obligations that do not bend for a research project. You cannot download the file, and you should not want to be able to. That looks like the end of open modeling in those places — you can encode their laws in the open but never open their data.

The answer we landed on is to stop trying to move the data and to move the referee instead. Give the institution that holds the microdata two things: the transferred population, and a pre-registered evaluation pack — the scoring code pinned to an exact version, the configuration frozen, and a scorecard schema that fixes, in advance, which numbers will be reported. The institution runs the pack inside its own walls, against its own protected records, and publishes only the scorecard. Because the scorecard was specified before the run, no one can fish the data afterward for a flattering statistic; the referee's questions are fixed before it ever sees the answers. No record leaves the building. The score does.

A scorecard is deliberately small — a handful of numbers, fixed in advance, saying how closely the transferred population reproduced the real one on the dimensions that matter: how the income distribution lines up, how many households the model places on each benefit, how far the tails diverge. It carries none of the underlying records and none of their risk, and it is exactly the kind of low-dimensional summary a verification chain is allowed to end in. Trust in a model of a country comes from a referee who has seen its microdata, a test written down before the answers were known, and a score published where anyone can read it. That is less than full openness, and it is enough.

Open microsimulation anywhere doesn't require open microdata anywhere — it requires an open referee.

Step back from the week and the experiments, and notice what quietly made every move legal. The rules, the data, and the prediction never had to travel together. Ghana's statutes were encoded in the open while Ghana's household records stayed home. A population built in Washington was shipped to Britain and graded there. A statistical office can run a referee no outsider is allowed to watch. Each move works only because law, population, and forecast are genuinely separable — built by different methods, checked against different ground truth, and best kept in different hands. So far in this book, that separation has been a technical convenience. The next chapter is about an organization that took it seriously enough to build itself around it.

Chapter 12: The decomposition

Organizations restructure for ordinary reasons — a merger, a funding round, a founder's exit, a board that wants cleaner reporting lines. It is rare for one to restructure around a claim about knowledge.

This one did.

The claim had been accumulating through the previous four chapters without being stated as one. Building the engine taught us that rules, data, and prediction are separable concerns; the SOUTHMOD week and the transfer experiment showed the separation doing real work, statutes traveling openly while microdata stayed home. For most of a decade the three had lived together inside one tool — one set of repositories, one brand, one habit of mind — and the joints stayed invisible because the bundle worked and nothing pulled on them.

Verification pulled on them. Once each part of a policy estimate must prove itself against ground truth, the parts stop resembling one another. A rule proves itself against the statute it encodes and against a reference calculator — TAXSIM, UKMOD, a SOUTHMOD model — that either reproduces its answer to the last cent or forces the difference to be explained, and it does so at the moment the rule is written, before it ever touches a household. A population estimate proves itself differently: against surveys, and against the administrative totals the government already publishes — a calibration you can check the day you build it. A prediction can prove itself in neither way, because the thing it describes has not happened yet. It waits, scored only when reality arrives and the official number lands.

Three clocks

Three kinds of claim, on three clocks. A rule's truth is settled today, against a document. A population's truth is settled today, against a count. A forecast's truth is not settled until some future date when a statistical agency posts a release. And the clocks imply the failure modes. A rule fails by contradicting the law — a bug against a fixed target, findable and fixable. A calibration fails quietly, by drifting away from a population it used to resemble. A forecast fails loudly and unarguably, by being wrong about the world — a miss you cannot litigate away, only record. You do not manage those three failures with the same reflexes, or the same people, or — this chapter's claim — inside the same institution.

Run the three clocks on a single tax credit and the difference stops being abstract. Whether the credit is encoded correctly is answerable this afternoon, against the code section and an oracle. Whether the households it reaches look like the country is answerable this afternoon, against the tax agency's published totals. How many families will actually claim it next filing season is answerable only next filing season, when the agency counts them. The first two are audits. The third is a bet.

Axiom states, Thesis predicts

The reshaping took its cue from that observation rather than from the forces that usually move an org chart. PolicyEngine divided along the seam verification had exposed, into two poles that are opposite in the most elementary way two claims can be — and the names were chosen to say so. An axiom is an accepted truth: not on trial, the thing you reason from. "What does the law say?" has that shape; the answer is fixed by a statute already written, and the only live question is whether an encoding faithfully matches it. A thesis is the mirror image — a proposition advanced in order to be tested, granted no standing until the evidence rules. "What will happen?" has that shape; no document settles it, and any answer stays provisional until the world supplies the verdict. Two words, two epistemic tempers. The reorganization is an attempt to stop housing them in one body.

The Axiom Foundation, chapter 9's subject, is the determinism pole. It encodes law exactly — federal statutes, state codes, agency manuals, the council-tax-reduction scheme of a single London borough — and asks its agents to state, never to guess. Its public launch is set for July 28, 2026. Everything it ships is meant to be right the way a proof is right: checkable line by line against a source, and wrong only if the source is.

The Thesis Institute is the prediction pole. Its product is a public docket of forecasts about government metrics — the first print of the unemployment rate, a program's caseload, a wait time — including forecasts made under explicit policy counterfactuals. One live page asks what Medicaid call-center wait times will be in March 2027 if the work-requirement deadline chapter 9 described gets delayed: a number that does not exist yet, about a world that has not happened yet, staked out in advance. Its public launch is set for around September 15, 2026. Everything it ships is meant to be provisional — a value with an interval around it, entered into the record early enough that reality can grade it.

The two poles compose, and the Medicaid page shows how. What the rule would be if the deadline slipped is an axiom: an encoded change to the law, exact, supplied by the determinism pole. What caseloads and wait times would then do is a thesis: a forecast, owned and eventually graded by the prediction pole. A policy counterfactual is precisely that stack — an exact *if* run forward into an uncertain *then* — and separating the institutions leaves the pipeline between them intact while making plain which half of every counterfactual is settled fact and which is open forecast. One honesty the docket enforces on itself: a conditional forecast is graded on the branch the world actually takes, and the branch not taken stays a projection, labeled as one.

The book's discipline thereby becomes a literal product feature. Every forecast on the docket carries a published interval and will be graded when the official number lands. As of this writing, none has resolved. The scoreboard is live; the grades arrive with reality — and there is nothing else honest to report about a forecasting institution whose first resolutions are still months away.

The engine, recomposed

What happens to the original tool? It goes to both poles, because it was always both, bundled tightly enough that nobody had to notice. For years one team shipped one release, and the rules, the data, and the projections rode out together; nothing forced the question of which was a proof and which was a guess. The oracles and the scoreboard forced it. PolicyEngine is being recomposed into the two layers that were underneath it the whole time: its rules — the thousands of encoded provisions that answer what a household owes and is owed — become part of Axiom, and the calibrated microdata beneath every population estimate becomes populace, the shared substrate both poles stand on. populace deserves one plain sentence and no more: it is the calibrated-microdata commons — households synthesized and reweighted until the totals they imply line up with the counts the government already publishes — and the entire stack rests on it, because an exactly encoded rule and an honestly scored forecast both still have to be applied to a representative population before they mean anything about a country. The model code that used to live in PolicyEngine's own repositories retires downward into those two layers. The product on top keeps its name, its interface, and its users.

That last point is where the recomposition matters to anyone outside it. The teacher checking a family's benefit cliff, the journalist scoring a bill, the congressional staffer running a distributional table — none of them sees a seam; the thing they use answers the same questions it always did. What changed is underneath. It now stands on two legs that can each be checked — a rules layer verified against statute, a data layer verified against administrative totals — rather than on one monolith whose internals you had to trust as a whole.

And the recomposition buys something the field has not had: a tally. Route the population layer into a forecast, post the forecast on the Thesis docket with an interval and a resolution date, and a policy estimate stops being a number a model asserts and becomes a claim the world will eventually grade. "This reform costs fifty billion dollars" turns into "this reform costs fifty billion dollars, here is the interval, and here is the date we will find out." The institutions that produce official estimates do grade themselves in places — CBO publishes retrospectives on its aggregate budget and economic forecasts, real self-scrutiny that Part IV will credit properly. No institution we know of keeps a standing, per-estimate ledger: a policy score is published, the vote happens, and nothing circles back to record what the world did to the number. The recomposition wires the tally in as the native output format of the prediction pole rather than a report card imposed from outside.

Wrongness, isolated

Why two institutions rather than two departments? Because the split is a firewall around one specific hazard: being wrong.

The Thesis Institute has to be wrong in public, and often. A docket that never missed would be a docket with its intervals padded uselessly wide, or one quietly declining the questions that are actually hard. The worth of a graded forecast comes precisely from the fact that some grades will be bad — visibly, on the record, with the interval printed beside the miss — because that is the only thing that makes the good grades worth anything. Public wrongness is the mission. A forecaster who cannot be caught out is not being honest, only vague.

The determinism pole cannot live that way for a moment. A calculator that a benefits screener consults, that a state agency wires into its eligibility workflow, that a tax product ships to customers, has exactly one job — to be right about what the law says — and its cardinal virtue is that it is boring: same input, same answer, matching the statute every time. Housed alongside the forecasting work, it would inherit a reputation it has no business carrying; every published miss on the scoreboard would raise the wrong question about the calculator. Technically the two share code and a population. Epistemically they make opposite kinds of claim, and the surest way to keep opposite claims from contaminating each other is to keep them under different roofs, with different promises attached to their names. One roof promises to be interesting and sometimes wrong. The other promises to be dull and always faithful.

A commons and its operators

A second design constraint predates the split. Chapter 9 argued that a layer this fundamental — the encoding of the law itself, and now the calibrated portrait of the population — is reference infrastructure, closer to a dictionary or a map projection than to a product, and is not well served by a private gatekeeper between the public and the statute. The 2026 structure gives that principle a two-sided form. On one side stand the nonprofit institutes stewarding the commons: the rules, the data, the scoreboard, all public. On the other stand commercial operators building on that commons the things production demands — managed runtimes, uptime guarantees, support contracts, applied products for the banks, agencies, and firms that need someone to answer the pager at two in the morning. That is honest work, worth paying for; a hospital system cannot run eligibility screening on a best-effort open endpoint.

The distinction turns on a single preposition. Operators build on the commons. They never own it. The foundation states the commitment plainly: the commercial service tier will always be a separate entity, so the institution stewarding law-as-code never finds itself competing with the builders who depend on it. A steward that also sells is a steward with a thumb on the scale. The operator chapter 9 mentioned is the live case — organized around graded simulation offered as a service, in formation, not yet public — and nothing about the commons depends on it. The rules and the data would stay open whether or not any company ever billed for a runtime built on them — which is the test of whether a commons is real rather than rhetorical. It must survive its operators, outlast any one of them, and owe none of them anything.

So who owns the whole? Nobody, and that is load-bearing design rather than evasion. A 501(c)(3) cannot be owned. The licenses on the rules, the data, and the scoreboard let anyone build without asking. No holding company sits above the poles, because the coherence lives in the pattern, not in any body that could be bought, captured, or pressured into changing course. Astronomy has a word for this. A constellation is official — there are eighty-eight, their boundaries chartered by the International Astronomical Union, one authority deciding where Orion ends. An asterism is the other thing: a shape everyone recognizes — the Big Dipper, the Summer Triangle — made of stars that belong to different constellations or to none, visible without any body ever declaring it.

This structure is an asterism, and typography even supplies the mark: , three stars in a triangle, one for each of the three questions — law, forecast, values.

The Greeks had a word too. When the villages of Attica joined into a single polis — the act tradition credits to Theseus — they called it *synoikismos*: separate communities becoming one society without dissolving into one another. Tradition connects Theseus's own name to *tithenai*, to place, the root that also gives us *thesis*.

The pattern underneath

Pull back from the org chart and a repeatable pattern comes into focus, with nothing intrinsically to do with taxes or benefits. Every project in this book's second half has the same three-part shape: find a corpus that is complete, public, and — the surprising part — un-computed, sitting in plain sight in a form no one has turned into anything queryable; point agents at it; ship the verified, open version of what was there all along. Axiom encodes the world's policy corpus. populace integrates the world's microdata. Thesis forecasts the government's metrics. Three instances of a single move: the analytic stack of government, rebuilt in the open, one verifiable layer at a time.

The move comes with a gate, and the gate is this book's whole discipline compressed into a test. A corpus qualifies only if its output can be checked, per unit, against ground truth — a rule against its statute and an oracle, a household against administrative totals, a forecast against the number reality eventually prints. Where that per-unit check exists, agents can work at a scale no institution could staff, precisely because each of their millions of outputs can be caught the moment it goes wrong; the check is what converts raw generation into trustworthy production. Where the check does not exist, the same agents at the same scale produce the exact opposite of knowledge — fluent, voluminous, confidently false. That is the line separating encoding the law from generating plausible law-shaped prose, and it does not move for enthusiasm or for compute.

What changed to make the pattern viable now was the unit cost of computing the corpus. For most of history the corpus of public life sat un-computed because computing it did not pay: encoding a single statute faithfully took a trained specialist days, and there are millions of provisions across thousands of jurisdictions; forecasting the budget of every county would take analysts nobody would fund. So law stayed on paper, microdata stayed in survey files, metrics went unforecast — public in principle, computed almost nowhere. An agent can encode a country's tax-benefit system in days, hold tens of thousands of provisions in view at once, draft a forecast for every metric on a docket, and never tire. The corpus was always open. The missing piece was a way to compute it that was cheap enough to attempt at scale and disciplined enough to trust, and the gate supplies the second half.

Run the test forward and the docket of candidate corpora is long, because most of public life shares the one property that matters — already written down, not yet computed. Local zoning codes are public, largely un-encoded, and checkable line by line against the ordinance text: a national map of what may legally be built where, assembled the way the tax code was. The fiscal futures of tens of thousands of county and municipal governments are forecastable and, in time, gradable against the budgets those governments actually adopt. Bills introduced in a legislature could be scored the day they drop, run through the same engine that scores enacted law, so a proposal arrives with its distributional consequences attached. Each of these is a direction rather than a promise, and each earns its place only if it clears the same gate — per-unit verifiability against something real.

The third question

Two of the three questions now have homes. What does the law say belongs to Axiom. What will happen belongs to Thesis. Between them they cover most of what a policy argument is made of — the facts of the statute and the facts of the forecast — but not the part that decides the argument once the numbers are agreed. The oldest question is still open: what do we want?

That one has no institute, and it may be the hardest of the three to anchor in ground truth, because no statute or first-print statistic holds its truth. It sits latent in a population's own preferences — harder to observe, easier to distort, and quick to punish anyone who claims to speak for it. The embryo exists: HiveSight, a survey-simulation tool that tries to elicit and forecast what populations think the way the other layers elicit rules and metrics. The later chapters take it up and benchmark it honestly, which is the only way this book is allowed to take anything up. Rules, predictions, values: the third primitive is at once the frontier of the whole program and its least finished piece — the place where a book about computing the government becomes a book about what the government is for. There is a version of this stack whose deepest payoff is not any single number but a shared, verifiable substrate everyone can reason against, so that public disagreement can finally be about ends rather than about whose figures are right.

That is the horizon. Before values, the book owes a debt it has been running up since Part I. Every estimate in every preceding chapter arrived as a single number — a cost, a poverty rate, a marginal rate — without an honest account of its own uncertainty. This reform costs fifty billion dollars: plus or minus what, and how would we know? The prediction pole exists to answer exactly that question, to turn a bare point estimate into a graded interval with a date on it. And so, before what we want, Part IV begins with how sure we can be of what we already claim to know.

Part IV: The prediction pole

Chapter 13: The uncertainty gap, and the scoreboard

In May 2017 the Congressional Budget Office published its analysis of the American Health Care Act, the Republican bill to repeal and replace much of the Affordable Care Act. The headline finding: by 2026, 23 million fewer Americans would have health insurance than under current law. Not “approximately 23 million.” Not “between 18 and 28 million.”

Just: 23 million.

The number did what headline numbers do. The bill’s supporters attacked it, its opponents carved it into granite, and for weeks the debate treated 23 million as a measurement — a fact to be defended or debunked — rather than what it was: an estimate of the gap between two futures, made nearly a decade before either would arrive. CBO had not erred. Twenty-three million was a defensible central estimate, produced by careful people using the best available methods. But a nine-year forecast of insurance coverage rides on economic conditions, behavioral responses, and a labor market nobody controls, and a single integer expresses none of that. It is 2026 now, the year the forecast pointed at, and the number can never be checked: the bill died in the Senate that July, so the world it described never ran. The estimate was precise, uncertain, and ungradable all at once, and nothing in how it was published let anyone see the second and third properties.

The 2002 film *Minority Report* built its dystopia on this exact suppression. Three “precogs” foresee murders; the police arrest people before they kill; the predictions display as facts, exact to the minute and the street corner, with no probability attached. The twist is that the system produces three forecasts that usually agree and sometimes diverge, and the dissenting vision — the minority report — is filed away to preserve the illusion of certainty. Every budget score that ships as a single number files away a minority report of its own. The interval exists, down in the sampling error and the contested elasticities. It just never makes the page.

Every result in this book so far has been a single number. A taper cut costs £2.8 billion. A carbon dividend cuts poverty 10 percent. A single mother faces an 80 percent marginal rate. That is not a defect of one model; the microsimulation paradigm, from Orcutt forward, was built to answer “what would happen if?” — it produces scenarios, not probability distributions. Nor is it ignorance: statisticians have known how to put standard errors on microsimulation output for decades. The headline scores that decide votes simply ship without them. This chapter is about what those clean figures hide, what it costs to expose it, and the institution I am helping build to put the hidden part on a public scoreboard.

Four kinds of not-knowing

Start with where the model's people come from. The Current Population Survey's March supplement asks fewer than ninety thousand households to stand in for a country of more than 130 million. For a national aggregate that is plenty; for the cell you actually care about — self-employed workers in Montana with investment income — it may be a handful of records, and the sampling error in each cell propagates when the model sums weighted impacts across the country. Calibration helps: the machinery chapters 6 and 7 described cut the Enhanced CPS's deviations from its administrative calibration targets by roughly 97 percent. But calibration pins down the questions the government already publishes answers to. It does nothing for the variance of a question nobody has posed. Call this input-data uncertainty: the survey is a sample, and every number computed from it inherits the sample's noise.

Second, the parameters. Tax brackets and benefit rates are usually known exactly — the statute prints them. Behavior is not. A meta-regression of 1,720 estimates from 61 studies put the mean elasticity of taxable income near 0.30, with individual estimates running from near zero to above 1.0 depending on method, sample, and country. The stakes are not academic: a reform raising the top marginal rate by ten points might add \$100 billion in revenue at an elasticity of 0.2, or \$60 billion at 0.5. Same reform, same model, opposite headlines. Parameter uncertainty is the honest name for a number the literature brackets and the model hard-codes.

Third, the model itself. Every microsimulation encodes assumptions about how programs interact, how households respond, how the economy adjusts — and different assumptions, written into code by different reasonable people, produce different results. Structural uncertainty is the possibility that the model's shape, not just its inputs, is wrong. It is the hardest kind, and it gets its own section below.

Fourth, the future. Any projection past the current year rests on forecasts of wage growth, inflation, and demographic change, which enter the model as fixed numbers and are, of course, forecasts. Future uncertainty is the oldest kind, and models swallow it whole.

Each layer compounds the ones before it. The tidy figure on the page is the sum of four kinds of not-knowing, reported as one kind of knowing.

Why the single number survives

False precision breeds false confidence. A number published without a range invites its audience to defend it to the decimal, which is how legislators end up litigating 23 million as if it were a census count. When a model reports that a reform costs £12 billion, the question that comes back is "is it worth £12 billion?", almost never "how sure are you?" Score two competing reforms at \$50 billion and \$48 billion and the \$2 billion gap reads as a real difference, even when each estimate carries ±$10 billion and the gap is noise wearing the costume of a finding. The blindness runs deeper than optics. A point estimate cannot say "there is a one-in-ten chance this costs twice the projection," and the downside case is often what a policymaker actually needs to weigh. And when two models disagree — Tax-Calculator says one thing, PolicyEngine another — a reader without intervals cannot tell whether one is wrong or both sit comfortably inside the same uncertainty.

So why does the single number persist? Because everyone downstream needs it. Reconciliation rules require one cost figure, not a distribution. A journalist needs a headline. An advocate needs a talking point — "this reform lifts two million children out of poverty" mobilizes in a way "between 1.4 and 2.6 million" never will. Producers of estimates know they are uncertain; consumers treat them as certain; producers, learning what gets quoted, stop reporting the spread. The equilibrium is stable and everyone in it is behaving rationally, which is exactly why no one inside it will fix it.

The producers are not oblivious, and the record here deserves stating fairly. The Penn Wharton Budget Model, one of the few groups to compare its projections to outcomes systematically, found its estimates generally accurate but with meaningful variance. CBO grades its own aggregate work in public: its June 2024 projection underestimated fiscal-year 2025 revenues by about 6 percent — $334 billion — in significant part because it had not anticipated a round of tariff increases. Across 1984 to 2023, its retrospectives show revenue is harder to forecast than spending and that errors compound with the horizon; its economic forecasts tend to be more accurate than the Administration's and the Blue Chip consensus, and run broadly similar to the Survey of Professional Forecasters. That is real self-scrutiny, rarer than it should be.

But notice what it is: an evaluation of the aggregate forecast, years later, in a study. No standing public loop grades each policy score, one by one, against what happened. Part of the reason nobody built one is deep rather than lazy: most policy scores are comparisons between the world that ran and a world that never will. The AHCA's 23 million was a difference between two branches, and the world ran one. You cannot look up a counterfactual. Hold that thought; the institution at the end of this chapter is designed around it rather than in denial of it.

Putting bands on the number

The methods for exposing uncertainty exist, and each buys a different slice of it. Monte Carlo simulation runs the model many times with inputs drawn from probability distributions and reports the spread. I use it in EggNest, a retirement-planning tool I built: instead of "you will have $1.2 million at 65," it runs ten thousand scenarios over distributions of market returns, inflation, and wage growth and reports "a 90 percent chance of between $800,000 and $1.8 million." The first formulation is what people want to hear; the second is what the arithmetic supports. For policy microsimulation the obstacle is cost. The engine already applies every bracket and phase-out to an entire simulated country; running that ten thousand times multiplies the work by four orders of magnitude, and seconds become hours.

Bootstrap resampling is cheaper and narrower: draw many samples from the microdata, reweight each to population totals, run the policy on each, and read the distribution of results. The computation is embarrassingly parallel: 500 replicates add minutes on a cluster, not hours. What it captures is input-data uncertainty, the fact that the survey is a sample rather than a census, and nothing else; a bootstrap says nothing about a wrong elasticity or a wrong model. But for the everyday question — how much should I trust this number? — input-data uncertainty is usually the leading term, which makes the bootstrap the right place for an honest program to start.

Beyond those two sit the deeper renovations. Bayesian methods treat a contested elasticity as a distribution — a prior centered near 0.30 that updates as evidence accumulates — instead of a constant, at the price of rebuilding models that hard-code the number. Squiggle, a probabilistic-programming language, propagates explicit distributions through a calculation; it suits a Fermi estimate and fights you across ten thousand lines of microsimulation. Scenario analysis reports a baseline, an optimistic case, and a pessimistic one, conveying sensitivity without pretending to know the probability of each branch.

To make the leading term concrete, I ran a small paired-subsample experiment in PolicyEngine US on March 31, 2026. I drew ten subsamples of 1,000 households each from the then-current Enhanced CPS, ran the same stylized reform to the Earned Income Tax Credit on each baseline-and-reform pair, and recorded the change in aggregate EITC and in household net income. On the full sample, the reform cut aggregate EITC by $12.6 billion. Across the ten subsamples, the same reform cut it by anywhere from $11.1 billion to $15.0 billion, with a mean of $12.9 billion; net income moved on the same pattern, a $13.9 billion full-sample drop against a range of $12.1 to $16.2 billion. Ordinary survey sampling alone — before any behavioral elasticity, any forecast, any modeling assumption — moved the national estimate from about 12 percent below the point estimate to about 19 percent above it. A swing that size carries a reform across budget thresholds the single figure made look safe.

The caveats, flatly: this is not a production-grade confidence interval. The subsamples are deliberately small and few, the exercise was a one-off run by hand, and it captures input-data uncertainty only — none of the other three kinds. What it demonstrates is narrower and still worth having: the number the model reports is the center of a distribution, not the result.

Making that demonstration routine instead of artisanal is a data-infrastructure problem, and it is one of the jobs populace — the calibrated-microdata commons Part II introduced — exists to do. The machinery that produced the imputations and the weights, from the quantile regression forests of chapter 7 to the synthesis across integrated sources and the calibration against administrative totals, ships inside certified release bundles rather than vanishing behind a finished file. That means imputation and synthesis error can be quantified and versioned instead of buried: a later analyst can re-derive the weights and measure how much they could have differed. The March 31 experiment took an afternoon of hand work. The point of the infrastructure is that it should take a flag.

The weather standard

One field solved this problem so thoroughly that we forgot it was ever a problem. Modern weather models run an ensemble — dozens of simulations from slightly perturbed starting conditions — and report the spread as a probability. "A 60 percent chance of rain" is a calibrated probability: across many such forecasts, it rains on close to 60 percent of the days so labeled.

Forecasters call that property *calibration*, and it is worth pausing on the word, because this book has been using it for something else. Data calibration — chapter 2's kind — adjusts survey weights until a dataset reproduces administrative totals that already exist. Forecast calibration is a property of stated probabilities: events assigned 60 percent happen about 60 percent of the time. The first is checked against the present. The second can only be earned against the future, by tracking whether the 60-percent days deliver rain at the promised rate and recalibrating when they drift. The loop is a procedure, not a metaphor: forecast, resolve, score, recalibrate. Weather forecasting has run it for decades, on millions of forecasts, and its skill has improved steadily and measurably because every forecast is eventually checked against a sky that either rained or did not.

Other fields run partial versions. Clinical trials report effects with confidence intervals, and "15 percent lower mortality (95% CI: 8–22)" is a different finding from "15 percent (95% CI: −2–32)," even though both center on 15. Financial risk management lives on Value at Risk and implied volatility; the 2008 crisis exposed how badly those models underweighted tail risk, and the response was better uncertainty modeling, not abandonment. Climate science runs large model ensembles and writes in calibrated language — the IPCC's "likely" means at least 66 percent, "very likely" at least 90. Election forecasting has absorbed the grammar too: a model that gave one 2016 candidate a 29 percent chance was not refuted when he won, because 29 percent is not zero. The structural parallel to policy modeling is exact — complex nonlinear systems, many interacting variables, forecasts people stake decisions on. The difference is that meteorology spent decades building the verification loop, and policy analysis has retrospectives but has never run the loop: no committed forecast, no fixed resolution date, no score, no recalibration.

When the model itself is wrong

Bootstraps and ensembles put bands around a model's answer. They cannot say whether the model asks the right question. You can propagate uncertainty through a wrong model and get a precise answer that is precisely wrong. Structural error is the uncertainty we do not know we have, and nothing inside the model catches it. Only reality does — which is a hard sentence for a modeler to type, and the evidence for it comes from the two natural experiments this field has actually been handed.

In 2017 and 2018 the Finnish government ran a randomized controlled trial of basic income: 2,000 unemployed people received €560 a month, unconditionally, measured against a control group of about 175,000. Microsimulation had a clear prediction. The design cut participation tax rates by 23 percentage points, and in the models, cutting participation tax rates raises employment. The first-year result: no statistically significant effect on days worked. The eventual published analysis found modest gains in the second year, concentrated in particular subgroups. The models had the incentive exactly right — participation tax rates really did fall 23 points — and the response wrong, because a permanent unconditional floor is not a marginal tax tweak, and the historical data the response parameters were fitted on contained nothing like one.

The Affordable Care Act's individual mandate tells the same story from the other direction. When Congress zeroed the mandate penalty in 2017, CBO projected 13 million fewer insured Americans by 2027, from models in which the mandate drove enrollment. The effect came in far smaller. People had enrolled for the subsidies, for the security, for the momentum of already being enrolled — and the historical record could not separate "enrolled because of the mandate" from "enrolled while the mandate happened to exist," so the models credited the mandate with behavior that belonged to everything around it. Finland's models overestimated a response to incentives; the mandate models overestimated a response to compulsion. Both were reasonable readings of history. Both were wrong in a way that no bootstrap, no prior, and no ensemble could have flagged in advance, because the flaw was in the model's shape, and only a committed forecast that missed by more than its interval allowed could have caught it.

The scoreboard

The last chapter split the stack into a pole that states and a pole that predicts. On the stating side the check is immediate: an encoded rule is right or wrong against the statute and the reference calculators the day it is written. The predicting side has no such settlement. The four uncertainties live in the gap between a deterministic calculation and a claim about the future, and no statutory exactness closes that gap. The only thing that closes it is the weather forecaster's loop — make the claim in advance, in public, and let the world grade it.

The Thesis Institute is that loop built as an institution: the last chapter's public docket of forecasts of government metrics, each locked with an interval before the number is known and graded when the official number lands. Remember the problem with grading policy scores: the counterfactual never prints. The docket is built to respect that rather than promise around it. Its unconditional entries — next month's unemployment print, next quarter's caseload under the law as it stands — resolve directly. Its conditional entries, the Medicaid wait-time page among them, resolve only on the branch the world takes; on the branch not taken, the entry expires unscored, and the ledger says so. The causal gap between branches — the thing the AHCA's 23 million actually claimed — never resolves at all, and the ledger labels which kind of object each entry is instead of letting a scenario impersonate a testable claim.

That is weaker than grading every policy score. It is also the honest version, and it still transforms the field it touches: each estimate the earlier chapters produced that names something reality will print — a caseload, a revenue line, the poverty rate the Census Bureau will publish — can now become a forecast with a published interval and a resolution date, held to the weather standard: verified probabilities, recalibrated on misses. The discipline this book keeps applying to other people's tools lands hardest here, on the institution built to enforce it.

There is no track record — only a mechanism for acquiring one, in public.

Reading a number until then

Grades take years to accumulate, and policy will not wait. So the practical posture, for however long the scoreboard stays young: treat every point estimate as the center of a distribution. Read "£12 billion" as "our best guess is near £12 billion, and the true value could reasonably sit twenty percent to either side, or further." Read two models' disagreement as information about the uncertainty rather than a fight to referee. Ask which assumptions move the answer most, because the answer usually has one or two hinges. Reserve the deepest skepticism for the most novel policies — Finland's lesson — where structural uncertainty dominates and no historical data has the answer's shape. And do not mistake imprecision for uselessness: "roughly \$50 billion" rules out \$5 billion and \$500 billion, which is often all a decision needed to know.

Here is what that looks like in practice, on a reform this book has touched before. PolicyEngine scores making the Child Tax Credit fully refundable — removing the refundable cap and the earnings phase-in that hold the poorest families below the full credit — against 2026 law. The static result: a cost of roughly \$23 billion for the year, with child poverty falling about 2.6 percentage points, from 16.6 to 14.0 percent. That is my own calculation, run at a specific model version on a specific data vintage, to be rerun before this book prints — not an official score. The uncertainty-aware version attaches a band to the cost, driven mainly by one behavioral unknown: how many currently non-filing families would actually claim. And it attaches a second band to the poverty effect, driven by the systematic undercount of income among low-income households in survey data. The central estimates do not move. What changes is what you can honestly say: the poverty reduction survives even the pessimistic branch, while the cost has real range. A legislator holding the banded version knows more than one holding a clean $23 billion — including knowing which part of the claim is sturdy and which part is a bet.

The scoreboard can eventually grade all of this, because all of it names things reality prints: an unemployment rate, a caseload, a call-center wait time, a child-poverty figure with a release date. But the docket has an edge, and standing at it you can see the next problem. What people believe about a policy, what they prefer, what they would trade away to get it — none of that has a first print. No administrative office certifies it; no future date resolves it. The next chapter steps past the edge of what ground truth can grade, into the simulation of opinion itself, carrying the same discipline into territory where the ground is soft.

Chapter 14: Simulating opinion

In 2023, researchers at Brigham Young University published a paper whose title runs the national motto backward: "Out of One, Many". They asked GPT-3 to adopt specific demographic personas — a 65-year-old Black woman from Mississippi, a 28-year-old white man from Oregon — to answer survey questions in character, then compared the answers against real survey data. Conditioned carefully on demographics, the model tracked human voting patterns across the 2012, 2016, and 2020 presidential elections. It captured the partisan divide by education. It reproduced regional variation. One model, conditioned a thousand ways, looked from the outside like a thousand respondents.

For a book that has spent thirteen chapters simulating what policies do to households, the question was unavoidable: could the same machinery simulate what households think about those policies? It is also the first question whose ground truth is only partial. A tax computation is right or wrong against the statute. An unemployment forecast meets its first print on a scheduled morning and takes its grade. An opinion has neither; no office certifies what a country believes, and no calendar date resolves it. The closest thing to ground truth is a survey, itself an estimate: sampling error, wording effects, a measurement of opinion rather than a certification of it. The scoreboard the last chapter built cannot operate on this ground, so its discipline has to be carried by hand — commitments registered before results, models tested on questions they cannot have memorized, misses printed beside hits. This chapter is what happened when I carried that discipline into the softest territory in the book, including into a tool of my own, half of which did not survive the grading.

An argument from the start

The field Argyle's paper opened has been an argument from its first year. The optimists had real evidence. Mei and colleagues put GPT-4 through six canonical behavioral-economics games: dictator, ultimatum, trust, and their kin. It reproduced human behavior across all six — impressive, and on reflection expected, because those games are among the most thoroughly described experiments in the literature the model trained on. Argyle's voting result has the same character. Decades of polling documented how demographics track the vote so completely that the model absorbed the correlation and replayed it on demand. For voting, partisan identity, and familiar consumer preferences — wherever the mapping from demographics to opinion is strong and well published — a language model generates plausibly human distributions.

The skeptics had evidence just as real. Santurkar and colleagues asked whose opinions these models reflect, and found that a base model's default view represents no real population: it skews toward the liberal, educated, Western profile of its training text, a tilt that fine-tuning with human feedback shifted but did not erase. Prompting could move the default, sometimes only as far as caricature. Bisbee and colleagues compared ChatGPT's simulated survey responses with real ones and found the silicon version too tidy — less variation than humans show, and regression coefficients that diverged, sometimes sharply, from estimates on human data, which are the numbers social scientists actually use. And Park and colleagues built the demo that captivated everyone: generative agents, AI characters inhabiting a simulated town, forming plans and relationships nobody scripted. The town proved nothing about measurement, which made it the cleanest caution in the literature. Fluency is not fidelity.

Both camps were telling the truth about the region they sampled. The practice soon had a name — silicon sampling, using a language model to stand in for human survey respondents — and an operating range: strongest for well-studied populations answering well-studied questions, weakest for underrepresented groups facing novel ones. The optimists tested familiar behaviors on well-covered groups; the skeptics probed the tails. Marketing researchers, who face the question with money on the line, converged on a modest assignment for the tool: pretest the instrument, pilot the study, and field the real thing.

Who gets to ask

The reason anyone bothers is the cost of asking. Companies and governments spend on the order of $140 billion a year on the insights industry: surveys, panels, focus groups. A well-designed survey of a thousand respondents costs tens of thousands of dollars and takes weeks to field. Reword one question and you pay again. Cut the results fifty ways and the sample thins exactly where the question gets interesting. Large institutions absorb all this; startups, nonprofits, local governments, and lone researchers often cannot, so their questions go unasked. When it costs $50,000 to learn what people think, only the well-funded ask

— information concentrates, and power with it. Cheap, approximate answers would change who gets to pose the question: a first-time candidate testing a message, a neighborhood association gauging support for a development, a graduate student chasing a hunch without a grant. It is the move that turned budget scoring from an institutional privilege into something a citizen can run, pointed this time at the survey.

The market was never going to wait for the methodology to settle. In February 2026 the Stanford researchers behind the generative agents emerged from stealth as Simile, calling itself "the simulation company," with a $100 million Series A led by Index Ventures and an argument that simulation, not prediction alone, is AI's next frontier. The field's most captivating demo had become a company. Synthetic respondents were going to be sold whatever the literature concluded, and for public-interest use the bar sits where the pitch rarely goes: are the numbers calibrated, is the uncertainty stated, and has anyone measured the operating range against ground truth the model could not have memorized?

A thousand personas

HiveSight is the tool I built to explore synthetic opinion, and its short history compresses the book's method into one product decision: benchmark against ground truth before believing the demo. The naive version fails on contact. Ask a frontier model whether it supports a $15 minimum wage and it returns a careful, hedged essay — the average of everything it has read, which is no individual's answer and no population's distribution. Turning up the sampling temperature adds noise, but human diversity is not noise. People differ systematically, by income and age and geography and a dozen correlated axes, so the repair is to condition the model on those axes. The obvious way to do that was the one the literature was already debating.

The first prototype built personas. From microdata it assembled a demographically detailed individual and asked the model to answer as her: you are a 45-year-old registered nurse in rural Ohio earning $62,000 a year, a weekly churchgoer, a 2020 Republican voter — do you support raising the federal minimum wage to $15 an hour? The nurse is invented, here and in the prototype; invention is what a persona is. Ask a thousand such personas, drawn from real demographic distributions, and the output has the shape of a survey: toplines, crosstabs, gradients by income and age. Watching it run feels like reading a poll of a country that answers in minutes. Benchmarked against known 2024 survey values, persona roleplay landed at roughly 25 points of mean absolute error, on toplines and subgroups alike — enough to put the majority on the wrong side of a 60–40 question.

It demoed like an oracle and scored like a guess.

Cells, not personas

The version of HiveSight that shipped in July 2026 keeps the microdata and retires the actors. Post-stratification is a workhorse of survey statistics: chop the population into demographic cells, estimate opinion within each cell, and reweight the cells to the population's known composition. It is the technique that lets a skewed online panel still call an election, and HiveSight keeps the whole apparatus, swapping a language model into the estimation slot. Roughly 150 cells per geography, drawn from populace, the same calibrated microdata the rest of the stack stands on; for each cell the model is asked not to be a person but to estimate a distribution — across people like this, how do the answers spread? — and the cell-level estimates aggregate under calibrated weights.

The mechanism explains the scores. A persona asks the model to collapse a group into one imagined individual, and the model obliges with the most probable, most agreeable, most legible version: a stereotype, or a hedge toward the center. A survey never measured an individual; it measures a spread, and the spread is what post-stratification requests, cell by narrow cell. In the prototype, the microdata had been doing the useful work all along — who is in the room, in what proportions — and the roleplay was the part adding error. The intuitive mechanism failed the benchmark; the boring one won.

The registered result

The benchmark that adjudicated all this froze a 63-item anchor bank — 52 attitude questions from the 2024 General Social Survey and 11 from the 2024 Survey of Household Economics and Decisionmaking — under a pre-analysis plan committed before a single answer was elicited: suites, estimator arms, and scoring rules fixed in advance. One control mattered more than the rest: the elicitation model's training cutoff predates every 2024 target. A model that has read 2024 is not forecasting 2024; it is remembering it.

The registered results, ties intact. The cell estimator scored about 9.2 points of mean absolute error on toplines and 9.8 on subgroups. A simpler contender — asking the model directly for the whole population's distribution, no cells at all — scored about 8.6 and 9.8. On toplines that is a tie, statistically equivalent within the registered ±1.5-point margin, and the paper prints the equivalence test instead of declaring a winner. Where the cells earn their keep is structure. They track the shape of opinion across subgroups better, rank correlation 0.62 against 0.48; they are roughly nine times steadier from run to run; they hold when the questions are reordered. A single national number comes as cheap from the direct question as from the cells. The shape of opinion — who supports, who opposes, and the same answer when you ask twice — is what the cells buy.

The state-level check came back harder. Four items scored across forty-nine states against a different anchor, the Behavioral Risk Factor Surveillance System, a health survey fielded state by state: the cells beat a naive baseline, but a national-constant guess — assume every state looks like the country as a whole — stayed competitive. On those four questions, the model's geographic signal is thin. Cheap national estimates are one thing; trustworthy fifty-state breakdowns are another, and HiveSight has earned the first, not the second.

The tie and the competitive baseline appear in the paper because the registration put them there. A pre-analysis plan is a promise, made before the numbers exist, not to relabel a loss as a win after they arrive; a tool built to be sold would have led with the subgroup victory and left the topline tie unmentioned. One caveat is mine to add: these are my numbers, from a benchmark I designed, scoring a tool I built. The registration constrains what I could shade, and the paper ships with the tool, but no independent team has rerun it. Shipping is not validation. What HiveSight has earned is a defined operating range with a measured error printed on the tin — and that is the product. A demo persuades by looking right; an instrument persuades by telling you how often it is wrong, and on which questions.

What is a tie good for? Suppose a city council is weighing congestion pricing, a $15 charge to drive downtown at peak hours, and a real survey would cost $30,000 to $50,000 and take three weeks. HiveSight can take the city's demographic composition, elicit a distribution for each cell, and return an estimate in minutes: say, 62 percent opposition, heaviest among middle-income commuters who cannot easily switch to transit, lighter among low-income residents who do not drive and high-income residents who shrug at the charge. The percentages are invented; the decision they inform is real. If simulated opposition runs 90 percent in every cell, shelve the idea and save the survey budget. If the split is close and structured, field the real survey, and go in knowing which subgroups to oversample. The estimate is a triage instrument.

Five ways to be systematically wrong

The errors that remain inside the operating range are not random, which is the good news hiding in the bad. Models train mostly on text from Western, educated, industrialized, rich, democratic populations (WEIRD, in the acronym psychologists use), so they simulate Americans better than Nigerians and college graduates better than high-school dropouts. The training data's gaps become systematic error for exactly the groups surveys already undercount.

Models also drift toward central tendencies and sand off the extremes. Real opinion has long tails, full of outliers and contradictions and people who defy their demographics, and silicon samples come out too clean — Bisbee's finding, now with a mechanism attached. A third limit is time. A model renders the world up to its training cutoff and not a day past it, so it cannot answer to the event it never saw — a pandemic, a crash, last week's viral video. The hard cutoff that made the contamination control possible makes the tool blind at the frontier, where opinion actually moves.

Safety tuning adds a fourth distortion. Trained to be agreeable, models hesitate to voice the extreme and unpopular views that some real, demographically matching humans hold: a sanitized public. And the fifth arrives with the prompt itself. Handed a demographic label, a model sometimes returns the caricature instead of the range; prompt it as a rural evangelical and you may get the cartoon.

Systematic is the operative word, because survey research already handles systematic error in its own instruments: correct it with weights learned against known answers. Measure where the model runs hot or cold on historical questions where both the demographics and the real responses are known, then adjust, and report the adjusted number with its measured error. The right output of a calibrated system is "58 percent, give or take eight, given this question type, this population, and our measured error on questions like it," never a bare "58 percent support this." This is the last chapter's forecast calibration pointed at an opinion model — estimate, resolve, score, recalibrate — with the next survey wave standing where the first print stood. The catch is generalization: a correction learned on last year's questions has to hold on next year's, or it is overfitting with extra steps, and the only honest check is grading on held-out items the correction never tuned against. The field is not there yet. The discipline is knowing that and saying so on the label.

The costume of a poll

The same cheapness that democratizes the question arms the people who abuse it. Simulate how a demographic reacts to a message and you can craft the message that moves it; a campaign can A/B test divisive framings by the thousand without exposing one real person, then deploy the winner on all of them. Silicon results wear the costume of real ones — tables, percentages, crosstabs — so a bad actor can pass a simulation off as a poll and manufacture the appearance of public support that no methodology earned. There is a slower failure too: if institutions act on synthetic opinion, and those actions reshape the information environment the next model trains on, the estimates begin to reflect opinions the tool helped create.

The quietest harm falls on dissenters. Because models default to the majority inside every group, the conservative professor, the working-class environmentalist, the minority view within a minority get rounded away — undercounted once by the training data and again by the central tendency. None of this is an argument against building; traditional polling gets manipulated too, and push polls did not wait for language models. But a real poll, however slanted, at least sampled somebody, and a simulation can conjure a consensus from no one and dress it in the same chart. The response available is the one this book keeps arriving at: build in the open, with the model, the cells, the calibration, and the error bars disclosed, and label every output an estimate, never a survey. Approximate knowledge, widely distributed and honestly labeled, can serve a public better than precise knowledge held by whoever can afford a poll. But "clearly labeled" has to be enforced with the same seriousness as the estimate is computed, because the label is all that separates a triage instrument from a counterfeit public.

The third primitive

This chapter's subject is harder than anything else in the book, and the reason is structural. A tax flows through rules; the rules encode exactly and check against an oracle to the dollar. An unemployment forecast names a number that will print on a schedule and grade it. An opinion emerges from a whole life, settles nowhere, and when a model writes "I strongly oppose this," no one is home who opposes anything. Even the anchors the benchmark graded against are surveys — noisy, wording-specific estimates of opinion, not certifications of it. The ground truth here is real but partial, and the discipline has to carry the difference: registration before results, cutoffs before targets, ties printed where they land.

Held to that standard, the tool supports two readings, both true at once. It predicts: given how people like these have answered questions like this, here is the likely distribution — the sample-to-population extrapolation survey research has always performed, with a language model standing in for the sampling weights. And it fabricates: a plausible artifact in the shape of opinion, with no opinion behind it. For deciding whether a survey is worth fielding, the first reading is enough. For anything that touches democratic legitimacy — or any group whose representation is thin and whose cost of being mis-measured is high — the second is the warning label.

Sized honestly, the work in this chapter is the embryo of a third primitive, beside the rules that state and the forecasts that resolve: values elicitation, machinery for hearing what people want. There is a version that plugs into everything this book has built — simulate a policy against the population, then simulate the public's reaction to it, the same household profiles carrying both the budget effect and the opinion, each wearing its error bars. But a distribution of what people think is only an input. What a society does with its opinions — how it counts them, weighs them, protects the dissenters the models round away, and turns the whole contested spread into one choice everyone must live under — has no benchmark at all. That question is the horizon, and the last part of this book walks out to it.

Part V: The horizon

Chapter 15: Simulating democracy

Two of the most persuasive books about voters argue that the voter this chapter needs barely exists. In *Democracy for Realists*, Christopher Achen and Larry Bartels take the textbook story — informed citizens evaluate policies, choose the candidate whose platform fits their preferences, and elections translate the public's will into government — and spend a book dismantling it as the "folk theory" of democracy. In their telling, people vote their social identities and group loyalties, roughly the way they inherit a religion. They punish incumbents for droughts. They reward economic growth that began before the incumbent took office. The rational, policy-weighing citizen of the civics textbook bears little resemblance to how democracy runs.

Bryan Caplan pushed past ignorance to something worse. *The Myth of the Rational Voter* set the public's economic views beside professional economists' and found the gaps were not random. Voters lean anti-market, anti-foreign, toward make-work, and toward pessimism — four systematic distortions. Random errors wash out in a large electorate; directional ones compound, and they push democratic outcomes away from what most economists would call welfare-improving policy.

I take the skeptics seriously enough to start from their side of the ledger. If identity and loyalty carry most of the vote, anything an information tool can touch is at best a minority share of the decision. So the claim in this chapter is deliberately modest. It is not that information fixes democracy; it cannot. It is that to whatever degree some slice of the vote responds to perceived policy impacts, the quality of those perceptions matters, and the quality is improvable. Even if only a fifth of the vote is policy-driven and the rest is identity all the way down, cleaning up that fifth still moves outcomes. The whole argument lives inside the fifth.

The perception problem

The last chapter estimated what people think: a distribution of opinion, estimated and benchmarked. A distribution is only an input. What a democracy does with it is aggregation — millions of noisy, partial pictures of the world compressed into one binding decision — and this chapter is about what the noise does on the way in.

Consider a voter I am inventing on purpose. Sarah teaches school, has two kids, and faces a choice between two candidates: one would expand the Child Tax Credit, the other would repeal her state's income tax. Which leaves her family better off? Without doing the arithmetic she has to guess. The state income tax stings visibly on every pay stub. She remembers the expanded pandemic-era credit helping. The ads on both sides are built to persuade her. Her eventual vote blends what she values, what she believes each policy would do, what her party and her neighbors support, and noise — a candidate's charisma, the morning's headlines. Even if her values are perfectly clear, her vote may fail to track them, because she cannot see the true impacts. She votes on a noisy signal.

Multiply Sarah by a national electorate and the aggregate inherits her noise: the outcome only loosely tracks what voters would have chosen with clear sight. Achen and Bartels would call even this too generous, since most votes never involve policy evaluation at all. Caplan would add that the voters who do evaluate lean predictably wrong. Grant both. The narrower question stands: where a signal exists, does its quality matter?

A toy, on purpose

To reason about that, I built a small simulation called Democrasim. The emphasis belongs on *small*. It is not a validated production system like PolicyEngine, and not a benchmarked estimator like the opinion work of the last chapter. It makes simplifying assumptions political scientists would rightly attack, and nothing it produces is a finding about the world. It is a thought experiment in code, built to make one relationship concrete enough to reason about, and I present it as exactly that.

Each simulated voter carries three things. First, a set of weighted preferences across policy dimensions — how much they care about the economy versus the environment versus social issues: their true values. Second, an accuracy level, which sets how close their perception of a policy's effects sits to the truth. Third, a bias — a systematic lean beyond random noise, such as consistently overstating a tax burden or understating an environmental cost. The whole model rests on one line: perceived impact equals true impact plus noise plus bias, with the noise shrinking as accuracy rises. Voters compare candidates on perceived impacts weighted by their preferences, and the majority wins. Then you vary the accuracy and watch what happens to the welfare quality of the winner.

To see the mechanism, give the toy an election worth arguing about, with round numbers invented for the purpose. Candidate A expands the child credit and pays for it with a small rise in the top marginal rate. Candidate B repeals the state income tax and pays for it by cutting food assistance. For a middle-income family with children and no food assistance, say the credit expansion is worth $1,500 a year and the tax repeal $700, while the offsetting rate increase never reaches them: they do better under A, even though "repeal the income tax" sounds like the bigger deal. An accurate perceiver picks A. A noisy one may pick B, against her own family's interest, on a vague sense of which candidate is the tax-cutter. Scale the comparison across an income distribution and it develops a shape — lower-income families do far better under A, upper-income families modestly better under B — and whether the election reflects that shape depends on how much of the vote is signal rather than framing.

What the runs show, in the only sense a toy can show anything: with accuracy high, outcomes track welfare, and the candidate whose policies would actually improve lives tends to win. With accuracy low, the link frays; good and bad platforms win indistinguishably, because the signal is too corrupted to sort them. The relationship behaves like a threshold — below a certain accuracy, elections become essentially random with respect to welfare; above it, the coupling strengthens fast. Bias bends the outcome differently than noise does. A population that systematically overvalues tax cuts relative to equivalent benefit increases elects tax-cutters regardless of the welfare arithmetic: Caplan's argument reproduced in miniature, with the reminder that it operates through perception, upstream of any voting rule.

I want to be careful not to overclaim. Democrasim's "true" impacts are numbers the model invented, so it can demonstrate a dynamic but never measure the world. Its voters maximize welfare through policy evaluation — the very folk theory Achen and Bartels took apart — and it ignores identity, loyalty, and nearly everything else that actually decides elections. What survives the caveats is one relationship: to the degree that any of the vote responds to perceived impacts, the quality of the perceptions governs whether elections can reward good policy at all.

Read backwards, the toy explains the book. If outcomes track welfare only where perceptions of policy impacts are accurate, then machinery that makes accurate perception cheap — rules encoded exactly, populations calibrated honestly, forecasts graded in public — stops being a convenience for analysts and becomes an input to whether self-government works.

The stack exists to raise the accuracy term in this model.

What the evidence shows

The real evidence on information and voting is mixed, and worth taking straight. Information moves votes when it reaches people. In Brazil, Claudio Ferraz and Frederico Finan studied federal corruption audits whose results were released before municipal elections: where auditors found corruption, voters punished the incumbents — but the effect concentrated in municipalities with local radio stations to carry the results. Information that existed but did not travel changed nothing. American nonpartisan voter guides sit at the weaker end of the same spectrum: they tend to raise political knowledge without much moving vote choice, because in a polarized electorate most voters have committed long before the guide arrives. Arthur Lupia and Mathew McCubbins showed that voters run on trusted-source shortcuts rather than full information, which cuts both ways — a good shortcut approximates an informed choice; a bad one entrenches identity over evaluation. And accurate information never competes on equal terms, because a well-crafted misleading claim travels farther and sticks longer than a dry correction. Democrasim treats noise as random. In the world, some of the noise is engineered by people with an interest in the distortion, which means the toy understates the problem.

Kansas ran the expensive version of the experiment. In 2012 the state enacted large income-tax cuts that the governor billed as a "real live experiment" in supply-side growth. Independent analysts projected large revenue shortfalls instead. The shortfalls came — years of budget crises and school-funding cuts — until June 2017, when a bipartisan legislative supermajority repealed the centerpiece cuts over the governor's veto. The experiment ran five years, and the tuition was paid by the citizens of Kansas in billions of dollars of lost revenue and degraded services. Better information infrastructure would not have blocked the enactment; the projections existed and lost the argument. What it might have done is shorten the loop.

The Affordable Care Act showed how long the loop can stay open. Chapter 3 told the forecasting side: the CBO got total coverage gains roughly right while badly missing the composition, overestimating the exchanges and underestimating Medicaid. Voters who opposed the law over predicted exchange disruption and voters who supported it for its Medicaid reach both experienced something different from what they had been told, and the gap between perception and reality persisted for years. In February 2017, seven years after passage, a Morning Consult survey found that 35 percent of American adults did not know the Affordable Care Act and "Obamacare" were the same law. No calculator repairs that. It measures how far real voter knowledge sits from the idealized version, and it is a reminder that much of the perception problem is transmission, not computation.

So the honest summary: better voter information is necessary and nowhere near sufficient. It is one lever among identity, institutions, and strategy — and the only one of the four that infrastructure can move directly, rather than through the generational work of changing culture.

Limits no information repairs

Some constraints sit deeper than perception. Kenneth Arrow proved in 1951 that no ranked voting system can satisfy a short list of reasonable fairness conditions at once; Gibbard and Satterthwaite later showed that any non-dictatorial voting system can be gamed by strategic voting. Better inputs repeal neither theorem. But the theorems constrain the aggregation mechanism, not the quality of what feeds it. An imperfect aggregator still delivers better outcomes from better-informed inputs; Arrow ruled out a perfect system, nothing more. If anything, the theorems sharpen the chapter's point: since the mechanism cannot be perfected, the inputs are where the improvement lives.

Three objections deserve direct answers. Maybe the preferences are the problem, not the perceptions. True, and orthogonal: voters who want harmful things and perceive accurately will get harmful things, which is a different failure from wanting good things and misperceiving. Better perception at least delivers people what they actually want, the precondition for holding them responsible for wanting it.

Maybe information won't reach the disengaged. Partly true: a non-voter will not open a policy calculator, and the citizens who would are already the relatively informed. But the interface is moving to where people already are. An AI assistant that volunteers a policy's household impact reaches past the self-selected, and upgrading the moderately informed still improves the signal.

Maybe calculated self-interest isn't citizenship. Fair, and the same engine computes poverty rates, inequality, and total budgetary cost as readily as one household's bottom line. The point is replacing perception with calculation, whatever a voter chooses to weigh.

The most radical proposal in this territory takes the split between wanting and knowing all the way to the constitution. Robin Hanson's *futarchy* would have democracies vote on values but bet on beliefs: elected representatives define what the country cares about — a national welfare metric, say a child-poverty rate — and prediction markets decide which policies achieve it. A legislature proposes a bill; markets price the welfare metric conditional on passage and on failure; if the market prices welfare higher with the bill than without it, the bill becomes law. No polity has adopted futarchy, and the reasons are visible from here. Deep pockets can lean on thin markets. Most policy questions never attract enough trading to price anything reliably. "Welfare" is exactly the contested term the mechanism pretends is settled. And democratic legitimacy depends partly on citizens feeling heard, which a price does not supply. But as a thought experiment, futarchy isolates the distinction this book is built on — values are the outcomes we want; beliefs are claims about which policies produce them. The first kind belongs to voters. The second kind can be checked.

Values are for humans; facts are for tools.

PolicyEngine works the beliefs side of that line and never crosses it. In Democrasim's vocabulary, it is an accuracy multiplier for whatever slice of a vote is policy-driven: it moves Sarah from "I sense this would hurt me" to "this changes my household's income by about this much," computed on her actual circumstances, free to anyone with a browser. The warrant runs deeper than "read the source code if you like" — the rules it applies are open encodings checked against reference calculators, chapter 10's machinery. And its limits are the ones this chapter has drawn. It does nothing for the identity-driven share of her vote, does not loosen her party's hold on her, does not repeal Arrow. It removes one source of noise from one channel. That is a smaller claim than "informed voters fix democracy," and it is the right size.

From toy to scoreboard

Democrasim can only toy with the perception problem, because its ground truth is invented: its voters misperceive effects that are themselves numbers the model made up. The version that terminates in something real belongs to chapter 13. Forecast a government metric — under the law as it stands, or conditional on a policy taking effect, like Medicaid call-center waits if a work-requirement deadline slips — publish the interval, and let the official number grade the forecast when it prints. That is the perception problem with a scoreboard attached: instead of asking whether voters guess policy impacts well, it asks whether anyone can state a policy-relevant number in advance and stand still while reality checks it. As I write, nothing on that docket has resolved, and its conditional entries will only ever resolve on the branch the world takes. The scoreboard has no scores yet; what it changes is the form of the question. A thought experiment is graded against numbers it invented. A forecast is graded by numbers nobody chose.

That reframes what all of this machinery is for. A tax calculator looks like a tool for people who want to trim their taxes. Built open, checked against the law, and pointed at an election, it becomes one component of informed self-government — an input to the accuracy term, offered to whatever share of the vote will take it. Democracy decides what we want; the tools in this book only sharpen the picture of what we will get. Which leaves the question the split saves for last. Facts are for tools and values are for humans — but values move, and the next chapter asks whether a simulation can see them moving before we can.

Chapter 16: Simulating values

Every chapter before this one obeyed the same rule. A simulation earned attention only where its verification chain ended in ground truth: an encoding checked against the statute and a reference calculator, a synthetic population calibrated to administrative totals, a forecast that an official number will eventually grade. This chapter asks the deepest question in the book's set — not what people want today, which surveys can measure, but what people would want after reflection, and whether that is something a model could forecast. The admission rule cannot follow the question that far. No office certifies reflective values; no future print resolves them. So read what follows as horizon, not foundation.

Here is what reaching past the rule looks like in practice. In 2024 I ran the first systematic test I know of on whether a language model can forecast value change, and the headline result was a miss. The miss is the center of this chapter — not the method, not the modest success that preceded it, and not the uses a mature version might someday have. The failure carries the lesson, and the two years since have mostly deepened it.

Why forecast a value

In 1996, Gallup began asking Americans whether same-sex marriages should be legally valid; 27 percent said yes. By the early 2020s the figure was about seven in ten. The change was not random. It followed exposure, generational replacement, and the slow crystallization of arguments, and in retrospect the trajectory looks almost legible — the kind of curve you feel you could have drawn in advance. Could you have, in 1996? And can you now tell which of today's contested values will look settled in a generation, and which will reverse?

This is not a search for moral truth. It is a forecasting question, the same shape as forecasting inflation or rainfall: a survey, a time series, a model, a calibrated interval, and a grade when the next reading lands. The apparatus of chapter 13, with an attitude on the vertical axis instead of a dollar. It matters because decisions get made against contested values constantly, and increasingly by automated systems. “Do what humans want” fractures the moment you ask which humans, and whether you mean what they say now or what they would endorse on reflection. Stuart Russell built his account of controllable AI on exactly that uncertainty: machines should stay unsure what humans want and learn it from behavior. But behavior reveals present preferences — precisely the ones most likely to move. If value change has structure, some of it might be forecastable, and the only way to find out is to forecast it and take the grade.

The backtest

The General Social Survey has asked Americans the same attitude questions since 1972, with wording stable enough to form real time series. I took seventeen of them — spanning sexual morality, drug policy, gender roles, race, religion, confidence in science, guns, capital punishment, and free speech — and gave GPT-4o each variable’s history through 2021, asking not for a point guess but for a spread: the 10th, 25th, 50th, 75th, and 90th percentiles of where the 2024 reading would land. The spread was there to force the model to express uncertainty rather than default to false confidence.

Two deliberately simple baselines set the bar. A linear trend fits a straight line to the history and extends it; a no-change baseline predicts the historical average. Any method worth having should beat both, and beating the straight line is the harder, more meaningful test. The model’s 2024 forecasts came in at a mean absolute error of about 4.8 percentage points, against about 7.2 for the linear trend and about 10.6 for no-change — roughly 1.5 times better than the straight line, roughly 2.2 times better than assuming nothing ever changes. Naming the baseline is most of the honesty here: “2.2 times better” is true only against the weakest opponent. The claim worth keeping is that the model beat a straight line by about half again, and that its advantage concentrated where trends bent: acceptance accelerating on some questions, change decelerating on others, exactly where a straight line goes wrong. Its raw intervals came out too narrow, the familiar overconfidence of language models, and had to be widened with a standard calibration step borrowed from weather forecasting before the 80 percent intervals covered 80 percent of the outcomes.

Read carefully, that is a modest, real result: on three-year forecasts of American social attitudes, a language model beat naive baselines, by more against the weaker one and less against the sterner one. Then came the variable I would have bet the house on.

The miss

Among the seventeen series, same-sex acceptance had the cleanest story. The GSS asks whether sexual relations between two adults of the same sex are wrong; the share answering “not wrong at all” had climbed for half a century, driven by generational replacement and exposure, with every hallmark of a one-way trend. It read roughly 57 percent in 2018, 62 in 2021, and 61 in 2022. It was exactly the series a forecaster feels safest about. In 2024 it fell to roughly 55, about six points below its 2022 reading.

Every method predicted up; the number went down.

The language model missed the reversal. The linear trend missed it. The no-change baseline missed the level and the direction both. Six points is not large by the standards of survey movement, and the reversal may yet prove to be noise, an artifact, or the leading edge of a genuine backlash; that question is still open. The lesson is the direction. On the one series where the trend looked most like destiny, everything pointed one way and reality went the other. Value change carries genuine surprises that no model, statistical or neural, reliably anticipates.

The break in the ruler

The GSS changed how it collects answers, and the change sits directly under this chapter's centerpiece. Facing the pandemic, the 2021 round moved from in-person interviewing to an address-based, push-to-web design, and its response rate fell to roughly 17 percent, against roughly 50 percent in 2022. Mode matters for exactly this kind of question: answering a screen instead of a person removes social-desirability pressure, and self-administration tends to raise the share giving the franker answer — including "not wrong at all." NORC, which runs the survey, cautions users to "carefully examine how a change they are analyzing relates to this methodological shift."

So part of the climb the model trusted — the 2021–22 peak — may be a mode artifact, and part of the 2024 reversal may be that artifact unwinding. Both the rise and the fall sit partly on a methodological break, not on pure attitude change. The caveat does not rescue the forecasts: none of the methods knew about the mode change either, and reality graded them on the numbers as published. What it does is make the miss doubly instructive. The model failed to anticipate the reversal, and the series it extrapolated was itself less solid than it looked. A forecaster's ground truth is a measurement, measurements are made by instruments, and instruments have seams. Anyone proposing to forecast humanity's values should first notice how much trouble we have measuring last year's.

The replication

The miss sent me back to rebuild the experiment properly, and the rebuild cut deeper than the miss. In July 2026 I pre-registered a replication before making a single model call. Twenty GSS items this time, survey-weighted; each model shown the series only through 2022 and asked for 2024; every training cutoff checked against the data's release date, so that no arm could be "forecasting" a number it had already read. And the baselines got sterner. The one that matters for a slow-moving series is persistence — the forecast that 2024 simply equals 2022.

Persistence beat everything.

Across the twenty items, the language models' errors ran 3.9 to 4.1 points against persistence's 3.15. A frontier reasoning model did no better than the cheap ones, and raising a model's reasoning effort made its accuracy slightly worse. The models' 90 percent intervals covered the truth about half the time; properly computed classical intervals covered at their stated rate. And the 2024 wave turned out to be stranger than one famous reversal: eight of the twenty items broke their pre-2022 trend, and four of them — approval of the school-prayer ban, same-sex relations, rejection of traditional gender roles, and marijuana legalization — posted the largest single-wave declines in their items' recorded histories. Every arm, model and baseline alike, missed the same-sex reversal again, with point forecasts between 60 and 66 against an actual of roughly 55; the only interval that contained the truth managed it by being twenty-one points wide.

Squaring the two runs is deflating and clarifying at once: the baselines are most of the result. The first experiment set the bar at a trend line and a decades-long average, and against an average taken over decades of much lower readings, any method that merely notices the recent level looks brilliant. Persistence is the bar that matters, and nothing cleared it.

Recall in a forecast's clothes

The rebuilt experiment included a control the earlier ones lacked, and it explains where much of the apparent forecasting skill in this literature comes from. Run a model on a series it can name, inside its training window, and it is not only extrapolating — it is remembering. Shown the same-sex series through 2010 with its identity visible, one model "forecast" the present at 78, a level the series has never reached in fifty years of asking. Shown the same numbers with the name stripped, the same model said 47.5. Thirty points, from the label alone.

An identified backtest inside a model's training window is a recall test wearing a forecast's clothes. The diagnosis reaches well past this chapter, because a growing literature evaluates language models by "predicting" surveys, elections, and experiments the models have already read about. Two protocols fix it: strip the identities, or forecast strictly forward, where no training corpus can help. The rebuilt experiment adopts both — its forecasts for the 2026 and 2028 GSS waves were registered before the data exist.

A stranger fix is coming into view. Language models trained exclusively on text from before 1931 now exist; such a model has read no poll it could parrot, because Gallup's first national surveys came in 1935, the GSS began in 1972, and the entire quantitative record of public opinion lies beyond its horizon. Early evaluations put their weakness at the elicitation layer rather than the knowledge layer — they can complete text they cannot converse about — which looks like an engineering problem rather than a wall, and stronger checkpoints at the same cutoff are expected within months. The eventual prize is ninety years of graded actuals: every reversal, plateau, and backlash in the polling record, out-of-sample by construction.

The same apparatus will also happily overreach, which is its own warning. Point it at 2050, or 2100, and it dutifully returns a tidy median and interval for each year. I once tabulated exactly those long-horizon numbers. After 2024, I cut the table. The only honest way to read such figures is as an illustration of what the method emits when asked, never as a finding about the future — and the further past the data you push, the more the interval, not the median, is the result worth reporting.

Heterogeneity is the target, not the noise

A tempting mistake runs through this whole subject: imagining that value forecasting converges on a single answer — the value system humanity is "heading toward." It does not, and it should not. Grant the method perfect foresight and people would still hold different things dear; the object worth forecasting is the whole distribution across a population. The reframing has teeth for anything that would consume the forecast. A point estimate invites a system to optimize toward it. A distribution refuses: it says that at the end of any plausible reflection there remain people who weigh liberty over welfare and people who weigh welfare over liberty, and that both are part of the answer rather than error bars around it. A system that took the distribution seriously would favor actions that score acceptably across several value systems over actions that score maximally within one, grow cautious where the value systems disagree most, and treat a move that catastrophically violates any large minority as disqualified rather than merely outvoted. That is value pluralism recast as an engineering constraint — closer to a robustness requirement than a moral theory.

The temptation to print the distribution anyway has to be resisted. An earlier draft of this chapter contained a table of invented population shares — tidy percentages for how much of a reflective humanity would land on which cluster of values. It was exactly the confident fiction this book stands against, dressed as a finding, and it is gone. The distribution is the thing to estimate, not to assume.

Two uncertainties

Any forecast of values carries two kinds of uncertainty that do not reduce to each other. Aleatoric uncertainty is the genuine spread of values across people: a perfect model of a population that will never agree still reports disagreement. Epistemic uncertainty is our uncertainty about what that spread even is: a distribution over distributions, which better data, longer histories, and better methods narrow. They demand different responses. Epistemic uncertainty is the part you can pay to reduce — more surveys, more countries, sterner calibration. Aleatoric uncertainty is the part you must design around, because it remains when the study is done. Both belong in the output, quantified separately: not "humanity will value X," but a probability that it does, an interval around that probability, and an account of how much of the interval is ignorance and how much is real variety. This is the same discipline chapter 13 applied to policy costs, turned on values — and it is harder here, because the ground truth itself is a survey with seams.

Reflection is not time

The deepest problem is not statistical. The appeal of forecasting values is the hope that where attitudes are heading is better than where they are — that the forecast points forward in judgment, not just in time. Nothing in the data licenses that hope. That support for same-sex marriage rose from 27 percent to about seven in ten within a generation shows that time passed and attitudes moved; it does not show that anyone reasoned more carefully. The engines of value change are sociological — generational replacement, media exposure, information cascades, tribal signaling. A model that predicts them predicts a social process, not a deliberative one. Some shifts plainly track moral progress: abolition, the widening of the franchise. Others may be drift with no progress in them. Others might be backlash to perceived overreach — one live reading of the 2024 reversal, and if that reading is right, a model that "correctly" projected continued acceptance would have been wrong in a morally loaded direction.

The philosophers who pressed hardest on this offer warnings more than solutions. Rawls distinguished reasonable from unreasonable comprehensive doctrines — not every value system deserves equal standing in political life — but "reasonable" resists being turned into a forecasting rule; you cannot regress on it. Habermas specified the conditions of undistorted deliberation, equal participation and the absence of coercion among them, and no historical stretch of attitude change is guaranteed to have met those conditions; most plainly did not. Sen and Nussbaum warned about adaptive preferences: values formed under oppression, stable and sincerely held, that it would be perverse to extrapolate forward as though they were considered choices. A person raised to expect little may sincerely want little. A society that has normalized an injustice may show stable, well-measured preferences for it. A forecasting model does not know the difference between a preference reasoned into and a preference ground into place — both are points in a time series — and predicting the second kind accurately would be an achievement in the service of nothing good.

We have no clean account of when temporal change tracks reflection and when it is mere drift. Predicting well is evidence of underlying structure; it is not evidence that the structure deserves honoring. That is a permanent reason for humility, not a bug a larger model removes.

Nor is there one trajectory to defer to. Everything above was measured on one country's attitudes. The World Values Survey, run across roughly a hundred countries since 1981, shows convergence and its opposite: on some dimensions, a broad drift toward self-expression values as societies grow wealthier and more secure — Ronald Inglehart's post-materialist pattern — and on others, religion and family structure and the norms around sexuality among them, persistent cultural zones in which different starting points produce genuinely different paths rather than the same path at different speeds. Recent years have added pointed reversals in several countries. A global forecast would have to model heterogeneity at the level of whole societies, and there is no neutral vantage from which to call one society "ahead" and another "behind."

The governance questions

Suppose the method someday worked: calibrated, replicated, graded forward for years. It would immediately become a technology of power, and who held it would matter as much as how accurate it was. A government could dress paternalism as foresight. A firm could steer preferences rather than serve them. A lab could align a system to projected values that happened to coincide with its own interests. And a forecast would not need a steering hand to move its own target: publishing that a shift is coming could accelerate the shift or suppress it. That feedback runs through all social science and is no reason to stop; it is a reason the adversarial questions would need answers before any of it earned authority.

Who generates the forecasts, and under whose incentives? A university lab answers to peer review, a statistical agency to its legislature, a company to its shareholders — different accountabilities, and none should hold a monopoly. Distributed production under open methods can at least be checked.

Which conditioning assumptions are on the table? A forecast conditioned on broad deliberation differs from one conditioned on polarized media, and the choice between them is itself a value judgment that must be stated where it can be argued with.

What is democratic input for? Forecasts should inform deliberation and never stand in for it; citizens must be able to examine the method, challenge the assumptions, and reject a forecast-based justification outright. How do disputes get resolved? Contested forecasts need appeals — a rival team rerunning the method, a public comment window, an independent review — not an oracle's say-so.

And the hardest one: what happens when the forecast and today's expressed values disagree? The honest answer stays uncertain, and the 2024 reversal is the standing cautionary tale, because a 2020 projection of continued acceptance, used to justify anything at all, would have been overconfident within four years. Whatever governs this work has to be built to absorb surprise and revise — built like the scoreboard, not like a prophet.

Interest without belief

If value forecasting ever earned admission on this book's own terms — calibrated across variables, countries, and horizons; leakage-controlled; adversarially tested; graded forward rather than backtested — it would bear on the largest open question in AI: how systems that act on our behalf should weigh contested human values. Even then its role would be bounded: one input, evidence about where considered preferences might sit, never an authority over them. Today it has earned no such standing. What exists is a research question with one suggestive backtest, one humbling miss, one pre-registered replication in which nothing beat persistence, and one methodological caveat sitting under the whole series. If the work matures, its institutional shape is already described in this book: value forecasting is at bottom a forecasting problem — a probability distribution over what people will come to value, scored against what actually happens — exactly the kind of claim chapter 13's docket exists to post with an interval and grade. The forward-registered forecasts for the 2026 and 2028 GSS waves would be its first entries; none has resolved. There is no track record here, only a method and a discipline for eventually being wrong in public.

One can imagine chaining this book's tools into a single machine — a forecast value distribution feeding a democratic simulation feeding a policy model feeding measured welfare. I have drawn that diagram. It is an unbuilt composition of mostly toy parts, and drawing a diagram is not building a machine.

What can honestly be said is smaller and sturdier than the grand version. Ask people what they value. Model the patterns. Project them forward with calibrated uncertainty, and stay loud about how uncertain the projection is. We cannot know what humanity would choose after reflection, and we should distrust anyone who claims they can. We might, someday, forecast it — badly at first, and a little better with each miss admitted in public. Everything else in this book earned its place by terminating in a check against the world; this chapter did not, and I have tried to keep the difference sharp, because it is the one place where I am reasoning past the edge of what can yet be verified. The correct posture toward it is interest without belief.

And nothing in the choice ahead waits on it. The fork the final chapter returns to — open or closed, graded or oracular, plural or singular — is being decided now, with the tools already built.

Chapter 17: Society in silico

"For the first time, history has an author."

The book opened on that line, and it has been waiting here ever since. Engerraund Serac says it with pride, standing in a control room while Rehoboam's predictions cascade across the displays — every life a trajectory, every trajectory arranged, a society optimized to one man's definition of optimal. He does not claim to see the future so much as to write it; in his machine, prediction and control have fused into a single act. The fiction earns its horror honestly. What it gets right is that the modeling of society was never the dangerous part — the danger was an author: one model, one owner, one objective, no one outside the room who could say it was wrong. Seventeen chapters later, the answer this book proposes fits in a sentence.

History should have auditors, not an author.

Both visions begin from the same recognition — that complex social systems can be modeled computationally, policies simulated before enactment, populations represented household by household, consequences estimated before anyone has to live them. They part on everything after. Serac's model is his alone; its predictions are hidden from the people living inside them; its objective is singular and imposed; it speaks in certainties; those without access are subjects. The open path inverts each property in turn: code anyone can read, predictions anyone can query against their own household, objectives contested among many stakeholders because no single optimizer gets to define "optimal," uncertainty printed as intervals rather than suppressed, and the capacity to analyze held as public infrastructure rather than private advantage. Both paths run on models. They differ in who can see the models, question them, and correct them — which is to say, they differ in everything.

What we built

A year ago, this chapter's job was to describe what somebody would need to build. It reads differently now for a reason the preface confessed: a chapter drafted in 2025 as a description of missing infrastructure had to be rewritten as a field report, the future tense overtaken by the past tense between one draft and the next. A book about the pace of change that was itself outrun by it is an awkward artifact, but an honest one — and the being-outrun is the most direct evidence for the claim that matters most here. The window in which this infrastructure gets built open or closed is not a generation away; it is now. So here is the ledger, scoped and dated as of July 2026: what exists, and then what does not.

PolicyEngine, the open engine of Part II, encodes thousands of federal and state tax and benefit rules and shows them from two angles — the household view, *what does this policy do to my family*, and the society view, *who gains, who loses, and what does it cost*. Its population data is calibrated to more than 9,000 administrative totals, its calculations are cross-checked against independent implementations through validation partnerships with the National Bureau of Economic Research and the Federal Reserve Bank of Atlanta, and benefit screeners such as MyFriendBen run families in four states across the same engine. It is still running and still free to anyone with a browser.

The rules commons is the agent turn made institutional. The Axiom Foundation launches on July 28, 2026, carrying the Part III work: roughly 3,000 US rule files spanning federal law and twenty-eight states, with companion repositories for the United Kingdom, Canada, New Zealand, and Belgium. Its stated mission is to encode all public policy — statute, regulation, agency manual, down to a single London borough's council-tax reduction scheme — with every monetary amount traced to a quoted excerpt of its governing source. In one week this July, its pipeline encoded five African tax-benefit systems and checked them against the SOUTHMOD reference models that UNU-WIDER and its national partners have maintained for a decade, reporting the findings back to the models' authors. On every surface compared against a reference model so far, every mismatch carries an evidence-backed explanation — zero unexplained on the tested surfaces, which is a narrower and more defensible sentence than "correct," and the right one. That week is the strongest evidence this book has for its thesis, and it happened while the manuscript sat open.

Beneath both views sits the data commons: populace, which assembles the calibrated population — synthetic households, tuned to administrative totals, that let a model represent anyone in the country rather than only the respondents a survey happened to reach. It is the successor to the enhanced survey files of chapter 6, and the answer to the asymmetry chapter 2 diagnosed: the government's confidential records were never coming out from behind Section 6103, so the open stack synthesizes its own people and proves them against the totals everyone can see.

The scoreboard is the youngest piece. The Thesis Institute publishes forecasts of government metrics — the first print of an unemployment number; a Medicaid call-center wait time under a policy that may or may not take effect — each carrying an explicit interval, each to be graded when the official figure lands. It launches publicly in mid-September 2026. Its status belongs in the record exactly as it stands: the docket is live, the grades arrive with reality, and not one forecast has resolved.

And one project exists to measure the boundary the others defend. PolicyBench puts today's best language models on a public board and scores them on complete household tax-and-benefit calculations. As of mid-2026, the best model still gets roughly one in nine of those calculations wrong by more than a dollar, and the typical model misses about one in four — family-level errors, a wrong Medicaid eligibility or a wrong SNAP amount, not rounding slips. That measured gap is why the deterministic layer has to exist at all, and the board exists so the claim stays checkable as the models improve.

The five entries are one structure pulled apart along its natural seams — three of the introduction's four questions, given institutional homes with different verification loops; the fourth stays where the introduction left it, with the public. PolicyEngine keeps its name and its users; underneath, it is being recomposed into Axiom, which holds what the law says, and populace, which holds who the people are, with Thesis as the third pole for what will happen: the decomposition chapter 12 argued for, with repositories and launch dates where the argument used to be.

What the split buys is a kind of accountability the closed stack never carried. Rules are checked against statute and reference calculators the day they are written; populations are checked against administrative totals; forecasts are checked against reality when the number prints. An estimate that names something reality will print — a caseload, a revenue line, a poverty rate with a release date — need no longer be a single number issued and forgotten; it can carry an interval and, on the branch the world takes, a grade. The CBO and its peers publish real retrospectives of their aggregate projections; what none of them runs is the per-estimate loop, commitments logged in advance and graded one by one, misses kept on the page. Building that loop is the reason to split the stack, and the loop has not yet closed on a single forecast.

What we haven't

The debit side of the ledger is long, and it belongs in the same list as the credits. Start with uncertainty, which is still mostly unquantified where ordinary users meet the tools. A microsimulation answers "this reform costs fifty billion dollars" when it should answer "fifty billion, give or take this much, and here is the assumption doing the work." Chapter 13 showed the banded version running on one reform; wiring it through production, so that every score arrives as a distribution instead of a point dressed as precision, is unfinished work.

Adoption is early. When Congress passed the One Big Beautiful Bill Act in July 2025 — Public Law 119-21, extending the 2017 tax cuts — the public argument ran on CBO scores and think-tank estimates. Scores mattered enormously: they shaped a live, enacted law worth trillions of dollars. But they were institutional scores, produced by the same closed machinery Part I described, and citizens running their own analyses played no visible part. The tools existed; the debate proceeded as if they did not. The shift from *available* to *expected* is a cultural transition, and weather forecasting — the book's favorite precedent — needed decades to turn "chance of rain" into something ordinary people plan around. Policy analysis has barely started that clock.

The prediction pole has no track record. The Thesis Institute has not resolved a single forecast. Its scoreboard is a set of promises with intervals attached; until reality grades them, they are predictions, not evidence. Saying this out loud, repeatedly, is the entire reason to build a scoreboard instead of a megaphone.

And the AI layer is half-arrived. The rules and the data now exist at production scale, and an AI assistant can call them today; that part is real and shipped. What has not happened is the habit. Ask most assistants a tax-policy question and the answer still comes from training data rather than from a calculation engine — the infrastructure is built, and the reflex of reaching for it is not.

Add it up and this book describes as much aspiration as achievement: the rules layer is further up the mountain than the last draft dared claim, and the prediction pole is barely off the trailhead.

We are partway up, and honest about the altitude.

Why the fork matters

If a society cannot reason about itself, it cannot govern itself. And the alternative to open simulation was never human judgment uncorrupted by models — it is the set of failure modes already in place. Agencies make black-box decisions with proprietary tools; the Joint Committee on Taxation produces the revenue estimates Congress votes on with models Congress cannot inspect. Debate runs on vibes: politicians assert, pundits assert louder, and numbers float past without sources or audit trails. AI systems answer policy questions from training data, confidently and often wrongly — the family-level errors PolicyBench measures, shipped at the scale of every chatbot. And analytic power pools where the compute and the data already are: a hedge fund can model the distributional impact of a tax reform in minutes while a community organization cannot, so the fund's version of the story arrives first and arrives armed. None of it is hypothetical — this is how policy is analyzed today.

Open simulation answers each failure in kind — auditable code where there were black boxes, reproducible results where there were vibes, tools an AI can call where it would otherwise invent, public infrastructure where there was private advantage. None of the substitutions is automatic, and each can be done badly. An open model can still encode a bad assumption; transparency does not make a mistake correct — it makes the mistake findable, which is the property everything else in this book compounds on.

The deepest version of the case is democratic. Democracy asks people to vote on policies without knowing their effects, to debate taxes without calculating impacts, to judge officials without any way to audit their claims — and for most of democratic history that ignorance was not chosen, it was structural: the models sat inside institutions, and expert analysis was expensive and late. What the open stack changes is concrete enough to picture. A candidate releases a tax plan; within hours, independent analyses appear — not from one think tank but from many people running the same open model under different named assumptions. A voter enters her own household and sees the specific dollar effect. A journalist checks the distributional claim against the calculation instead of against a rival quote. The disagreement that survives that process is the real one, about values, argued from a shared factual floor. Chapter 15 sized this claim honestly — information is one lever among identity, institutions, and strategy — but it is the lever infrastructure can move, and now the option to know exists where it did not.

The machines will do the asking

As AI systems become the interface to everything else, more people will meet policy through them than through any government website or newspaper table. Someone asks what a child-credit expansion would do for them, or how a carbon tax lands on a family like theirs, or which candidate's health plan leaves them better off. The system answering can do one of two things. It can synthesize plausible text from training data — inventing eligibility rules, hallucinating statistics — which is what happens today, and why the best frontier models still complete fewer than a third of full federal returns correctly and still miss household calculations at the rate PolicyBench records. Or it can call a tool: look the parameters up in an open encoding, run the calculation deterministically, and return an answer with an audit trail.

The value of the second path compounds. A microsimulation model that takes ten minutes to learn serves one researcher at a time; the same model, reachable by an AI agent, serves anyone who can ask a question in plain language. Every hour spent encoding a rule exactly, tracing it to its source, and racing it against an oracle pays off in proportion to how many agents eventually call it — and that number is climbing. Reliability here is a mechanism, not a promise. When PolicyBench's own explanatory notes were caught inventing the derivations behind scores they described, the fix was not better prose discipline; it was mechanical grounding — regenerate every explanation from the engine's actual internals and reject any that cannot be traced. Even the audit layer needs ground truth.

Nor is the architecture a private bet. The same shape — an AI front end over a deterministic, checkable back end — keeps appearing wherever institutions try to let people ask real questions: the IRS has put AI agents to work summarizing cases and searching guidance, and California's Poppy assistant helps state employees navigate dense policy catalogs. None of those systems is PolicyEngine, and that is the point. The pattern converges because it is the shape that works. The alternative, at the scale of every assistant on Earth, is more access to less reliable information — which no one should mistake for progress.

Brittle and improvable

In the fiction, Serac's machine fails. Rehoboam cannot handle the hosts — beings who do not fit its model, whose choices it cannot predict — and the closed system shatters on contact with genuine novelty. Television resolves these things too tidily, but the lesson underneath is sound: closed systems optimize for their own assumptions, and when the world moves, they break in ways their designers never anticipated. The government modeling monopoly of the 1980s was a mild, real version of the same disease. When the Joint Committee on Taxation produced an estimate, you could disagree with it, but you could not demonstrate the error — models that cannot be challenged cannot be corrected, and so they weren't. The open-source movement changed this not by producing better models, but by making models *improvable*. And in the 2026 stack, openness is machinery rather than gesture: the same encodings anyone can read sit behind gates where tested coverage may only rise and unexplained mismatches may only fall, so the worry that open code quietly rots has an operational answer instead of a hopeful one.

Run the through-line once, end to end. Guy Orcutt proposed simulating a nation household by household in 1957. Government agencies realized the idea behind closed doors, and for decades the machinery answered only to its owners. Think tanks broke the monopoly; open-source projects put the tools in browsers; and agents have now made them buildable in days and checkable line by line against the models governments already trust. Every step in that sequence widened the circle of people who could take part in the argument about what policy does, and to whom. The last step widened something else: who can *build* the argument's instruments, and how fast an error anywhere in them can be found.

An invitation

The fork, restated one last time, is not science fiction. Both paths are live institutional choices being made right now — in code, in procurement contracts, in the defaults of AI assistants, in decisions that look technical and are not. One path rebuilds the analytic machinery of government behind glass: closed, singular, speaking in certainties. The other rebuilds it in daylight: inspectable, plural, graded when reality arrives. The difference was never sophistication — Serac's machine had plenty — but whether anyone outside the room can check it, correct it, and build on it.

So the invitation is quiet, because the work does not need believers; it needs checkers. The code is public. The encodings can be read, the models run against your own questions, the forecast docket watched while it earns or fails to earn a record — the admission rule chapter 3 stated and every chapter since has enforced, now in your hands. Find something wrong and file it; the findings ledgers already run in both directions — divergences dispositioned against our encodings, bugs filed upstream against the reference models themselves. Or reject the premise entirely: argue that these tools entrench technocracy rather than open analysis, that what is measurable will crowd out what matters. I think that argument is wrong, and I would rather lose it in the open, against specifics anyone can examine, than win it by default behind glass.

What the tools actually do is narrow, and it is enough. They cannot tell a society what to value, cannot guarantee that insight gets used wisely, cannot stop a bad-faith actor from trying to game them. What they can do is make the invisible visible: the benefit cliff buried in a formula, the uncertainty a confident headline hides, the interaction between programs that no single-program analysis reveals, the distributional table folded into a bill's fine print. A voter who can see that a plan raises her family's income by twelve hundred dollars may still vote against it, for reasons of value or identity or loyalty — that is her right, and nothing in this book has an opinion about it. The difference is that she chooses with open eyes.

Guy Orcutt died in 2006, forty-nine years after the paper almost nobody read. He lived to see microsimulation become institutional infrastructure, but not public infrastructure; the browsers, the open engines, and the agents that build and check the rules one household at a time all came after him. His insight was per-unit — model society one household at a time, and a picture emerges that no aggregate equation can hold. The agent era supplied the half he never got to see: per-unit verifiability, each household's answer checked against a statute, an oracle, or a real determination, the discipline the whole stack now rests on. The arc from his idea to here bends toward openness — not inevitably, not smoothly, but detectably: monopolies gave way to competition, proprietary models to open code, expert-only interfaces to anyone's browser, and now to machines that must answer for their arithmetic. The next turn of the arc is the one this book was written inside, as AI becomes the way millions of people meet what policy does to them. Whether that turn expands access or expands misinformation depends on a choice that is still open, and on who shows up to check the checkers.

Serac's machine had an author and no auditors, and the show was right about where that ends. The machinery can simulate the futures on offer; only the public it serves can decide which one is worth building. Society in silico is not a destination. It's a method.